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CHAPTER 10

Quantum Confinement: An Ultimate Physics of Nanostructures

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1. INTRODUCTION

There is plenty of room for practical innovation at the bottom, which is an idea that Fennymann addressed. Scientists are now trying to understand the unique physics at the bottom that govern matter and antimatter at the atomic and subatomic levels. It is noteworthy to mention that the detailed knowledge at the nanoscale level will be at the heart of future nanotechnology. Trillions of dollars have been pumped into this research at a feverish rate since the start of the National Nanotechnology Initiative in the United States. Nanotechnology has begun to seep into the national and international consciousness and awareness and is being spoken about as a revolutionizing technology that will change everything from basic building materials to computers to space travels.

The term “nanotechnology” was in use as early as 1974. It was defined by Professor Norio Taniguchi in a paper titled “On the Basic Concept of Nanotechnology” [1]. Here nanotechnology is referred to as the ability to engineer materials precisely at the nanometer level. Similarly Eric Drexler used “nanotechnology” interchangeably with “molecular technology” (now referred to as “molecular manufacturing”) in his book “Engines of Creation” [2]. Drexler placed his nanotechnology focus on molecular machining saying that the “new technology will handle individual atoms and molecules with control and precision.” These definitions clearly distinguish the difference between nanotechnology and bulk technology. In principle, nanotechnology is the application of nanoscience. In turn, nanoscience is the science that relates to objects with at least one dimension (1D) between 1–100 nm. The ability to work with atoms at nanoscale—at the molecular level—with atomically precise design, promises to open up a lot of areas of technological development. Breakthroughs are being developed for nanostructure metals, exactly shaped ceramics and polymers without machining, nanostructure sensors, and nanoelectronics for embedded health systems.

This definition of nanotechnology is useful in outlining a philosophical difference between the previous technologies and the concurrent developments. However, it should be expanded in order to include some aspects of nanotechnology as it is practiced today. In recent years, it has been discovered that many materials have extremely unique properties when they are developed at nanoscale. Many materials also configure themselves in different crystallographic structures not seen in a bulk form of the same material. Many of these properties are different regardless of whether they were formed from the bottom-up or from the top-down approaches.

Over the past few years, the scientific media have devoted a great deal of attention to nanotechnology, yet for many people a clear idea of this rapidly growing field remains elusive. Many herald reports revolutionizing manufacturing, medicine, and energy based on advances in nanoscale science and technology. While the products of nanoscale science and technology are already showing up in our daily lives, broad agreement exists that the major impacts from these advances are still years to decades away. It is the children of today who will inherit as adults the majority of both the benefits and the costs of our current early explorations into nanoscale science and technology.

1.1. Research Trends

What then is the current dominant direction of research? The answer is the development, characterization, and functionalization of nanomaterials. Nanomaterial development represents a critical component in achieving the goals of nanotechnology. As the material used becomes central to the makeup of the device, the properties of that material become central as well. Indeed, in nanotechnology, material, structure, and device are virtually indistinguishable from each other.

Nanomaterials have actually been important in some sense in the materials field for some time. For example, nanosize carbon black particles have been used to reinforce tires for nearly 100 years. Another common example is precipitation hardening [3]. This accidental discovery in 1906 allowed for significant improvements in the strength of aluminium (Al). At that time, researchers could not image the precipitates, but it was discovered later that nanoscale precipitates were the source of this hardening. The advent of the electron microscope allowed for a better understanding of the structure of the precipitates and, thus, allowed for improvements through composition selection and intelligent processing techniques. Today, this tradition continues, but with more engineered techniques. Therefore, clearly the current direction of nanotechnology research focuses on materials research and development.

Nanoscience and nanotechnology are still in their early stages. Envisioned to change almost everything about how we manufacture and approach technology, it is a technological development that is well anticipated and significantly hyped. The current explosion of interest in the field began with the characterization of carbon-based nanotubes (CNTs) in 1990. This discovery set off a flurry of study on other 1D nanostructures—materials that have one growth direction significantly faster than all other growth directions. Examples of 1D nanostructure include nanotubes (NTs), NWs (NWs), nanobelts (NBs), and nanorods (NRs). Two-dimensional (2D) nanostructures have two fast growth directions, so that only one direction is contained to nanoscale dimensions. Examples of these include nanosheets and self-assembled monolayers. Zero-dimensional (0D) nanostructures have all directions confined to nanoscale dimensions. These include quantum dots (QDs) and nanoparticles. What makes all of this interesting is that the materials exhibit significantly enhanced or altered properties because they are confined to the nanoscale. Due to this change in properties, developing control over these materials has been the significant thrust of research in nanotechnology for the past decade. The basic physical and chemical concepts that govern the interactions at play with nanotechnology are not fully understood. This is why the bulk of the research aims at discovering novel properties, unique structures, and morphologies, or basic commercial device fabrication using nanostructures. If nanoscience and nanotechnology are to continue to make a lasting scientific impact, the fundamental concepts and basic theoretical constructs governing materials design and property must be unearthed.

Positioning and patterning nanostructures will play an important role in incorporating these structures into device fabrication, as they provide for a significant amount of ease in scaling up production of nanostructures for industrial use and design of specific patterns and architectures of nanomaterials. Some progress has been made on the positioning of nanomaterials; however, rational control and design of their properties has proven somewhat difficult. This is a case where one of the greatest strengths of nanotechnology is also one of its greatest detriments. Changing the size of a material by only a few nanometers can have a significant impact on the properties of that material, allowing for a great variety and uniqueness to new designs. However, a deviation in the size of a material by only a few nanometers can have a significant impact on the properties of that material, strongly limiting the margin of error allowed.

When measuring properties of a material on a nanoscale, there is a strong correlation between the dimensionality of the material and the physical (or chemical) properties. For example, a small change in the size of a nanostructure (~ 5 nm) can shift its luminescence from the visible light spectrum to the ultraviolet region. Therefore, the precision required to controlling the dimensionality within a few nanometers or less is necessary for the development and use of nanomaterials. In part motivated by a desire to understand these basic concepts, this chapter aims to demonstrate, address, and understand the basics of these nanomaterials and the consequences in the quantum confinement effects (QCE) for the common II–VI and III–V semiconducting nanostructures.

1.2. Nanomaterials

In the natural world, the impact of design on the nanoscale is well known, and nature has evolved some very interesting uses for nanomaterials. For example, some bacteria

have magnetic nanoparticles inside of them that are used as a compass and help provide a sense of direction to the bacterium [4]. Even larger creatures have taken advantage of nanoscale design. Gecko foot-hair, which is nanoscale in size, has shown to be central in the gecko's exceptional ability to climb rapidly up smooth vertical surfaces. The individual hair operates by van der Waals forces, allowing for great adhesive forces [5]. Even the most basic building blocks of biological things are an example of nanoscale design; most forms of movement in the cellular world are powered by molecular motors that use sophisticated intramolecular amplification mechanisms to take nanometer steps along protein tracks in the cytoplasm [6].

Not just used by nature, nanomaterials have also unwittingly have been used by artisans for centuries. When gold (Au) is significantly reduced in size, it no longer has the yellow-metallic appearance that is most familiar but can take on an array of colors [7]. Chinese artisans discovered this when crushing Au to form the red paint that appears on many vases. Separately, medieval artisans in Europe discovered that by alloying gold chloride (AuCl) into molten glass they could create a rich ruby color. Though the cause was unknown at the time, the tiny Au spheres were being tuned to absorb and reflect the sunlight in slightly different ways, tunable to the size of these particles. Figure 1 shows the different Au and silver (Ag) nanoparticles with different colors in terms of shape and size. In continuing this size reduction technology, one of the early nanomaterials of 0D QDs was studied. QDs were first theorized in the 1970s and initially synthesized in the early 1980s. If semiconductor particles are made small enough, QCE begins to assert them. These effects limit the energies at which electrons and holes can exist in the particles. Because energy is related to wavelength of the resulting photon, this means that the optical properties of the particle can be finely tuned depending on its size and shape.

In terms of quantum mechanics, only certain discrete energy levels are allowed within a single atom. If two identical atoms are held at large distances from each others' electrons within, each will have the exact same energy. As those two atoms are brought closer to one another they interact, and no two electrons with the same spin can have the same energy. This governing principle is called the Pauli Exclusion Principle. When a large number of atoms are brought together to form a solid, the discrete allowed energy levels of the individual atoms becomes a continuous energy band. This band structure directly impacts electronic and optical properties of the nanomaterials—the pathway to studying

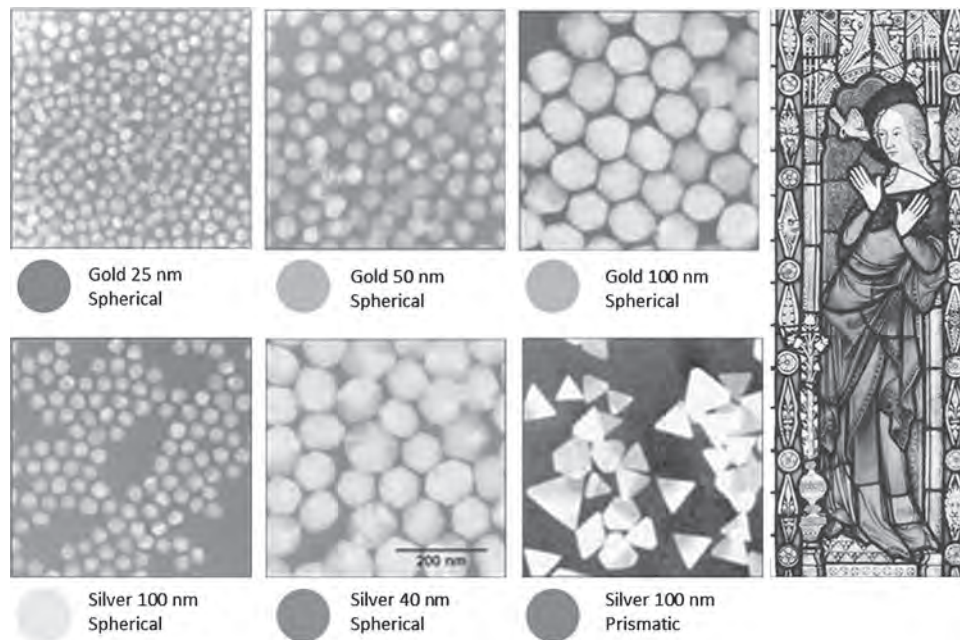


Figure 1. Nanomaterials on glass substrates by grinding up Au and Ag nanoparticles to small sizes. These are SEM images of Au and Ag nanoparticles with sizes ranging from 25 nm to 100 nm.

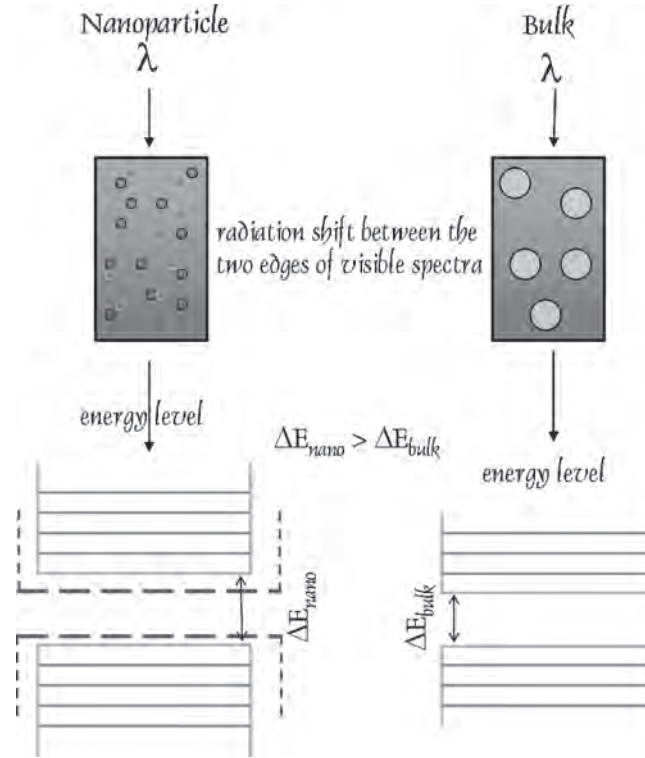


Figure 2. Schematics show nanoparticle shapes and energy states differences in between the bulk and nanoparticles. Nanoparticle has higher energy than the bulk counterpart ($\Delta E_{\text{nano}} > \Delta E_{\text{bulk}}$).

structural, optical, morphological, and electrical properties. Figure 2 clearly shows the energy states in between the bulk and the nanoparticles.

In bulk structures, these properties can be altered only by adding constituents to create defects, interstitials, or substitutions in the material. The impressive phenomenon in QDs is that the optical and electronic properties can be precisely tuned by changing the size of dots in addition to adding dopants. The electronic and optical properties that were thought to be inherent to a material were transformed when the material was formed in a nanoscale. As the dimensions of a material decrease in size, QCE begins to occur on the order of or smaller than the exciton Bohr radius of its constituent compound. This restricts each atom's movement, resulting in the above-mentioned discrete energy levels, represented with a schematic in Figure 2, and differences in material properties. Figure 3(a) shows the size-dependent properties of CdSe QDs. In this image, each vial is filled with a solution of monodispersed CdSe QDs with the particle size getting smaller from left to right. Figure 3(b) shows core/shell CdSe/CdS nanostructures with a quantum yield (QY) from 60–80%. This color difference reflects that the size and shape have a significant impact on the optical and electrical properties of CdSe nanoparticles.

Scientists at Bell Labs were some of the first to determine the direct relationship between QCE in 0D CdSe QDs and the induced higher energy shift in the electronic band structure [8]. It has been demonstrated that when CdSe is reduced in 1D–3D to sizes that are on the nanoscale, the energy bands reconfigure to a band structure that resembles individual atoms [8–10]. That is, with a small enough number, groups of atoms act as a single atom. For this reason, QDs are often referred to as artificial atoms. The QDs then emit light through luminescence or fluorescence, allowing the detection and tracing of the biological targets inside the body [11–13]. The advantage of using QDs for this application as opposed to organic dyes, which are currently used, is that QDs are brighter and more resistant to photobleaching [14].

The wire- or rod-like shape of 1D nanostructures has triggered them to be the source of somewhat intensifying research over the past several years. In particular, their novel

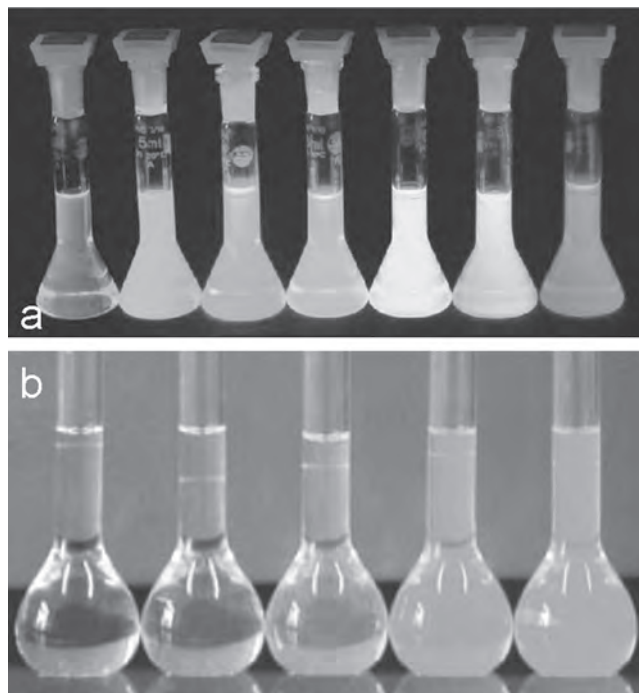


Figure 3. (a) The vials contain solutions with different sized monodispersed CdSe quantum dots ranging from blue to red color. (b) Photograph of core/shell CdSe/CdS nanocrystals with core sizes ranging from 1.2 to 1.5 nm, exhibiting luminescence quantum yields of 60–80% (under room light). Reprinted with permission from [8], D. V. Talapin et al., *J. Phys. Chem. B* 106, 12659 (2002). © 2002, American Chemical Society.

electrical and mechanical properties are the subject of intense research. The category of 1D nanostructure consists of a wide variety of morphologies such as whiskers, NWs, NRs fibers, NTs, nanocables, and nanotubes, among others. Whiskers and NRs are essentially shorter versions of fibers and NWs. 1D structures with diameters ranging from several nanometers to several hundred micrometers have been referred to as whiskers and fibers in early literature, whereas NWs and NRs are more recent and refer to 1D nanostructures whose width is <100 nm.

1.2.1. NWs, Nanorods and Nanotubes

The term NW is widely used to represent 1D nanostructures that have a specific axial direction while their side surfaces are less well-defined [15, 16]. Typically, NWs have a radius that is negligible in comparison to their length. A NR is typically much shorter in length than a NW. NRs show more side facets than NWs; what tends to differ NWs and NRs from NTs is the geometry of their cross-section. NTs are hollow and have a cross-section resembling the perimeter of a circle or the outside of a rectangle. NWs and NRs have solid cores and their cross-section resembles a filled-in circle or hexagon. For the NWs, the cross-section is typically so small relative to its length that any faceted structure the cross-section may have is negligible. NRs, on the other hand, have a length closer to the order of magnitude of their width. In general, NRs are a short NW and are therefore considered a simple subgroup of this nanostructure. As such, the term NWs will be used to refer to both NWs and NRs. NWs have been successfully synthesized out of a wide range of materials, including ZnO [17], GaN [17], GaAs [18], InP, TiO₂ [18], In₂O₃ [18], and indium tin oxide (ITO) [19].

1.2.2. Nanobelts and Nanorings

NBs are confined to a nanoscale, with the 3D being relatively very long. It exhibits faceted side surfaces; the cross-section is rectangular. Pan et al. [20] first reported the transparent semiconducting oxides synthesized in a belt-like manner. The reported materials used

to synthesize these NBs are ZnO, CdO, In_2O_3 , Ga_2O_3 , and SnO_2 [20]. These materials are all transitional metal oxides, ranging over three different elemental groups (II–VI, III–VI, and IV–VI) and at least five types of crystallographic structures. Since this first report, NBs have also been synthesized in non-oxide semiconductors such as ZnS [21], CdS [22], CdSe [23], and ZnSe [24].

NBs have several unique properties that make them amenable to study and for use in nanotechnology. In general NBs can be synthesized as single crystals that are relatively long (1–2 mm). The width of the NBs can range from as much as $100\text{ }\mu\text{m}$ to as little as 6 nm [25]. Though the stacking faults may be present, the NBs are also essentially dislocation and defect-free. This allows for tuning of properties and of catalytic surfaces, and can have a profound effect on the structure and properties of the synthesized material. Many NBs are made of ceramic materials, notorious for their rigidity. However, at the nanoscale NBs are very flexible, enduring great strain without breaking. This strain is also fairly reversible; because of the lack of presence of dislocations, NBs should be extremely resistant to fatigue and failure. Figure 4 is a schematic summary of 1D nanostructures that have been already reported: (a) NWs and NRs, (b) core/shell structures with metallic inner core, (c) nanotubules/nanopipes and hollow nanorods, (d) heterostructures, (e) nanobelts/nanoribbons, (f) nanotapes, (g) dendrites, (h) hierarchical nanostructures, (i) nanosphere assembly, and (j) nanosprings.

1.3. Chiral Vector

CNTs are made up of a hexagonal network of C atoms forming a crystalline graphite sheet. This sheet is rolled up to form a tubular structure. If the tube consists of only a single C sheet that meets end on end, then the CNT is referred to as a single wall NT. However, if the NT consists of multiple sheets rolled up coaxially or if the NT rolls up somewhat spirally, then the CNT is referred to as a multiwall NT. The other important

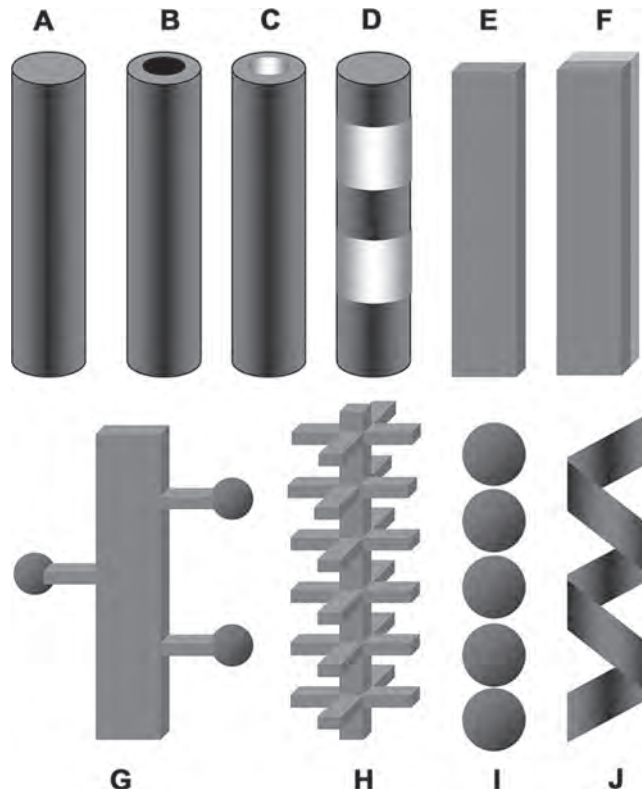


Figure 4. Schematics of (A) NWs and nanorods, (B) core/shell structures with metallic inner core, (C) nanotubules/nanopipes and hollow nanorods, (D) heterostructures, (E) nanobelts/nanoribbons, (F) nanotapes, (G) dendrites, (H) hierarchical nanostructures, (I) nanosphere assembly, and (J) nanosprings.

characteristic of the NT is the chirality, or the direction the NTs are rolled. The chirality has a large impact on the physical properties [26, 27]. There are three distinct types of NTs based on their chirality:

- (1) chiral,
- (2) armchair, and
- (3) zigzag.

The difference between the three is understandable considering the idea of a chiral vector and angle. The chiral vector is defined by

$$C_V = na_1 + ma_2 \quad (1)$$

where a_1 and a_2 are unit vectors and n and m are integers. The chiral angle θ is measured relative to a_1 , shown in Figure 5. Chiral angles between 0° and 30° are known as chiral NTs. Armchair NTs are formed when $n = m$ and the chiral angle is 30° . It is called an armchair NT because the pattern C lattice forms when it is rolled this way. Zigzag NTs are formed when either n or m is zero and the chiral angle is 0° .

Figure 5 shows a 3D schematic view of each of these types of CNTs. Chirality is directly related to the electrical conductivity of CNTs. Armchair NTs have an electronic conduction closely resembling that of a metal [27]. In fact, the standard rule is that for a given (n, m) NT, if $2n + m = 3q$ (where q is an integer), then the NT is metallic. In theory, metallic NTs can have an electrical current density more than 1000 times stronger than metals such as silver and copper [27]. Alternatively, zigzag NTs tend to have the same electronic properties as a semiconductor, where electrons must overcome a band gap in order to enter the conduction band. It is this wide range of electronic properties that make NTs interesting in the field of nanoelectronics.

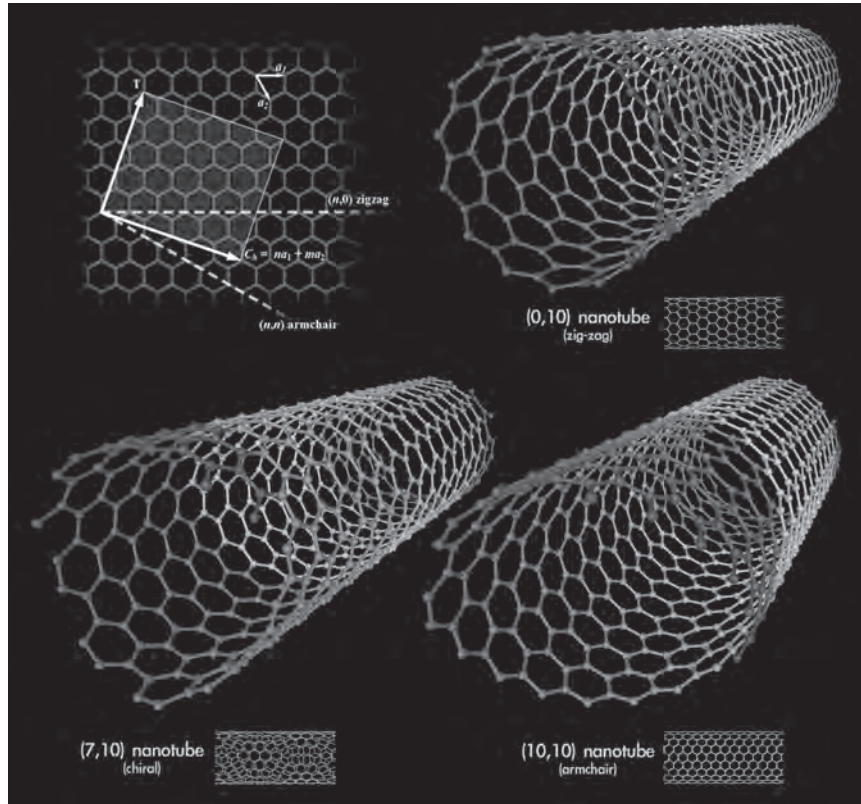


Figure 5. The 3D models of different chirality nanotubes. Model of a graphene sheet illustrating the lattice vectors a_1 and a_2 and the chiral vector C_V . This model helps describe different types of nanotubes.

One interesting phenomenon associated with the metallic-conducting NTs is ballistic conduction. Ballistic conduction allows electrons to flow through the NT without collisions [28]. Therefore, it has a quantized conduction band and no energy dissipation. The lack of energy dissipation means that it generates no heat when conducting electrons from the valence band to conduction band [29]. The impact of having electronic components that have quantized conduction and generate no heat would be significant for electronic packaging applications. The electronic packaging industry is rapidly approaching the limits of current technology and heat management as miniaturization packs more and more components with increasingly fine-feature sizes into ever-shrinking devices.

2. QUANTUM CONFINEMENT

Recent trends in the synthesis of nanostructures covers a wide range of materials, including the III–V and II–VI semiconducting materials. New science waits for discoveries in nanomaterials research, in particular to the quantum physics, such as quantum confinement effect (QCE) at nanoscale. It is expected that control of the nanostructure's dimension will facilitate the study of QCE since the diameter of nanostructures will be beyond the exciton Bohr diameter. We have considered several parameters in this chapter that affect the QCE in nanostructures, such as barrier layers and band offsets. Until today, however, there have been few nanostructures that have been synthesized down to this subatomic level. Self-assembled nanostructures basically do not show any blue shift, but the core/shell nanostructures exhibit a significant blue shift, leading to the QCEs. In this chapter, we will discuss and address these II–VI and III–V nanostructure properties toward the studies of QCE—the ultimate physics of nanomaterials.

Figure 6 shows a schematic potential well as the region surrounding a local minimum of potential energy. Energy captured in a potential well is unable to convert to another type of energy because it is captured in the local minimum of a potential well. Therefore, a body may not proceed to the global minimum of potential energy, as it would naturally tend to due to entropy. Energy may be released from a potential well if sufficient energy is added to the system such that the local minimum is surmounted. In quantum physics, potential energy may escape a potential well without added energy due to the probabilistic characteristics of quantum particles; in these cases, a particle may be imagined to tunnel through the walls of a potential well.

QCE can be observed once the diameter of the particle is of the magnitude of the wavelength of electron wave function. When the materials are so small, their electronic and optical properties deviate substantially from those of bulk materials. A particle behaves as if it were free when the confining dimension is large compared to the wavelength of the particle. During this state, the band gap remains at its original energy due to the continuous energy state. However, as the confining dimension decreases and reaches a certain

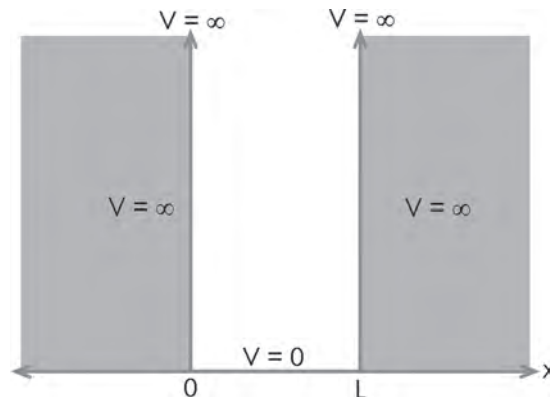


Figure 6. A simple schematic of a potential well surrounding a local minimum of potential energy. Energy captured in a potential well is unable to convert to another type of energy because it is captured in the local minimum of a potential well.

limit, typically in nanoscale, the energy spectrum turns discrete. As a result, the band gap of the nanostructures becomes size and shape dependent. This ultimately results in a blue shift in optical illumination as the size and shape of the particles decreases. Specifically, QCE describes the phenomenon results from electrons and holes (excitons) being squeezed into a dimension that approaches a critical quantum measurement, called the exciton Bohr radius.

In order to understand the QCE, we need to know the basics of quantum mechanics; namely, the role of a particle in a box. Quantitatively, the energy in a box can be defined by $E_n = n^2 \hbar^2 / 8mL^2$, where n = quantum number, m = mass, and L = the size of a particle. In the case of semiconductors, this simply means that the band gap, starting from the bulk value, increases as the size of the nanocrystal decreases. In the bulk solids, the energy levels are closely spaced, and thus form quasi-continuous bands. At nanoscale, the energy-level separation increases and discrete energy levels are observed, as shown in Figure 7. Calculations on different systems show that QCEs are observable at sizes <10 nm for most of the nanomaterials. The onset of QC depends on a number of parameters, such as the dielectric constant of the semiconductor and effective masses of the charge carriers.

In principle, the QCE keeps excitons trapped in a small area and occurs only when the nanostructure dimension is very small, which approaches the size of an exciton in bulk crystal, called the Bohr exciton radius, defined by $a_B = \hbar / m_e c \alpha$, where $\hbar$ is the Planck's constant, m_e is the electron rest mass, α is the fine structure constant, and c is the speed of light. The electronic and optical properties of nanomaterials are affected by size and shape. Well-established technical achievements, including QDs, were derived from size manipulations and investigation for their theoretical corroboration on QCE. The major part of this theory is the behavior of the exciton ensembles more like an atom as its surrounding space shortens. A rather well approximation of exciton behavior is the 3D model of a particle in a box. The solution to this problem provides a sole mathematical connection between energy states and the dimension of space. It is obvious that if volume or the dimensions of the available space decreases, the energy of the states increase. The schematic band diagram shows the change in electron energy level and band gap between nanomaterial and its bulk state. The relationship between energy level and dimension spacing can be described by

$$\psi_{n_x, n_y, n_z} = \sqrt{\frac{8}{L_x L_y L_z}} \sin\left(\frac{n_x \pi x}{L_x}\right) \sin\left(\frac{n_y \pi y}{L_y}\right) \sin\left(\frac{n_z \pi z}{L_z}\right) \quad (2)$$

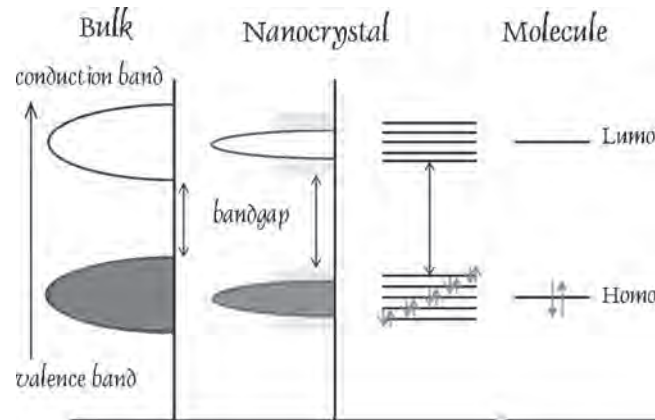


Figure 7. Band schematics for the bulk, nanocrystals, and molecules. The density of states clearly shows differences for the different crystals dimensions in conduction and valence bands.

where

$$E_{n_x, n_y, n_z} = \frac{\hbar^2 \pi^2}{2m} \left[\left(\frac{n_x}{L_x} \right)^2 + \left(\frac{n_y}{L_y} \right)^2 + \left(\frac{n_z}{L_z} \right)^2 \right] \quad (3)$$

This is the energy level in the quantum level of materials/atoms. Due to the 1D confinement, the electron energy levels can be found from the solution of the Schrödinger equation for an infinitely deep potential well with the form

$$E_n = \frac{\hbar^2 \pi^2}{2\mu L^2} n^2 \quad n = 1, 2, 3, \dots \quad (4)$$

where μ is the effective mass of the electron and the dependence of the energy levels on L^2 is the quantum size effect. The dispersion relation for the conduction band can be written as (for confinement in the z direction)

$$E(k) = E_c + E_{nz} + \frac{\hbar^2 (k_x^2 + k_y^2)}{2\mu} \quad (5)$$

where E_{nz} is given by Eq. (4). For the conduction band, the density of states per unit area in a 2D QW system in the sub-bands can be expressed as (for energies greater than $E_c + E_{nz}$) $g(E) = \mu^2 / \pi \hbar^2$. Thus, for each quantum number nz the density of states is constant and the overall density of states is the sum of these for all values of nz , which results in a staircase-type distribution with a step height given by $g(E)$.

In a 1D QW system, the carriers are confined in two directions (z and x), and the dispersion relation for the conduction band can be written as

$$E(k) = E_c + E_{nz} + E_{nx} + \frac{\hbar^2 k_y^2}{2\mu} \quad (6)$$

In each of the sub-bands, the density of states can be described as a function of energy by $g_e(E) \propto E^{-1/2}$. In a 0D QD system, the carriers are confined in three directions, and the dispersion relation for the conduction band can be expressed as

$$E(k) = E_c + E_{nz} + E_{nx} + E_{ny} \quad (7)$$

In this case, the density of states is described by a set of discrete δ -functions as shown in Figure 8. Thus, QDs are generally referred to as artificial atoms due to the similar discrete energy-level structures. It is noteworthy to mention that the density of states in the valence band has a similar distribution to that in the conduction band state.

In the past few years, the quantum heterostructures composed of group III-N such as GaN, AlN, InN, and their ternary compounds with hexagonal crystal wurtzite structures and wide band gaps covering from ultraviolet to red have attracted much attention due to their potential device applications in electronics and optoelectronics and interesting physics [30]. The high-brightness blue/green light emitting diodes and laser diodes have been reported in experiments based on the wurtzite nitrides QW heterostructures [30].

Recently, on the basis of the great progress of crystal-growing technology, the wurtzite GaN and ZnO NW quantum structures, also referred to as quantum well wires (QWW), have been synthesized [31]. The 1D structures of wurtzite nitride QWW with wide band gaps are ideal electronic confined systems for the fundamental studies of their physical properties and for the fabrication of new optoelectronic nanodevices [32]. For example, the self-organized ZnO and GaN QWW ultraviolet lasers have revealed a narrow emission line width and relatively low threshold [31]. The Raman scattering spectra of quasi-1D wurtzite QWWs [33] also showed more complicated phonon vibration properties than those in zincblende QWs or quantum wires. Both the theoretical and experimental works revealed that the phonon modes and electron-phonon interactions in the wurtzite QWs are obviously different from the cases in cubic lattice QW systems due to the anisotropy of wurtzite crystal [34]. To the best of our knowledge, there are few works

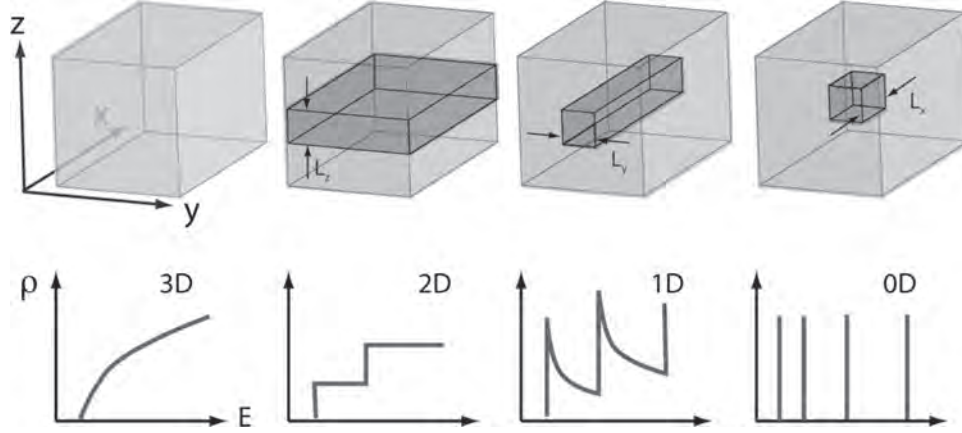


Figure 8. Schematics of density of states as a function of energy in bulk, quantum wells, quantum wires, and quantum dots. E_c and E_v indicate the bottom of the conduction band and the top of the valence band, respectively.

published that have investigated the polar optical phonon modes and their electron-phonon interactions in the wurtzite QWWs. It is well known that at elevated temperatures the scattering of electrons by optical phonons plays a dominant role for various electronic properties. The electron-phonon interactions and scattering also govern a number of important properties of quantum heterostructures, including hot-electron relaxation rates, interband transition rates, and room-temperature exciton lifetimes. The optical and transport properties of wurtzite quantum heterostructures can also be strongly influenced by the electron-phonon interactions. On the other hand, because of the reduction of the dimensionality and the anisotropy of wurtzite structures, the properties of optical phonon modes in wurtzite QWWs have more distinct phonon branches [35].

In the previous studies of the polar optical phonons in wurtzite planar heterostructures, the dielectric continuum (DC) model and Loudon's uniaxial crystal model have been widely adopted [6]. As pointed out by Wendler et al. [36], the validity of the DC model is given by two basic facts. First, the Frohlich type of electron-phonon interaction plays an important role only in the center of the Brillouin zone because only long-wavelength optical phonons produce large polarization fields. And, second, the results of microscopic calculations of the optical phonons are in good agreement with those of the DC model [37]. For example, Butcher et al. [38] calculated the electron-optical phonon scattering rates on the basis of microscopic description of the phonon spectra; results reveal that the DC models agree fairly well with the microscopic calculation. Bordone and Lugli [37] and Dharssi and Butcher [38] have also clearly indicated that the electron transport in QWs is well described by the DC approximation. Furthermore, Zhu [39] compared the potentials of LO-phonon modes in superlattices and found that the LO modes determined by the DC model agree perfectly with results calculated from the microscopic model. In fact, the Raman scattering spectra [40] and the microscopic calculations [40] also clearly showed that the DC model well describes the LO phonons in quantum-confined systems. In general, the microscopic models of optical phonon spectra dealt with the quantum-confined systems with a few monolayers (ML) to tens of MLs [41]. On the other hand, the experimental results of angular dispersion of polar phonons and Raman scattering in wurtzite planar heterostructures are proven to be in good agreement with the calculations based on Loudon's uniaxial crystal model [42].

It is necessary to account for the origin and significance of the wurtzite QWW model. In 2002, Lauhon et al. and Dayeh et al. [43] reported Si and Ge core/shell and multishell NW heterostructures. On the basis of the structures brought forward, Zeng et al. [44] predicted the single-electron sub-band properties and transport properties in multi-layer GaAlAs/GaAs systems, and the systems exhibit some interesting and unique behaviors unexpected in other nanostructures. Tangarife et al. [45] and Aktas et al. [46] investigated the hydrogenic impurity states in coaxial GaAs/AlGaAs QWWs with applied magnetic

fields and electric fields, respectively. Results reveal that there are two peaks in the curve of the 1s-like state binding energy versus QWW radius in magnetic fields [45]. A sharp decreasing behavior of the donor binding energy has been observed in the electric-field-biased coaxial QWWs [46]. The motivation behind the study of these composite QWWs mainly lies in the evident fact: transition energies and optical dynamics in these structures can be precisely designed by changing the internal core diameter and thickness of each shell [44]. For instance, due to their inner configurations for electrons and holes, a long life-time exciton and large dipole moment may be anticipated, which consequently enhances their nonlinear optical properties in a weak field [47]. Though the coaxial multilayer QWW structures made by wurtzite GaAlN/GaN materials in an experiment have not yet been reported, as predicted by Zeng et al. [48], these structures may be achieved from GaAlN/GaN materials in the near future. Figure 9 shows a schematic of wurtzite AlN/GaN/AlN cylindrical QWW and the corresponding potential profile. Schematic view of wurtzite AlN/GaN/AlN cylindrical shows quantum well wires and corresponding potential profile. The axial direction of the quantum well wire is taken along the c -axis of the wurtzite crystal materials.

2.1. Nanoscale

2.1.1. Basic Views

“Nano” comes from the Latin word “nanus” meaning dwarf and is used to describe very small quantities. The prefix “nano” indicates a scaling factor in exponential notation of 10^{-9} m. Thus a nanogram, nanosecond, and nanometer are one billionth of a gram, a second, and a meter respectively. Some sense of just how small these quantities are can be gained by considering the speed of light. While in one second light travels 186,000 miles, in a nanosecond it travels just 11 inches. Put another way a nanometer, abbreviated nm, is about 1/10,000th—1/100,000th the diameter of a single human hair.

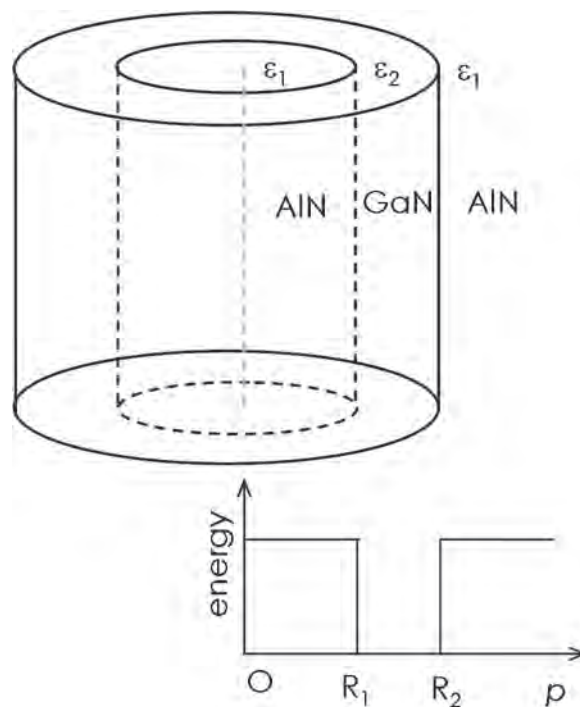


Figure 9. (a) Schematic view of wurtzite AlN/GaN/AlN cylindrical quantum well wires and corresponding potential profile. The axial direction of the quantum well wire is taken along the c -axis of the wurtzite crystal materials. Reprinted with permission from [48], L. Zhang and J. Shu *Semicond. Sci. Technol.* 20, 592 (2005). © 2005, IOP Publishing.

The terms nanoscale science and technology, nanoscience, and nanotechnology are used somewhat interchangeably in casual conversation and in the popular media. In general, nanoscience may be thought of as the study of chemical and physical consequences of manipulating materials on a 1–100 nm length scale. In turn, nanotechnology is concerned with the developing of tools for characterizing and manipulating materials at this scale and exploiting these tools for the development of new products or processes. As the underlying science and derived applications are so often intermingled in this field, the more general term of nanoscale science and technology, covering both of these aspects, is often to be preferred.

2.1.2. Materials Perspectives

Materials' properties—physical, chemical, and electrical—depend on how their atoms and molecules interact with each other and with the other atoms and molecules near them. These interactions, in turn, depend in part on the quantity of material. In bulk form—that is, billions or more atoms, the scale on which we normally consider and confront matter—only a tiny fraction of the atoms are typically exposed on the surface of the material. Most atoms are inside and interact with only others of their own kind. Obviously this can vary depending on what shape the material takes; if hammered into a thin foil or drawn out into an elongated wire, for example, more of the atoms or molecules will be on the surface. But as a first approximation, the proportion of atoms on the surface varies inversely with the numbers of atoms.

In nanoscale quantities, a much higher percentage of atoms are exposed on the surface than is the case for bulk quantities. If you consider the case of atoms packing together in regular cubical or spherical arrays, then simple “ballpark” calculations reveal a marked difference in the percentage of atoms exposed on the surface of nanoscale versus bulk quantities. A quantity of a billion to a trillion atoms typically will have less than one-half of one percent of its atoms exposed on its surface. But in quantities of less than 1000 atoms, over 50 percent are likely to be exposed on the surface, available for interaction with the surroundings.

These exposed atoms are thus available to interact with atoms from the surroundings. Moreover, these atoms exposed at the surface are often bonded less stably to others of their own kind than are the more deeply buried atoms. Also the arrangements of their electrons are changed, making them more likely to interact with the other atoms surrounding them. For these reasons, the shift from bulk to nanoscale quantities can have major impacts on properties. For example, in bulk form Au is a shiny golden-colored, non-reactive metal; it does not readily tarnish or oxidize, making it ideal for use as jewelry as it does not interact with the oils, moisture, or acids from our skin. And, far from being chemically non-reactive, nanoparticles of Au may bond with certain other elements, becoming excellent catalysts. Such dramatic shifts in physical and chemical properties are not specific to Au, but rather are typical on reducing the particle size of many different types of materials to nanoscale dimensions. Materials that are normally electrical insulators may become semiconductors or conductors when reduced to nanoscale-sized particles.

As we gain knowledge about how atoms and molecules actually interact and what types of environments and situations promote or inhibit certain types of interactions, we are moving toward more accurate and comprehensive understanding of our universe. This kind of progressive learning about the universe lies at the core of science. Enhanced knowledge of matter derived through nanoscale investigations then has the potential to reveal new laws of science and to reshape our conceptions of some core aspects of scientific understanding. The advances of nanoscale science and technology are illustrating in dramatic ways that the frontiers of our knowledge often lie at the intersections of these traditional fields, and require knowledge, perspective, and methodologies from several of these simultaneously.

2.1.3. Applications

It is noteworthy, for example, that while materials scientists and chemical engineers may use the most sophisticated and complex of their modern tools to carry out relatively crude nanoscale manipulations, each and every cell in the world—ranging from individual microorganisms to those of all of our human tissues—routinely carries out much more complex operations on an ongoing daily basis. In this realm, nanoscience and nanotechnology are already a part of our everyday lives and becoming more deeply intertwined with them with each passing day. Current applications and products of nanotechnology are already in use, for example, in the following commercial areas:

- **Electronics.** Computers, cell phones, etc. depend on ever smaller and therefore faster circuitry to carry out their increasingly sophisticated operations. Transistors have shrunk to a few tens of nanometers in size, and research is ongoing to create switches and other circuit components as single molecules.
- **Textiles.** Nanoparticle coatings on synthetic fibers have produced fabrics with multiple desirable properties: appealing texture (i.e., feels soft and luxurious), breathable but largely waterproof, stain resistant, long-lasting.
- **Autos.** Beyond the electronics increasingly integrated into our cars, lighter and stronger side panels and safer tires have been developed through nanotechnology, together with recreation tools smaller such as DVD players.
- **Sport.** Tennis balls with a better (nanotechnology-based) seal and therefore longer life complement tennis rackets infused with nanoparticles to provide improved functionality.
- **Paints.** The ability to now create color without pigments, based on size-sorted nanoparticles and enabling permanent, non-fading paints, is reshaping the industry.
- **Cosmetics/sunscreen.** Better penetration and balance of coverage and transparency is being offered by reducing the size of cosmetic constituents to nanoscale dimensions.
- **Solar cells.** There exist many potential renewable energy technologies in the form of solid-state devices such as solar cells, which convert solar energy in the form of light to the more practical form of electricity. In addition, a large collection of condensed matter phenomena involve the conversion of energy from one form to another, and some proceed with efficiency near unity. Consequently, the study of energy conversion in materials is a field full of opportunities for practical and socially significant applications.

Devices made from semiconductor materials are the foundation of modern electronics, including radio, computers, telephones, and many other devices. Semiconductor devices include the transistor, solar cells, many kinds of diodes including the LEDs, the silicon controlled rectifier, and digital and analog integrated circuits. Solar photovoltaic panels are large semiconductor devices that directly convert light energy into electrical energy [48]. In a metallic conductor, current is carried by the flow of electrons. In semiconductors, current can be carried either by the flow of electrons or by the flow of positively-charged “holes” in the electron structure of the material.

Si is used to create most semiconductors commercially. Dozens of other materials are used, including Ge, GaAs, GaN, ZnO, and SiC. II-Os and III-Ns and their low-dimensional quantum structures have attracted considerable attention because of their potential applications in the high-performance optoelectronics and information processing [49]. It has been known that the built-in macroscopic polarizations, which consist of the spontaneous polarization and the strain-induced piezoelectric polarization, play a critical role in carrier recombination in the wurtzite GaN- and ZnO-based quantum structures. These polarization charges generate a built-in electric field directed along the c axis (along the [0001] direction), leading to the so-called quantum confined Stark effect (QCSE). On the other hand, it has been discussed that the carrier localization formed in the plane of GaN- and ZnO-based QWs, e.g., InGaN QWs and CdZnO QWs (with potential fluctuation) can suppress lateral carrier diffusion, so the probability for carriers to be

trapped by nonradiative recombination centers will be dramatically reduced [50]. Therefore, GaN and ZnO can be an ideal active system to develop ultraviolet emitters with high radiative quantum efficiency and better performance due to the lack of adapted substrate for epitaxial growth of nitride materials and to the subsequent structural defects acting as nonradiative recombination centers.

3. CANDIDATE MATERIALS

Nanoscience is concerned with nanomaterials and systems whose structures and components exhibit novel and significantly improved physical, chemical, and biological properties, phenomena, and processes. Nanoscience aims to understand the novel properties and phenomena of nano-based nanoentities. Reducing the dimensions of structures leads to nanoentities with novel properties, such as CNTs [27], nanomaterials, DNA-based structures, and laser emitters. Such new forms of materials and devices herald a revolutionary age for science and technology, provided that we can discover and fully utilize the underlying principles.

A lot of optical and electronic materials have been investigated, such as Si [51], GaAs [52], InAs [53], InP [54], and GaN [55], including CNTs [56] that will be used for nanoelectronic, optoelectronic, biomedical, and spintronic devices. However, for the viability with the environment and study of QCE—the ultimate physics of nanostructures—we need a material that has both the outstanding physical properties and application in nanoelectronics with biocompatibility that will not harm the environment. In this regard, ZnO and related oxide materials (SiO_2 , TiO_2 , CdO , and BeO) are promising since they are biomaterials and have the largest exciton and biexciton binding energies in the semiconducting materials family of ~ 60 and ~ 15 meV, respectively; the driving force for the QCE. The exciton and biexciton binding energies of ZnO are two-and-a-half and two times larger than that of GaN and the thermal energy (25 meV), respectively. With exciton binding energies greatly exceeding $k_B T$, illumination of excitonic solar cells generates tightly bound electron-hole pairs. In order to separate the charge, the exciton must diffuse to the junction without recombining. In parallel, no oxidation will occur in contact to device operation, which is a notable drawback in the III-Vs materials. In addition, oxide materials have significant advantages for the following reasons:

- (1) abundant mineral sources of Zn, Cd, and Mg,
- (2) natural O source and
- (3) commercially available of larger area and higher quality single crystal substrates [57–58].

The hydrothermal method has been employed to grow ZnO single crystals which are transparent and already in the market.

In addition, the II-O semiconductors band gap energy can be engineered, which can cover the full color crystal display; these materials can cover the broad spectrum of studies and applications, ranging from the fundamental physics to nanoscience and full color crystal display to energy harvesting to biomedical devices. The band gap energy of ZnO can be engineered from 2.24–7.60 eV by alloying with CdO and BeO materials. These wide band gap II-O materials will be used for the fabrication of exciton- and polariton-based laser diodes (LD) and light-emitting diodes (LED) [59–60]. In addition, oxide materials will also be used for applications in varistors, transparent high-power electronics, surface acoustic wave devices, photocatalysts, piezoelectric transducers, chemical and gas sensors, dye-sensitized solar cells, security alerts, hydrogen-selective sensors for use with proton-exchange membrane and solid oxide fuel cells for space craft and other applications. The fundamental properties of the most common III–V and II–VI materials are plotted in Table 1.

3.1. Common Nanomaterials

Nanomaterials encompass a wide range of materials including nanocrystalline materials, nanocomposites, nanoparticles, CNTs, and QDs. The common link between all these

Table 1. The most common III–V and II–VI materials with their bulk properties.

Material	Crystal structure	Lattice constant (Å)	Band gap (eV)	Density (gm/cm ³)	Dielectric constant	Melting point (K)
GaAs	Zincblende	5.653	1.42	5.32	12.46	1240
GaP	Zincblende	5.450	2.27	4.13	11.10	1730
GaSb	Zincblende	6.096	0.75	5.63	15.70	985
InAs	Zincblende	6.058	0.35	5.69	14.60	1215
InP	Zincblende	5.869	1.35	4.81	12.40	1335
GaN	Wurtzite	3.189	3.44	6.09	10.40	2227
InN	Wurtzite	3.544	0.80	6.81	15.00	1373
AlN	Wurtzite	3.112	6.28	3.25	8.50	3273
ZnS	Zincblende	5.410	3.61	4.11	8.90	2196
CdS	Zincblende	5.832	2.50	4.87	8.90	1750
ZnSe	Zincblende	5.668	2.69	5.26	9.10	1790
CdSe	Zincblende	5.007	1.74	5.65	9.30	1350
ZnTe	Zincblende	6.104	2.39	5.65	7.40	1570
CdTe	Zincblende	6.482	1.43	5.86	10.20	1041
ZnO	Wurtzite	3.245	3.37	5.66	3.87	2250
BeO	Wurtzite	2.713	7.60	3.01	7.74	2547
CdO	Rocksalt	4.686	2.24	8.15	6.20	1430
MgO	Rocksalt	4.524	6.20	3.58	9.90	3125

materials is that they all have microstructural features at nanoscale. By virtue of its structure, nanomaterials exhibit different physical, chemical, electrical, and magnetic properties than conventional materials.

Nanomaterials are materials possessing grain sizes of the order of a billionth of a meter. They manifest extremely fascinating and useful properties, which can be exploited for a variety of structural and nonstructural applications. All materials are composed of grains, which in turn, comprise many atoms. These grains are usually invisible to the naked eye, depending on their size. Conventional materials have grains varying in size anywhere from 100s of μm to mm.

Particles, complexes, tubes, coatings, active surfaces, and devices are being explored on the nanoscale. Assembly of nature's building blocks (e.g., carbon, nucleic acids, lipids, and peptides) along with the combination of different materials (e.g., CdSe, ZnS, Au, Ag, Si, ZnO, and thin films) are leading to insightful understanding and the creation of new scientific tools. Chemists and physicists have been manipulating matter on the molecular level for centuries. When one looks at the absolute elegance of the nanometer scale biological system, however, one is compelled to create, understand, manipulate, and control systems with equal elegance.

Linear nanostructures such as NWs, nanotubes, or nanorods can be generated from different material classes, e.g., metals, semiconductors or carbon by means of several production techniques. Today there are many compound nanostructures that cover II–VI and III–V semiconductors. Briefly, we have addressed some of those nanomaterials and their properties here.

3.1.1. GaAs Nanostructures

Epitaxially III–V NWs have been fabricated in a potentially scalable, flexible, and well-controlled manner by a VLS assisted-growth mechanism [60–61]. Among semiconductor NWs, GaAs NWs have significant merits associated with the direct band gap and high electron mobility. The GaAs material is used extensively in the electronics and optoelectronics industries. Accordingly GaAs NWs are prime candidates for electrically and optically active NW devices as lasers and photodetectors [62]. Furthermore, integrated NW device architectures [63] generally demand straight NWs with well-controlled orientation. Therefore, crystallographic defects, notably twinning [64], and NW kinking, are two major growth issues to overcome. Additionally, the unintentional incorporation of dopant impurities, which gives rise to electronic energy levels within the band gap, can adversely affect the optical and electronic properties of these NWs.

Recently high-quality GaAs NWs have been produced by controlling the substrate (growth) temperature [65] and the temperature dependencies of GaAs NW growth [66, 67]. It has been reported that an intermediate V/III ratio can lead NWs that are free from crystallographic defects. These NWs also give strong excitonic emission with minimal impurity-related emission, indicating their high purity. For GaAs NWs, the most common acceptor impurity is C. Further, these effects of V/III ratio yield important information on the mechanisms of kinking, twin formation, and C impurity incorporation. Figure 10(a) shows the twins and kinks, while Figure 10(b) shows no defects that have been synthesized at the intermediate III/V ratios. This means that by tuning the III/V ratios in GaAs NWs, the defects can be manipulated depending upon materials engineering. It will require more time to finalize these properties for commercial marketing.

3.1.2. InP Nanostructures

1D growth of InP NWs often occurs preferentially in the hexagonal wurtzite phase, in contrast to the zincblende structure (thermodynamically stable crystal phase) ubiquitous in bulk InP [68, 69]. Such a dramatic structural change should also be reflected by changes in the band gap energy, the electron and hole masses, and recombination selection rules. It has been reported that the zincblende NWs are 2–7 μm long with an average diameter of 80 nm. In contrast, wurtzite InP NWs are 1 μm long tapered with the growth axis along the (0001).

Figure 11(a) shows the photoluminescence (PL) spectra from the zincblende and wurtzite InP NWs. The zincblende InP epilayer exhibits a strong and narrow emission line centered at the free exciton recombination energy of 1.418 eV [70]. The zincblende NWs display a single broad (20 meV) PL peak, which is approximately centered at 1.418 eV. Bao et al. [70, 71] reported the NW PL from the same growth substrate that varies substantially from NW to NW, and the most intense emission is observed for InP NWs at 1.418 eV. These zincblende NWs also display lifetimes of 500 ps–1 ns, which are comparable to that observed in the zincblende epilayer (1.7 ns) [70]. The emission width and energy position is strongly affected by the state filling, and so it is important to measure the NW at the lowest possible excitation intensities [70]. In contrast, the wurtzite InP NWs display a somewhat narrower emission line (17 meV) at 1.49 eV, 80 meV higher than for zincblende InP NWs. It is noteworthy to mention that there is a band gap energy difference in between the wurtzite and zincblende phases as the crystal structure and lattice are different in the third nearest neighbors at the lattice matrix.

3.1.2.1. Polarization in Crystal Phases. The polarization of PL emission from the zincblende and wurtzite NWs also displays distinctive differences. To measure this, circularly polarized laser excitation was used to excite all the linearly polarized electronic

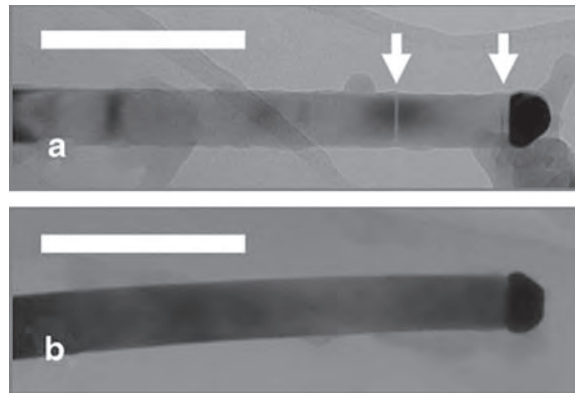
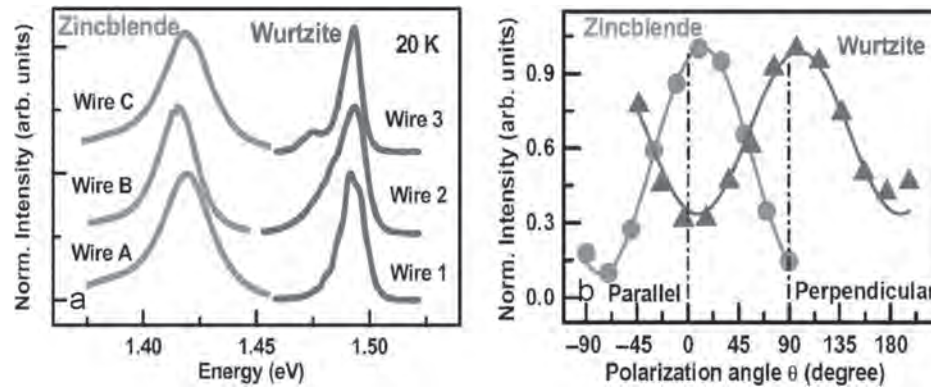


Figure 10. Transmission electron microscopy images of GaAs quantum wires with (a) twins and kinks and (b) apparently no defects that have been synthesized at the intermediate III/V ratios. Reprinted with permission from [52], H. J. Joyce et al., *Adv. Funct. Mater.* 18, 1 (2008). © 2008, Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.



states with equal probability, while the emitted PL was analyzed for linear polarization. Figure 11(b) (the circles) shows the variation of the PL integrated intensity from zincblende NW as a function of the polarization angle, with the solid line a fit to $\cos 2\theta$. The zincblende InP NW emission is strongly polarized (82%) parallel to the NW axis [72, 73] due to the strong dielectric contrast between the approximately cylindrical NWs and the surrounding air. In the bulk InP, the recombination in zincblende InP is between the *s*-like conduction band electrons, and so the emitted light is completely unpolarized [74, 75]. Zincblende NW emission is polarized because of the dielectric contrast throughout the wavelength (energy).

In contrast, a hexagonal wurtzite crystal exhibits a completely different symmetry. The s -like conduction band has Γ_7 symmetry, while the p -like hole bands split into three separate hole bands due to a combination of the spin-orbit interaction and crystal field splitting [74, 75]. Figure 12 shows the band structure of wurtzite and zincblende materials. The lowest energy hole states (highest lying energy bands) having a symmetry of Γ_9 (the so-called A band), while the two higher energy hole bands (B and C bands) have Γ_7 symmetry [74, 75]. The lowest energy exciton state should be between a Γ_7 electron and a Γ_9 hole, which means that the $\Gamma_7 \rightarrow \Gamma_9$ recombination is dipole allowed only for E perpendicular to the c axis of the NW, and forbidden for E parallel to the c axis [76, 77]. Thus the lowest energy exciton transition in wurtzite InP should be strongly polarized perpendicular to the wurtzite c axis and, thus, perpendicular to the NW axis.

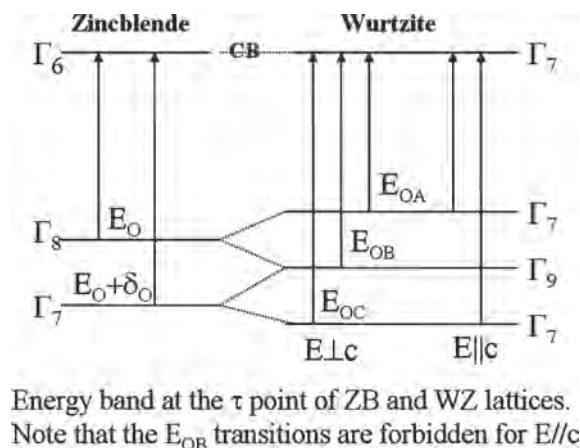


Figure 12. Schematic band diagram for the wurtzite and zincblende phases of materials at their Γ point. Here conduction band (CB) and valence band (VB) are marked. The spin-orbit and crystal-field splitting are also shown.

Polarization analysis of the emission from wurtzite NW shows that the PL is polarized (49%) perpendicular to the NW axis, consistent with growth of the NW along the (0001) axis. The degree of polarization in the wurtzite NW is weaker than the zincblende NW because the dipole selection rules of the wurtzite crystal are modulated by the dielectric contrast inherent for all the semiconductors NWs [78].

3.1.2.2. Strain Kinetics in Terms of Material's Phases. Since the increase in band gap energy due to the wurtzite (wz) phase is higher in the InP than in the InAs material system, it follows that the difference between the a lattice constant of the two materials in the wz phase is larger as compared to the zincblende (zb) case. Hence the lattice mismatch between the two materials is higher when they are in the wz phase and, consequently, the blue shift in energy due to strain will be higher when InAs and InP are in the wz phase. We denote by $\Delta_{\text{strain}} = E_{\text{strain}}^{\text{wz}} - E_{\text{strain}}^{\text{zb}}$ the increase in the InAs emission energy due to this effect. We can summarize all the mentioned contributions to the emission as follows:

$$E = E_{\text{bulk}}^{\text{wz}} + E_{\text{strain}}^{\text{wz}} + E_{\text{QC}} \quad (8)$$

where E is the measured emission energy, $E_{\text{bulk}}^{\text{wz}}$ is the band gap of the bulk InAs in the wz phase and E_{QC} is the contribution given by QCE. Indeed, for each diameter, the dependence of the measured emission energy E and that has been calculated using eight-band strain-dependent $\mathbf{k} \cdot \mathbf{p}$ theory without the inclusion of QCEs for the zb phase ($E_{\text{calc}} = E_{\text{bulk}}^{\text{zb}} + E_{\text{strain}}^{\text{zb}}$) can be written as

$$E - E_{\text{calc}} = \Delta_{\text{strain}} + \Delta_{\text{gap}} + E_{\text{QC}} \quad (9)$$

Hence the energy separation of the two peaks is reproduced well (within ~ 10 meV). Moreover, since the calculations are performed for zb InAs and InP materials, the difference between the calculated and measured (at low excitation power density) values is given by $\Delta_{\text{gap}} + \Delta_{\text{strain}}$, which therefore amounts to [79]

$$\Delta_{\text{gap}} + \Delta_{\text{strain}} = 148 \text{ meV} \quad (10)$$

If we use the GW value for the difference in energy gaps ($\Delta_{\text{gap}} = 55.7$ meV), we obtain $\Delta_{\text{strain}} = 92.3$ meV [79]. This clearly indicates that the strain kinetics in the wz and zb phases are different, and play a significant role in the band gap renormalization. In particular to the InAs material, the strain-induced band gap energy shift is estimated to be 92.3 meV [79].

3.1.3. GaN Nanostructures

GaN is a direct band gap semiconductor material (with 3.4 eV at room temperature) with wurtzite crystal structure, used in optoelectronic, high-power, and high-frequency devices. Its sensitivity to ionizing radiation is low, making it a suitable material for the photovoltaic solar cell arrays for satellites [80]. Because GaN transistors can operate at much hotter temperatures and work at much higher voltages than GaAs transistors (exciton binding energy in GaN is higher than GaAs), they make ideal power amplifiers at microwave frequencies. A GaN-based violet laser diode is used in the Blu-ray Disc™ technologies, and in devices such as the Sony PlayStation® 3. The InGaN or AlGaIn alloys with a band gap dependent on ratio of In or Al to GaN allows the building of LEDs with colors that can go from red to blue [81].

Nanostructures of GaN are of particular interest since their band edges fall in the ultra-violet region, and a number of optoelectronic devices are currently being made out of this material. Various research groups have reported the synthesis of GaN NWs using different techniques such as CNTs-confined reactions [82], arc discharge [83], laser ablation [84], sublimation [85], and chemical vapor deposition (CVD) [86]. In all these studies, the average diameter of the synthesized GaN NWs was larger than the reported Bohr exciton radius of GaN (11 nm) [87] and thus no QCE have been observed. Recently Bae et al. and Kuykendall et al. [88] have demonstrated GaN NWs of mean diameter 8 nm although



Figure 13. Photoluminescence images of individual NWs: spectra from left to right with peaks around 380, 450, and 525 nm. It shows the photoluminescence of individual NWs. The corresponding images of the NWs are marked with their wavelengths observed in photoluminescence measurements. Reprinted with permission from [88], T. Kuykendall et al., *Int. J. Photoenergy* 1, (2009). © 2009.

they did not report any optical properties of those NWs. However, colorful luminescence has been demonstrated in individual GaN NWs that cover from 380–525 nm, shown in Figure 13. The corresponding images of the NWs are marked with their wavelengths observed in PL measurements.

3.1.4. CdS Nanostructures

The control of size, morphology, and structure represents some of the key issues in nanotechnology fabrication because these parameters are some of the key elements that determine their properties. Lots of nanomaterials with various morphologies, which include not only nanotubes, NWs, and nanobelts, but also some special nanostructures such as nanobridges, nanonails, nanodiskettes, star-shaped and flower-like, have been fabricated [89, 90]. Many effects have been employed to synthesize various morphologies of CdS nanomaterials, including NWs, nanorods, nanotubes, hollow spheres, peanuts, and nanocables by various physical and chemical solutions [91, 92].

CdS nanomaterials have been investigated intensively due to their interesting optoelectronic and physical properties [93]. The effect of the compartment arrangement on the yield of CdS NWs is evident in Figure 14(a). This compartment arrangement was deployed to trap the CdS vapor at a specific temperature, enabling the growth of CdS nanostructures at a specific temperature and pressure.

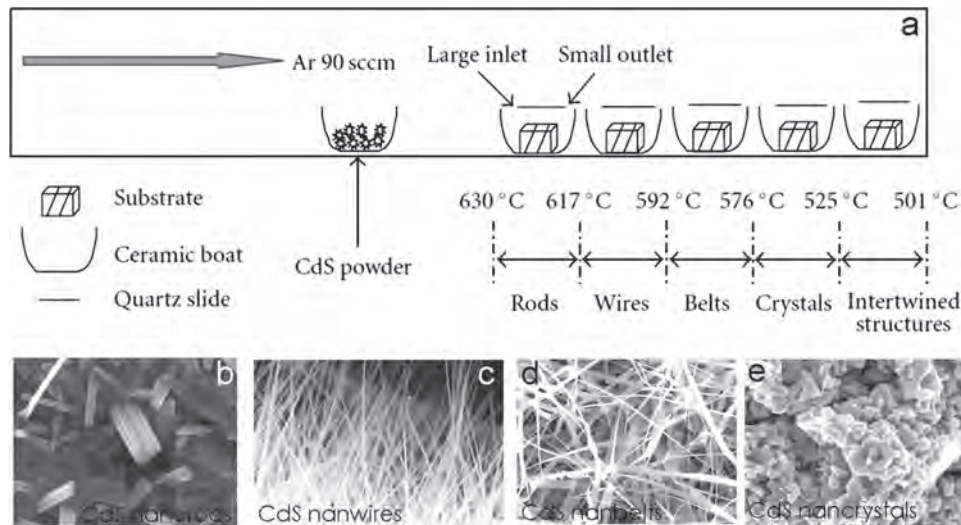


Figure 14. (a) CdS NWs growth compartments; demonstrates the much higher yield of CdS NWs with the compartment arrangement. This compartment arrangement was deployed to trap the CdS vapor at a specific temperature, enabling the growth of CdS nanostructures at a specific temperature and pressure. It shows CdS (b) nanorods, (c) NWs, (d) nanobelts, and (e) nanocrystals. Reprinted with permission from [93], J. M. Green et al., *J. Nanomaterials* (2008). © 2008.

Even though the different CdS nanostructure morphologies depend upon several factors, such as pressure, catalyst size, temperature, and flow rate, it has been shown that vapor nucleation is extremely sensitive to temperature and pressure conditions [94], and temperature plays a key role in determining the morphology of the different CdS nanostructures. Therefore, to determine the temperature at different spots inside the furnace shown in Figure 14(a), a thermocouple mapping was employed; this clearly showed the specific temperature at which each different nanostructure morphology is formed. There are CdS (b) nanorods, (c) NWs, (d) nanobelts, and (e) nanocrystals. These are absolutely the materials engineered in the reactor chamber by varying the synthesis parameters and the VI/II ratios.

3.1.5. CdSe Nanostructures

Brus [95] and his co-workers Alivisatos et al. [96], and Bawendi et al. [97] were some of the first scientists to correlate the QCE in CdSe nanostructures to an induced higher energy shift with decreasing size. This led to a surge of research investigating the size-dependent luminescent properties of nanomaterials [98]. The most profound phenomenon of the nanosaw structure is the growth of asymmetric saw teeth on one side while the other side is straight and smooth. From the structure model shown in Figure 15(a), the (0001) surface of wurtzite can terminate with different atom planes. Take CdSe as an example, the (0001) surface is terminated with Cd and the $(0\ 0\ 0-1)$ is terminated with Se. To quantitatively determine the polarity of the surfaces, convergent beam electron diffraction (CBED) pattern was recorded from the ribbon, as given in Figures 15(b–c). The intensity distributions in the (0002) and the $(0\ 0\ 0-2)$ disks are significantly different, which is due to the non-central symmetric structure of CdSe, which can uniquely determine the polarity of the CdSe nanosaws [99]. Quantitative interpretation

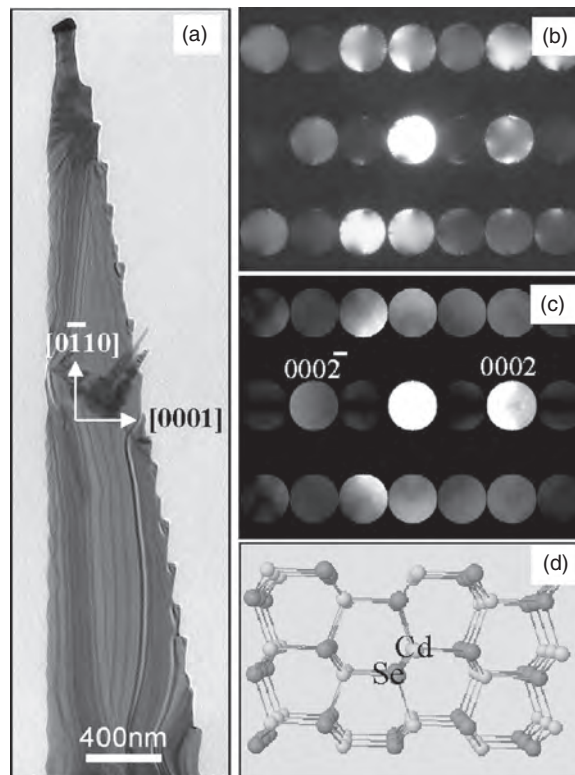


Figure 15. (a) Transmission electron microscopy (TEM) image of a CdSe nanosaw where the (0001) surface is terminated with Cd and the $(0\ 0\ 0-1)$ surface is terminated with Se, (b) experimental CBED pattern, showing the polarity of the different surfaces, is compared, favorably, with the (c) theoretical CBED pattern and (d). Reprinted with permission from [99], C. Ma et al., *Int. J. Nanotech.* 1, 431 (2004). © 2004, Inderscience Enterprises Ltd.

of the CBED relies on dynamic simulations, which were performed using an improved version of the Bloch wave program [100]. A comparison of the experimental pattern with a theoretically calculated pattern (Fig. 15(d)) indicates that the saw teeth grow out of the (0001)-Cd surface. Therefore, the growth of the saw teeth is due to the catalytically active Cd-(0001) surface, while the Se-(0 0 0 - 1) is relatively inactive, producing the asymmetric growth morphology.

3.1.6. ZnS Nanostructures

Wurtzite ZnS has a band gap energy of 3.54 eV at room temperature [101], a high refractive index of 2.2, and a high transmittance of light in the visible to ultraviolet wavelength region of the spectrum. The hexagonal crystal structure of the wurtzite ZnS is looking at this type of growth, as it asserts itself in larger structures in fairly obvious ways. In the physical vapor deposition (PVD) system, the ZnS columns were synthesized using 1.5 grams of ZnS as a source, as shown in Figures 16(a–c) [102]. As is seen, the columns synthesized maintain the six-fold symmetry of the wurtzite crystal structure. Many of the columns are capped with a pyramid-like top. This growth probably occurred when the furnace was shut down and the system was pulled to a vacuum, purging the environment of growth species.

Figures 16(b and c) show the layer-by-layer growth of ZnS columns as one layer grows upon another. First, each layer grows with a hexagonal profile individually, which helps to create the hexagonal column. Second, each layer is approximately 50 nm thick and grows within the profile of the layer below it, implying that it grows by having the incoming vapor species deposit onto the layer and incorporate into the crystal structure at a growth site such as a step or dislocation.

It is useful to divide 1D ZnS nanostructures into two different categories, depending on the type of growth, shown in Figures 17(a–d). The first category is nonpolar surface-dominated nanostructures. These are more normal morphologies, such as a plain nanowire or nanobelt shape. The second category is the polar surface-dominated nanostructures [103]. When the polar crystal surface that is present in the wurtzite crystal structure dominates some aspects of the growth, some interesting growth phenomena occurs. Although the original intent in studying ZnS was to synthesize this material into the nanobelt morphology, a self-assembly was observed when four types of morphologies were collected in the as-deposited material: nanobelts, nanocombs/nanosaws, nanowindmills, and NWs. The length and width of each of these morphologies could

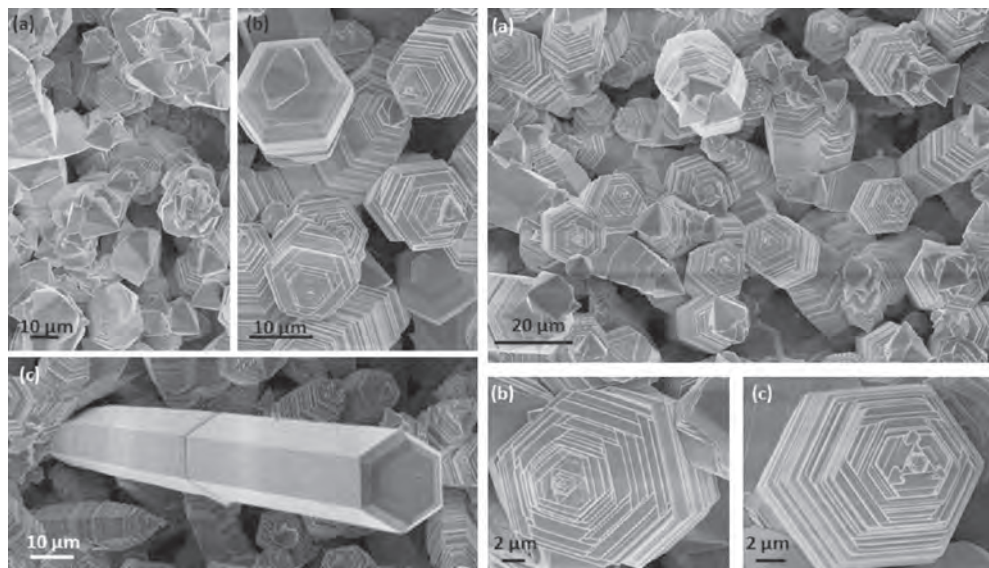


Figure 16. Hexagonal ZnS columns. (a) Hexagonal ZnS columns growth is shown, (b–c) hexagonal layer-by-layer growth of the columns are shown.

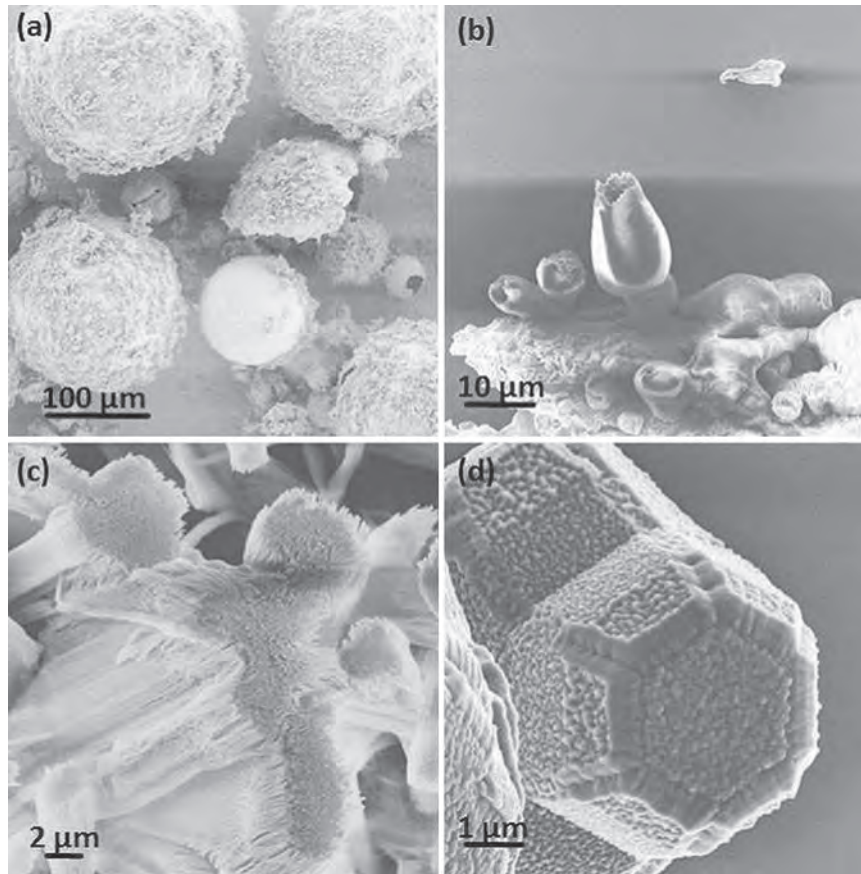


Figure 17. Various ZnS nanostructures synthesized using the simple vapor deposition system. (a) nanocages, (b)–(c) nanoflowers, and (d) nanoplucks. Reprinted with permission from [103], Z. L. Wang, ESI Special Topics 2005; <http://esi-topics.com/fmf/2005/july05-ZhongLinWang.html>.

be several micrometers in length, however the thickness of the structures is consistently in the nanometer range. The most typical structure observed is ZnS nanobelts, with a uniform cross-section along their length and with a typical width of 2–30 μm , extending to over 100 μm in length. An examination at the root of the group of the nanobelts was grown from a site which is the seed for the nucleation of the nanobelts. Numerous belt- and saw-like structures grow from a single nucleation site. The comb-like structure is found in the product and also large sheets have also been identified Ref. [103].

3.1.7. ZnSe Nanostructures

The fabrication of nanocrystals with controlled morphologies and phase structures is always potentially important in the preparation of materials suitable for optoelectronic and luminescent applications. As one of the most important II–VI group semiconductors, ZnSe with a room-temperature bulk band gap of 2.7 eV has attracted intense attention in recent years because of the potential applications in LEDs, photodetectors, full-color display, and room-temperature exciton devices [104, 105]. The synthesis of nanocrystalline ZnSe powders with tunable phase, morphology, and size provides alternative variables in tailoring the physical and chemical properties [106, 107]. Therefore, tremendous efforts have been made to control the shape and phase structure of ZnSe nanocrystals, and various methods have been reported for the synthesis of ZnSe-based nanostructures [106, 108, 109] especially the 1D ZnSe nano/microstructures [103, 110, 111].

Ma et al. [111] reported that 1D bundles of ZnSe NWs could be prepared by the solvothermal method (SVT), shown in Figure 18. This kind of method can be termed an amine-assisted SVT, since amine is used as the templating agent, and it is a crucial parameter for controlling the morphology of the nanoparticles. It remains a challenge on how

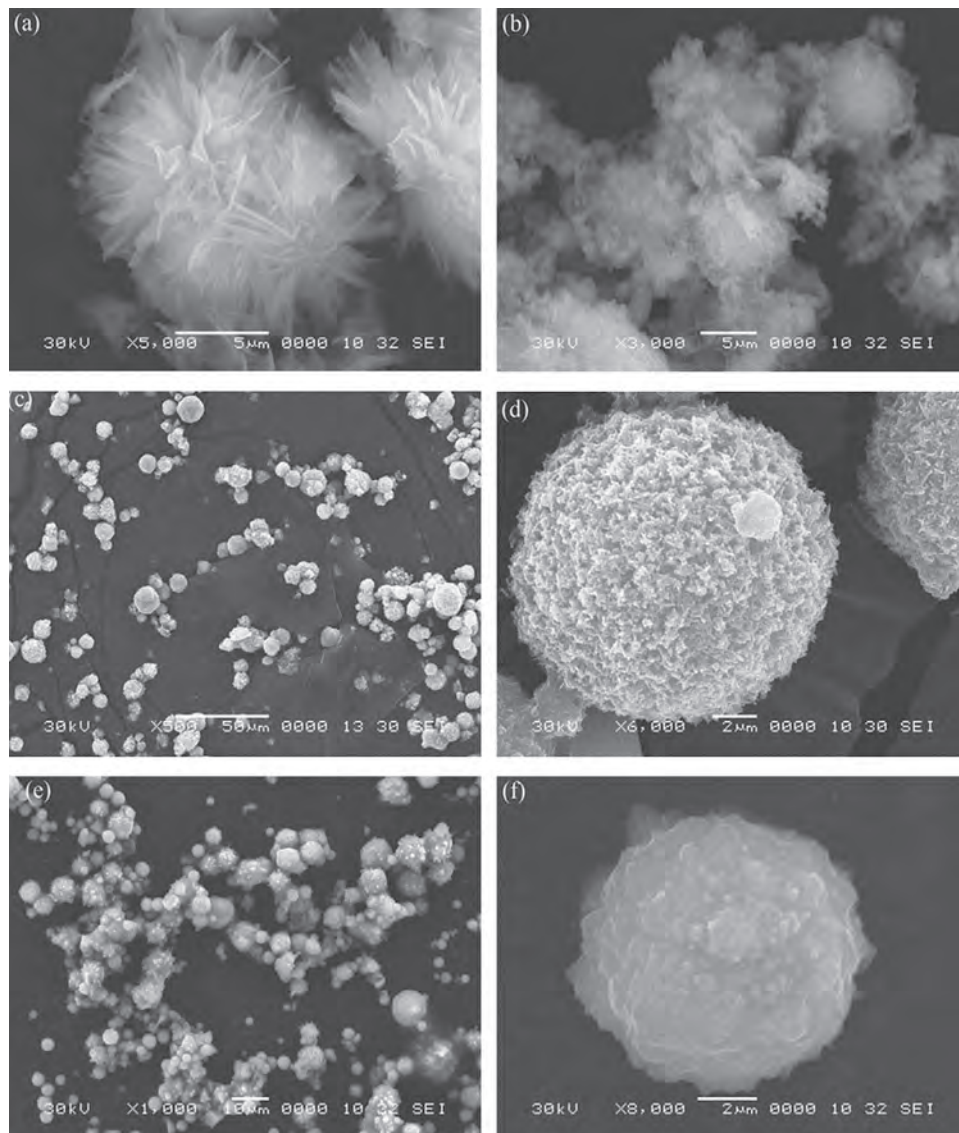


Figure 18. Bundles of ZnSe nanostructures prepared by the solvothermal method. Reprinted with permission from [111], C. Ma et al., *J. Am. Chem. Soc.* 126, 708 (2004). © 2004, Inderscience Enterprises Ltd.

to design and develop new amine-assisted solution systems to prepare novel nanostructures of ZnSe and other semiconductors and to understand the controlling mechanism.

When at least 1D of the material becomes smaller than the responding de Broglie wavelength or Bohr radius, a QCE phenomena occurs that can change the properties of the material [112]. The ability to control both the shape and size of semiconductor nanocrystals is essential for precise tuning of their optical and electronic properties, as well as their overall functionality, to enable their use in practical applications [113]. These nanostructures have potential applications in nonlinear optics, solar cell technology, and in the miniaturization of electronic systems leading to novel optoelectronic devices [114]. Here the simultaneous size and shape control of ZnSe nanostructures using liquid crystal templates formed by the self-assembly of a poly(ethylene oxide)–poly(propylene oxide)–poly(ethylene oxide) (PEO–PPO–PEO) amphiphilic block copolymer in the presence of water and *p*-xylene were used. The ZnSe nanocrystals were synthesized in the aqueous phase of the PEO–PPO–PEO/water/*p*-xylene templates, shown in Figure 19.

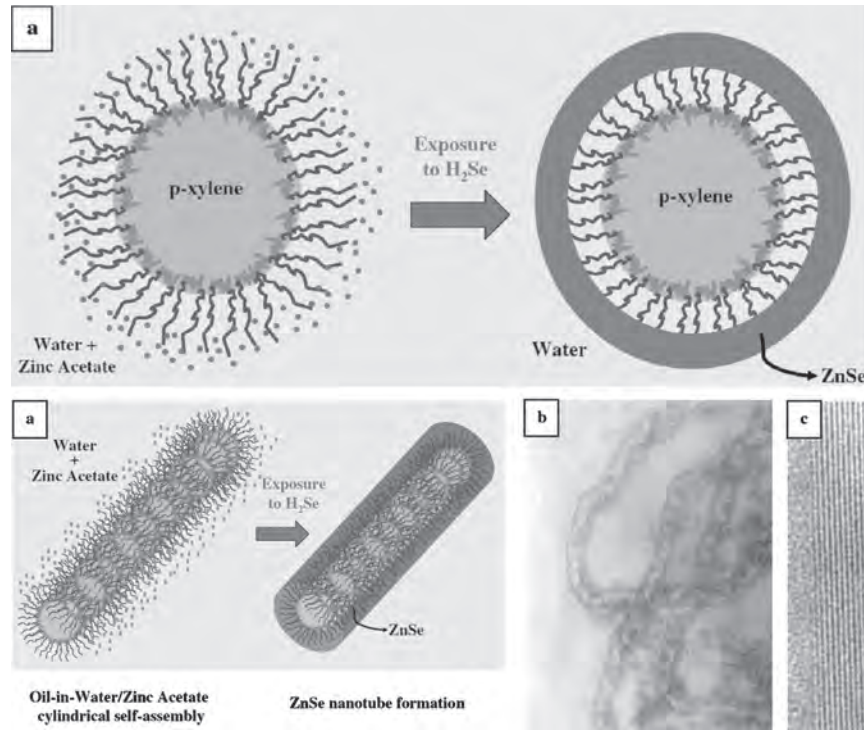


Figure 19. (a) The ZnSe nanocrystals synthesized in the aqueous phase of the PEO-PPO-PEO/water/p-xylene templates. (b–c) TEM images of ZnSe nanocrystals with crystalline orientation. Reprinted with permission from [114], G. N. Karanikolas et al., *Nanotech.* 17, 3121 (2006). © 2006, IOP Publishing Ltd.

3.1.8. ZnO Nanostructures

Recently much effort has been made to fabricate semiconductor nanostructures on different material bases, including ZnO. If self-organized nanostructures are obtained in terms of position, composition, and doping they can then eventually be the basis of very interesting novel nano-optoelectronic devices. The possibility of obtaining stimulated emission in ZnO nanostructures [115] indicates that their optical quality can be high, and that they would be suitable for the realization of blue/UV lasers. Since QW LEDs have been proven to have relatively low-threshold current [116] it would be promising to obtain QW structures on the basis of self-organized ZnO nanorods, since such structures would be of very high crystal and optical quality.

Gu et al. [117] reported the QCE within ZnO nanorods of 1.10 nm width synthesized through a colloidal method. The observed QCE was attributed to the nanorod size, smaller than the bulk ZnO exciton Bohr radius 1.8 nm. However, if such nanorods are to be used in vertically emitting laser structures, the diameter of the nanorods has to be in the range 100–200 nm, since only then is efficient wave guiding guaranteed. Several attempts have been made to grow ZnO-based coaxial nanorod heterostructures [118, 119] but very few of these works were extended to the realization of coaxial QWs. For example, Jhang et al. [121] reported on catalyst-free ZnO/Mg_{0.2}Zn_{0.8}O multishell nanorod heterostructures. Figure 20 shows the schematics of nanostructures, their strain compensation schemes, and their axial and radial core/shell nanostructures. QCE had earlier been reported in catalyst-free ZnO/Zn_{0.8}Mg_{0.2}O vertical heterostructures, although the QW peak was relatively broad [120] probably as a consequence of the limitations in producing sharp heterostructure interfaces.

3.1.8.1. Nanotubes. ZnO nanotubes have been synthesized epitaxially with hexagonal ZnO morphology with the same in-plane orientations. The well-defined hexagonal shape is an indication of epitaxial growth and consequently single-crystalline nature of ZnO nanotubes [121, 122]. The density of hexagonal ZnO nanotubes has been estimated to be $0.6\text{--}4\ \mu\text{m}^{-2}$. At the initial stage of growth, the diameter of the nanotube increases largely

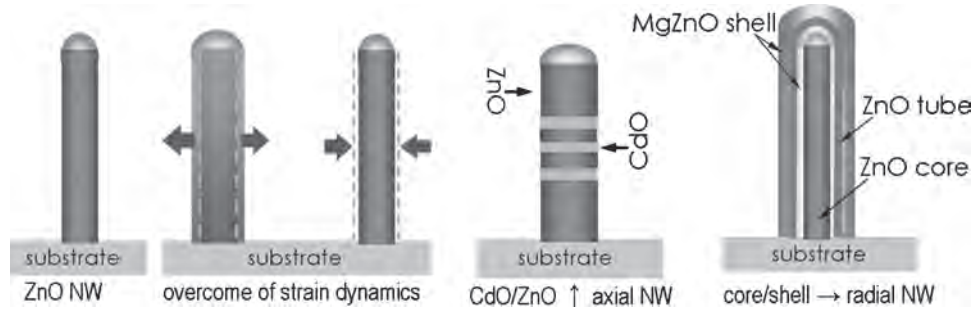


Figure 20. Schematic of catalyst-assisted CdO/ZnO and ZnO/MgZnO NWs. These axial and lateral core/shell NWs show the strain compensation mechanism and overcoming strategy.

within a length of 1–2 μm . The overall nanotubes are uniform with cross-sectional variation of the inner/outer diameter from 0.1/0.5 μm at the bottom to 0.4/0.7 μm at the top, except in the nucleation stage. Therefore, the diameter on an average of ZnO nanotubes ranged from 0.1 to 0.5 μm . Figures 21(a–d) shows the hexagonal ZnO nanotubes/rods. The hexagonal facet feature of nanorods provides evidence growing along the [0001]. On an average, the length of ZnO nanorods is about 5 μm , and the diameters are in the range 110 ± 10 nm. It is interesting to note that all the ZnO nanorods have the same

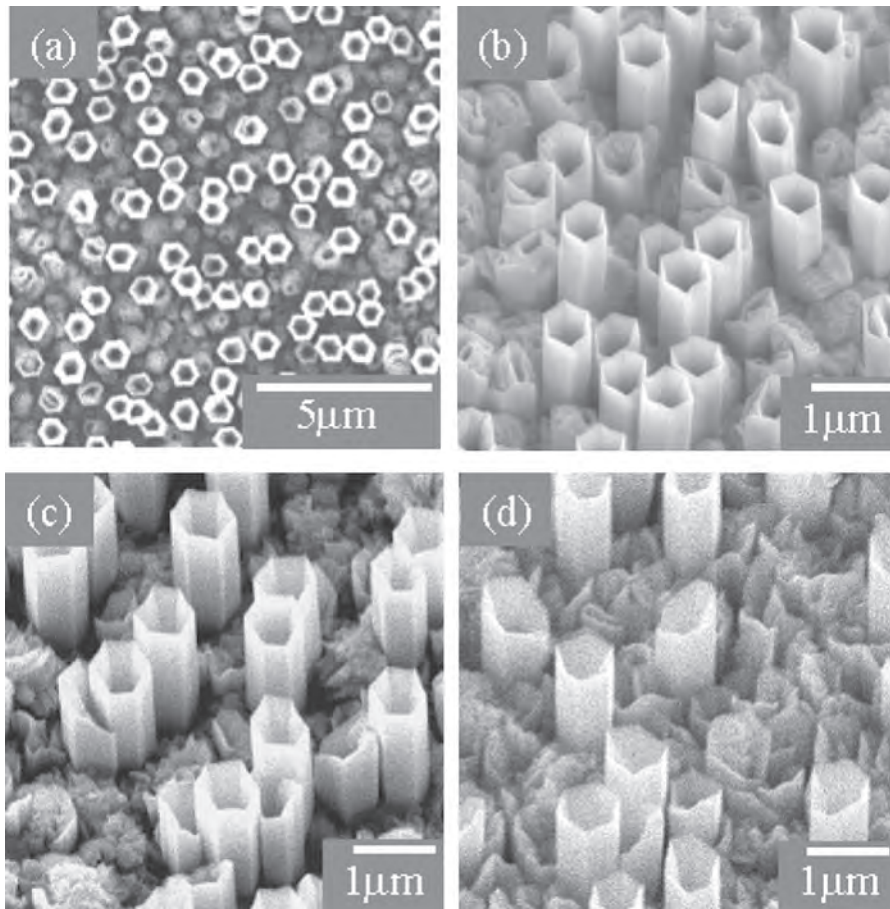


Figure 21. Hexagonal ZnO nanotubes/rods with facet feature of nanorods provide evidence growing along the [0001]. On an average, the length of ZnO nanorods is about 5 μm and the diameters are in the range 110 ± 10 nm. Reprinted with permission from [121], B. P. Zhang et al., *Appl. Phys. Lett.* 84, 4098 (2004). © 2004, American Institute of Physics.

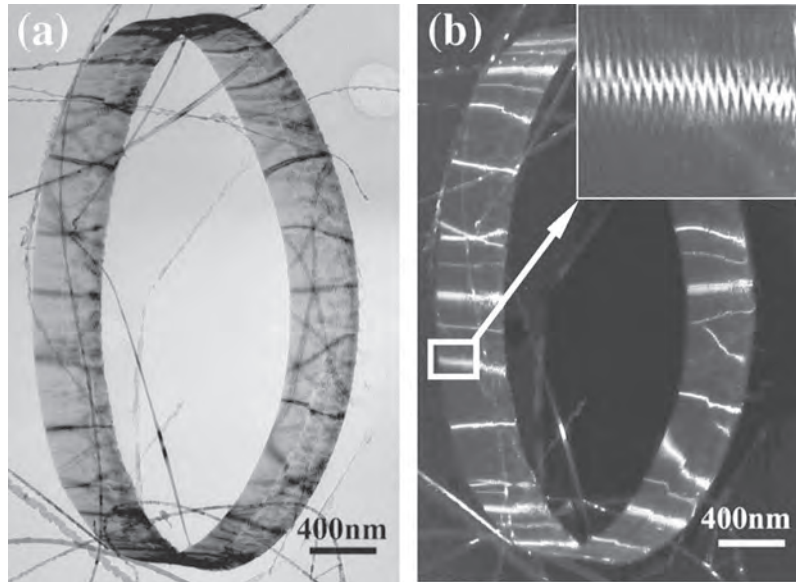


Figure 22. Typical ZnO nanorings with the dimension of ~ 15 nm width and ~ 10 nm thicknesses. The nanobelt has polar charges on its top and bottom surfaces, resulting in a dipole moment interaction. Reprinted with permission from [123], Y. Ding et al., *Phys. Rev. B* 70, 235408 (2004). © 2004, American Physical Society.

length and diameter, implying that they are synthesized homogeneously by the same growth mechanism with an equal growth rate.

3.1.8.2. Nanorings/Nanobelts. Figures 22(a–b) shows the typical ZnO nanorings with the dimension of ~ 15 nm width and ~ 10 nm thicknesses. The nanobelt has polar charges on its top and bottom surfaces, resulting in dipole interaction. As the growth continues, the nanobelt may naturally be attracted onto the rim of the nanoring due to electrostatic interaction, extending parallel to the rim to neutralize the local polar charge and reduce the surface area, resulting in the formation of a self-coiled, coaxial, uniradius, multi-looped ZnO nanoring structure [123].

3.1.9. CdO Nanostructures

CdO nanostructures, such as nanobelts, nanorods, and NWs, have a wide and direct band gap of 2.27 eV and a narrow band gap of 0.55 eV [124]. The difference of band gap originates from Cd and O vacancies and strongly depends on the synthetic procedures [125]. In recent years, many researchers have focused on the synthesis of CdO with various structures for application to the field of optoelectronic devices such as photovoltaic cells, phototransistors, photodiodes, and transparent electrodes. Owing to these diverse applications of CdO, various synthetic methods have been explored to prepare CdO nanocrystals, most of which are based on physical and chemical techniques, such as sol-gel [126], spray pyrolysis [127], sputtering method [128], chemical bath deposition [129], activated reactive evaporation [130], and CVD [131].

In this context, a simple method is used for the synthesis of nanocrystalline CdO by directly heating Cd samples in an O torch flame at atmospheric pressure, shown in Figure 23(a). Due to the high temperature of the microwave plasma torch, a few seconds were enough for the Cd granules to evaporate, creating CdO nanocubes through the oxidation in the oxygen plasma. This is a self-catalytic method of metal based on VSS growth mechanism by making use of an explosive oxidation reaction in the microwave plasma flame. The method is seedless, surfactant- and template-free, without all the subsequent complicated procedures for the removal of seeds, surfactants, or templates. The advantage of rapid and catalyst-free synthesis meets the requirements of the various device productions and of practical applications. However, with Au catalyst for $20\sim 30$ Å, Figure 23(b) shows a synthesized nanostructures appeared needle-shaped, with sharp

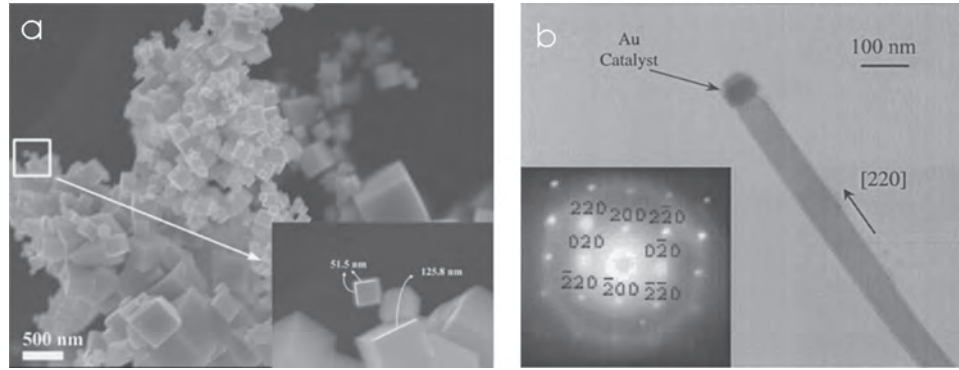


Figure 23. (a) CdO nanocrystals synthesized by directly heating Cd samples in an oxygen torch flame at atmospheric pressure and (b) TEM cross-section and transmission electron diffraction (TED) pattern of CdO nanocrystals. Reprinted with permission from [125], J. H. Kim et al., *Jpn. J. Appl. Phys.* 46, 4351 (2007). © 2007, Japan Society of Applied Physics; X. Liu et al., *Appl. Phys. Lett.* 82, 1950 (2003). © 2003, American Institute of Physics.

tips about 40–100 nm in diameter and wide butts of several hundred nanometers to several micrometers at the other end, while the lengths of the nanoneedles normally ranged from 2–20 μm . The inset of Figure 23(b) shows a magnified view of the butt end of one nanoneedle, where a square cross section can be clearly seen.

4. SIZE AND SHAPE OF NANOMATERIALS

QCE becomes important as the size of nanostructures is reduced, while the shape is less appreciated. New data show that shape matters as much as size does. Kan and colleagues [132] show that the electronic structure and optical properties (that is, the QC effects) of nanocrystals depend sensitively on the ratio of their length and diameter. In function, performance, and behavior, shape matters.

The normal size of an exciton in a large crystal provides an approximate dimension for the onset of QCEs. When an exciton is squeezed into a nanocrystal, the effective band gap energy of the semiconductor material increases. The smaller the nanocrystal, the larger the effective band gap, and the greater the energy of optical emission resulting from electron–hole recombination. Typical exciton Bohr radii range from 2.2 nm (ZnS) to 7.5 nm (CdTe) in the common II–VI semiconductors, and from 11 nm (InP) to 60 nm (InSb) in the III–V semiconductors [133]. So the magnitude of QC is known to depend on nanocrystal size and composition, but what about the nanocrystal shape?

The nanostructures are large in 2D, and confined only in the third dimension. Particle-in-a-box approximations find that 3D confinement is stronger than 2D confinement, which in turn is stronger than 1D confinement. These conclusions predict that the change in the band gap energies in nanostructures of a given composition and thickness (or diameter) should be the greatest in dots, followed by rods, wires, and finally wells. But experimental confirmation of this prediction is largely lacking. So far no systematic experimental comparison has been made of the relationships between band gap energies and size in systems that differ only in the geometric dimensionality of confinement; that is, 1D against 2D, 2D against 3D, or 1D against 3D. InAs QDs with uniform 4-nm diameters and a mixture of lengths were separated into various length fractions having length/diameter ratios ranging from 2.3 to 5.7 [134]. The band gaps of the rods were found to decrease with increasing length/diameter ratio. As the rod length increases, the band gap decreases towards the value expected of true quantum wires.

The excited-state dynamics in QDs are apparently quite distinctive. The results of Kan et al. [132] indicate that the luminescence quantum yield (QY) decreases as the QDs lengthen, which is probably because of an increasing spatial separation of electron and hole wavefunctions. Reducing the wave function overlap should decrease the probability of electrons and holes recombining radiatively, and thus reduce PL efficiency. The work

convincingly establishes that both the magnitude and character of QCE in semiconductor nanostructures are highly shape dependent.

Potential applications are now emerging that exploits the anisotropic shapes of NRs and NWs. The pseudo-1D morphologies of nanorods and NWs increase carrier transport in light-harvesting cells [135] and aid their assembly into complex crossed NW devices [136]. Polarized emission supports applications, such as orientation-sensitive fluorescence labels, and allows linearly polarized lasing [137]. Ensembles of nanorods or NWs could have collective properties, such as the efficiency of interparticle energy transfer, that are different from and perhaps better than those of close-packed QDs.

The size dependence of the band gap is the most identifiable aspect of QCE in semiconductors; band gap increases as the size decreases [138]. The band gaps in 1D-confined wells, 2D-confined NWs, and 3D-confined QDs should evolve differently with size as a result of differing dimensionality of confinement [138]. InP QWs have diameters in the strong-confinement regime and a comparison of their band gaps. Theoretical evidence established that the QCE observed in InP NWs is weakened to the expected extent, relative to that in the InP QDs, by the loss of 1D confinement. Figure 24 shows theoretical and experimental InP QDs and QWs data plotted as ΔE_g versus $1/d^n$, for $n = 1.35$ (dot) and 1.45 (wire). The dotted lines are least-squares fits to the theoretical data, yielding the slopes $B_{\text{dot}} = 3.30 \pm 0.02$ eV 1.35 nm and $C_{\text{wire}} = 2.49 \pm 0.02$ eV 1.45 nm, and the theoretical slope ratio $C_{\text{wire}}/B_{\text{dot}} = 0.75 \pm 0.01$ nm [139–140].

Control of the properties of objects of nanomaterial science is connected with spatial ordering of similar structures. As a rule, self organization of nanostructures occurs as a result of disturbance of fractal symmetry and (or) the dimension of an object. Here it is necessary to determine quantitative indices of the adaptability of a material structure towards external action, including the dimensions of nano-objects. Self-organizing systems make adaptation possible to a dominant type of surroundings, i.e., they react to changes in the surroundings. The principles of harmonic self-organization connected with the system of golden ratios, determines a new interdisciplinary direction of contemporary science. Various areas where nanoscience and nanotechnology are expected to have a significant impact on energy and environmental systems are shown in Table 2.

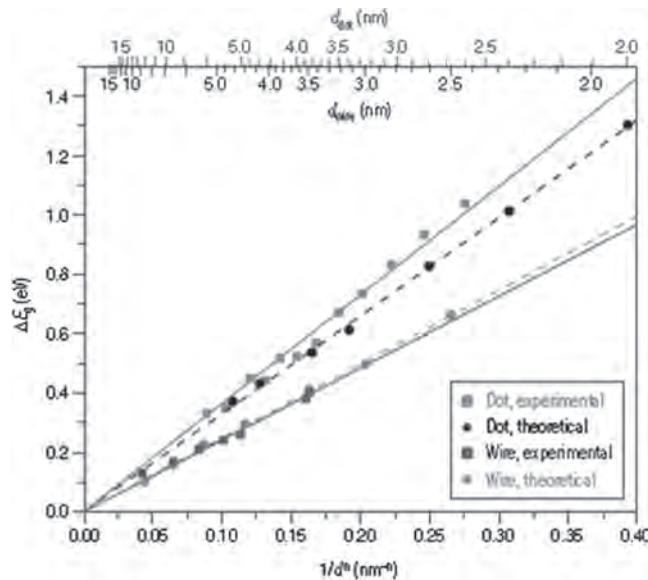


Figure 24. The size dependence of the band gap increases as the size decreases. The band gaps in 1D-confined wells, 2D-confined NWs, and 3D-confined QDs should evolve differently with size as a result of differing dimensionality of confinement. Reprinted with permission from [138], A. D. Yoffe, *Adv. Phys.* 42, 173 (1993); A. D. Yoffe, *Adv. Phys.* 50, 1 (2001). © 2001, Taylor and Francis.

Table 2. Various areas where nanoscience and nanotechnology are expected to have a significant impact on energy and environmental systems.

Area	N&N aspects
Energy sources	
Renewable and/or unlimited	
Solar	
Solar thermal	Materials
PV	Light harvesting, materials
PEC, solar hydrogen	Light harvesting, materials, catalysis
Solar power satellites	Materials
Wind	Materials
Wave	Materials
Hydro	Materials
Biomass	Catalysis
Nuclear fusion	Materials, phase (fuel) separation
Geothermal	Sensors, materials
Gravitational	Materials
Extraterrestrial feedstock	Materials
Nuclear fission	Materials, phase (fuel) separation, nanofluids
Non-renewable	
Oil	Catalysis, materials, sensors
Gas	Catalysis
Coal	Catalysis
Unconventional fossil fuels	Catalysis
Energy storage	
Water dams	Materials
Super gyro; pneumatic	Materials
H ₂ storage	Materials, catalysis, kinetics
Batteries	Materials
Supercapacitors	Materials, nanolayered structures, electrolytes
Energy conversion	
Combustion engines	Materials, diagnostics
Turbines	Materials, diagnostics
Fuel cells	Catalysis, diagnostics, electrolytes
Magneto-hydrodynamic generators	Materials
Thermo/piezoelectric materials	Nanolayered structures
Energy use	
Industrial/mining	Sensors, phase separation
Residential	Non-thermal light sources, materials (smart windows), nanofoams
Transportation	Materials
Emission cleaning	
Gas/air	Catalysis
Water	Catalysis

4.1. Shape Matters

The union of distinct scientific disciplines is revealing the leading edge of nanotechnology. Fifteen to 20 years ago, the interdisciplinary activity of geneticists, biologists, immunologists, and organic chemists created a more diverse toolbox now known as life science. Bioconjugates were created to help us move from outside the cell to the inside. Enabling technologies brought about the ability to create, identify, and specifically manipulate genetic maps to engineer designer proteins [141]. In parallel, physicists, chemists, polymer chemists, and engineers were creating the foundation for the small world of nanomaterials science. In less than a decade, materials science and life science together are today unraveling the mysteries of controlling, on a molecular level, the structure of matter.

Particles, complexes, tubes, coatings, active surfaces, and devices are being explored on the nanoscale. Figure 25(a) shows the schematics of nanowell, NW, nanorod, and nanodot. The assembly of nature's building blocks (e.g., carbon, nucleic acids, lipids, and peptides) along with the combination of different materials is leading to insightful

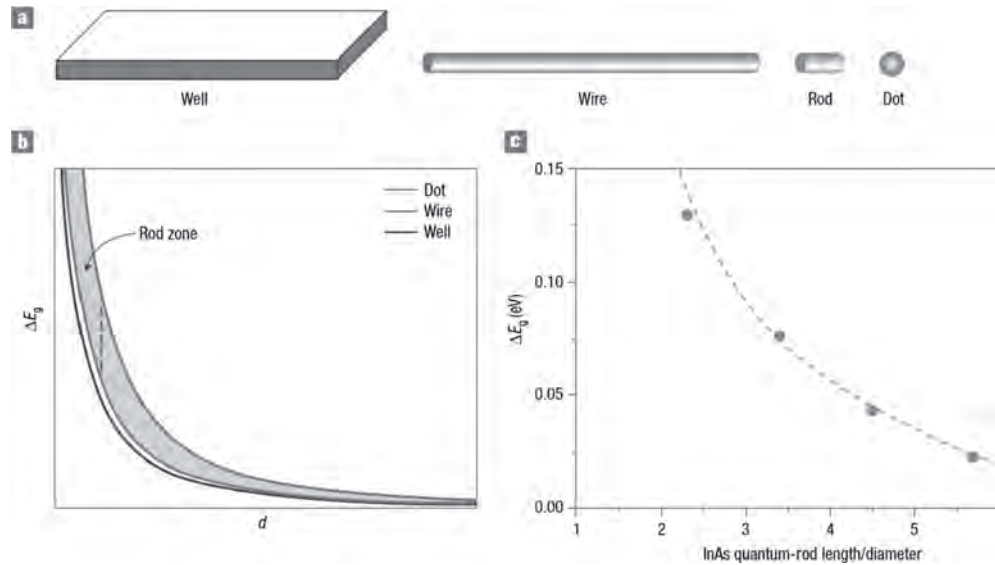


Figure 25. Comparisons of quantum wells (QWs), NWs, nanorods and quantum dots (QDs). (a) Geometries of the different structures. (b) Plots of ΔE_g against d (diameter) for rectangular QWs, cylindrical quantum wires, and spherical QDs obtained from particle-in-a-box approximations. The vertical dotted line and points qualitatively represent the expected variation in the band gap for InAs QDs of varying length/diameter ratio. (c) A plot of ΔE_g against length/diameter ratio for the InAs QDs showing the dependence of the band gap on the shape of the quantum rods. The dotted line represents the variation expected from a particle-in-a-box approximation. Reprinted with permission from [141], W. E. Buhro and V. L. Colvin, *Nat. Mater.* 2, 138 (2003). © 2003, Nature Publishing Group.

understanding and the creation of new scientific tools. Chemists and physicists have been manipulating matter on the molecular level for centuries. When one looks at the absolute elegance of the nanometer-scale biological system, however, one is compelled to create, understand, manipulate, and control systems with equal elegance. This high level of activity and promise has attracted private and government funding with considerable economic impact and growth.

Science is beginning to set free nanotechnology—soon the only limit will be the imagination. In 1995, III–V semiconductor NWs were grown in solution from In nanoparticle catalysts by a process analogous to the VLS mechanism, and by analogy was named “solution-liquid-solid” (SLS) mechanism [142]. The nanoparticle-catalyzed VLS mechanism and its solution-phase variants have emerged as popular, widely practiced, general methods for the synthesis of semiconductor NWs [143]. Au nanoparticles are presently by far the most commonly employed catalysts.

“Form ever follows function,” asserted architect Louis Henri Sullivan, provided a maxim for designers of all persuasions. His protégé, Frank Lloyd Wright, preferred “Form and function should be one.” Such adages express the widely held conviction that shape and utility or properties are inextricably linked. Kan [141–144] showed that the electronic structure and optical properties (that is, the QCEs) of rod-like semiconductor nanocrystals depend sensitively on the ratio of their length and diameter. Thus, bodybuilders, distance runners, architects, and quantum mechanics agree: in function, performance and behavior, shape matters.

So far no systematic experimental comparison has been made of the relationships between band gap energies and size in systems that differ only in the geometric dimensionality of confinement. The work makes an important step in that direction by exploring the transition from 3D to 2D confinement in InAs quantum rods.

The authors prepare InAs quantum rods with uniform 4-nm diameters and a mixture of lengths, which were separated into various length fractions having length/diameter ratios ranging from 2.3 to 5.7. The band gaps of the rods were found to decrease with increasing length/diameter ratio (the points in Fig. 25(c), as predicted by Fig. 25(b). As the rod length increases, the band gap decreases towards the value expected of true quantum

wires. Similar length dependence was previously demonstrated in CdSe quantum rods by Alivisatos [145]. So both the II–VI and III–V semiconductor nanostructures have now been shown to have shape-sensitive band gaps.

The excited-state dynamics in quantum rods are apparently quite distinctive. The results indicated that the luminescence quantum yield decreases as the quantum rods lengthen, which is probably because of an increasing spatial separation of electron and hole wave functions [141]. Reducing the wave function overlap should decrease the probability of electrons and holes recombining radiatively, and thus reduce PL efficiency. Finally, unlike isotropic InAs QDs, the InAs quantum rods show polarized luminescence (along their long axes). This phenomenon has been previously demonstrated in semiconductor quantum rods [142] and NWs [143]. The work convincingly establishes that both the magnitude and character of QC in semiconductor nanostructures are highly shape-dependent.

4.2. Blue Shift

In the recent years, exclusive research has been focused on the fabrication of 1D~3D nanostructures [144]. In principle, a 1D nanostructure is anisotropic with the larger aspect ratio (length/diameter) and diameter of ≤ 100 nm. There are a large number of opportunities that might be realized by making new types of nanostructures, or simply by down-sizing the existing nanostructures into the regimes of 1–100 nm. At this scale, in principle, some of the classical laws of physics no longer apply and gives way to properties defined by quantum physics. So, nanomaterials research not only will be a gateway to a realization of the high-performance nanoelectronic and optoelectronic devices but also will explore the novel approaches of quantum physics. The most successful example is provided by microelectronics, where smaller has meant greater performance ever since the invention of integrated circuits, more components per chip, faster operation, lower cost, and less power consumption. It is generally accepted that QCE of electrons by potential wells of nanometer-sized structure will provide one of the most powerful means to control the electrical, optical, magnetic and thermoelectric properties of solid state materials.

Semiconductor nanocrystals are of intense scientific and technological interest. Their electronic structures can be tailored by size and shape, leading to many new applications, including lasers, biological cell labeling, and solar cells. Previously, the study of QWs attracted less interest than QDs because of the technical difficulties in synthesizing them. Recently, however, high-quality semiconductor QWs have been fabricated by a solution-liquid-solid approach in wet chemistry. The diameter of the QWs synthesized by this method is small enough to show strong QCEs.

According to an overly simplified particle-in-a-box effective-mass model represented in Figure 6, the band-gap increases of QDs and QWs from the bulk value are $\Delta E_g = 2\hbar^2\zeta^2/\mu d^2$, where d is the diameter. For spherical QDs, $\zeta = \pi$ is the zero point of the spherical Bessel function, whereas for cylindrical QWs, $\zeta = 2.4048$ is the zero point of the cylindrical Bessel function. Thus, the ratio of band gap increases between QWs and QDs should be $\Delta E_g^{\text{QW}}/\Delta E_g^{\text{QD}} = 0.586$. From this outcome, there are two questions about this simple but effective mass model as follows:

- (1) Does the ΔE_g follow the $1/d^2$ scaling? (If not, do the QW and QD ΔE_g have the same scaling?)
- (2) Is the real ratio of $\Delta E_g^{\text{QW}}/\Delta E_g^{\text{QD}}$ close to 0.586?

4.2.1. II–VI Nanomaterials

Synthesis and properties of ZnO nanostructures have extensively been investigated, and several reviews on current progress in this field have been published in Ref. [145]. A wide range of synthesis routes has been developed that have led to the production of ZnO nanoparticles on a large scale of volume. From the 1960s, synthesis of ZnO thin films has been an active field because of their applications as sensors, transducers, and transparent

conductors. In the last few decades, especially since the nanotechnology initiative led by the United States, Europe, and Japan, the study of 1D nanomaterials has become leading edge in nanoscience and nanotechnology. By reducing the size and shape, novel electrical, mechanical, chemical, and optical properties have been demonstrated which are largely believed to be the result of surface and QC effects.

To focus this basic study and application in devices, a lot of work has been done to control the molecular and atomic scale, including surface modifications. The most common ZnO nanostructures are nanorods [146], NWs [147], nanorings [148], nanocombs [149], and nanonetworks [150]. The nomenclature of these nanomaterials is based on the size and shape of the individual nanostructure. However, growth mechanism and control of these nanostructures, such as shape, size, uniformity, mesoscopic properties, and reproducibility as well as optical and structural properties are still a challenge. In this section, therefore, we will focus on the morphological evaluation of ZnO nanostructures and the subsequent growth mechanism both theoretically and experimentally.

It is a fact that although all the ZnO nanostructures are promising with their surface properties, the optical properties are poor in terms of QCE. No QCEs have been observed yet since the ZnO nanotubes and nanorods diameter are >100 nm, which is several times larger than the Bohr radius of ZnO material. The ZnO exciton Bohr diameter is 3.60 nm. The single-crystalline ZnO nanorings/nanosaws/nanocombs showed the emission energies of 3.21 eV, 2.81 eV, 2.58 eV and negligible green band at room temperature. Several sharp peaks emerged at photon energies around 3.18 eV with the line width of ≤ 4 meV when the sample is excited above a certain threshold [151]. This random lasing in ZnO nanosaws results from the multiple coherent scattering in randomly oriented ZnO nanosaws. A strong exciton absorption peak from nanoflowers at 3.48 eV has been observed, which has a large blue shift compared to that of the bulk ZnO material. The strong background below the peak should be due to the strong scattering of nanoflowers as their sizes are comparable to the wavelength. Although the blue shift has been observed from the nanoflowers, we believe more experiment are necessary since there are many nanostructures even under <10 nm diameter that have no QCE.

Figure 26 shows the variation of emission energy (ΔE) as a function of ZnO nanocrystal diameter [152]. The variation of ΔE of the ZnO nanocrystals has been calculated from the free exciton transition energy at RT. It shows the blue shift with the decrease of ZnO nanoparticle size. The variation ΔE of ZnO nanocrystals has been estimated roughly to be 40 meV and occurred as a result of changing the diameter from bulk to 24 nm

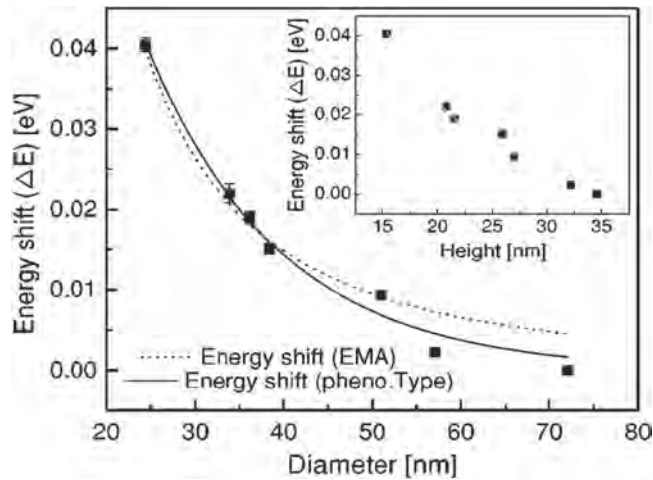


Figure 26. Emission energy (ΔE) variation as a function of ZnO nanocrystal diameter. The variation of ΔE of the ZnO nanocrystals has been calculated from the free exciton transition energy at RT. The variation ΔE of ZnO nanocrystals has been estimated roughly to be 40 meV, and occurred as a result of changing the diameter from bulk to 24 nm (height: 15 nm). Reprinted with permission from [152], T. B. Hur et al., *Appl. Phys. Lett.* 86, 193113 (2005). © 2005, American Institute of Physics.

(height: 15 nm). The first explanation for the size dependence of the electronic properties of the nanocrystals was the effective mass approximation (EMA) in the infinite confining potential by Efros and Efros [153]. In this model, $\Delta E_g = \mu/d^2$, where the d is diameter of nanocrystals. The size dependence of electronic properties in a finite-size system was proposed by Brus and co-authors [154]. This expression is composed of a kinetic energy term, a Coulomb attraction term between electron and hole, and a polarization term. The size-dependent function of Brus was useful in the strong confinement regime and below roughly ~ 14 nm in diameter with ZnO parameters such as effective masses and static dielectric constant. Therefore, the data on the size dependence of the emission energy shift in Figure 26 have been fitted by an EMA and phenomenological simple exponential functions. The QCE in the Volmer-Weber type self-assembled ZnO nanocrystals were apparently revealed for the smaller sizes than 57 nm in diameter (height: 32 nm). Therefore, to correct the different values, the detailed studies on the surface polarization effects and geometrical effects of nonspherical-type nanocrystals are necessary.

4.2.2. III–V Nanomaterials

Semiconductor nanostructures have been successfully incorporated into a variety of nano-scale devices as both active elements [155] and device interconnects [156]. However, important questions remain about the optical properties of these structures, particularly concerning photoexcited carrier dynamics and relaxation, which are crucial for the design optimization and ultimately the success of NW-based optical devices. Moreover, because such nanostructures are not small on a quantum level, they can support excitation of a degenerate density of electrons and holes, so that many-body effects can be quite prominent.

Electrons and holes in InP are particularly insensitive to surface states, which cause extremely short recombination lifetimes and low quantum efficiencies in other NW systems including CdS, GaAs, and GaAs/GaAlAs [157]. The surface recombination velocity, S , one metric often used to characterize the effect of nonradiative surface (or interface) states, is notably different in different semiconductors. The surface recombination velocity in as-grown, undoped epilayers in InP is $S = 10^3$ cm/s [158], while GaAs exhibits velocities nearly a 1000 times higher: $S = 10^6$ cm/s [159]. Such a difference should have a dramatic impact on the nonradiative lifetime in NWs. Thus equivalent 40 nm diameter GaAs or InP NWs should exhibit nonradiative lifetimes differing by a factor of 1000, from 4 ps in GaAs to 4 ns in InP [160]. We therefore expect InP NWs to exhibit significantly higher quantum efficiencies, which will allow us to measure the intrinsic exciton lifetime in single NWs as well as to enable us to probe many-body interactions at high electron-hole carrier densities.

Unlike the case of state filling in QDs and related quantum-confined systems where the density of states is discrete along at least 1D–3D, the NWs studied here have diameters significantly larger than the exciton Bohr radius, and therefore no QCEs are expected. Under high excitation density conditions, emission from electron-hole plasma (EHP) should be observed with Fermi energies that show significant occupation at higher energies at early times. One characteristic signature of such large electron-hole carrier densities are many-body interactions, which induce a renormalization of the band gap. The band gap renormalizes, moving to lower energies as a result of the increased exchange and correlation energy of the carriers. Using a theoretical estimate of the band gap renormalization (BGR), we will show consistently that, for the measured EHP spectra, the BGR scales with the increased Fermi energy expected for an EHP of a particular density.

The blue shift of the optical band gap allows estimating the particle size of GaN. In a simple model, in which electrons and holes are individually confined in small particles, the energy of the lowest optical transition in semiconductor nanoparticles is given by

$$E^* = E_g + \frac{\hbar^2}{8\mu R^2} \quad (11)$$

where R is the radius of nanoparticle. Using the band gap estimated from absorption coefficient and Eq. (11), the mean diameters ($2R$) can be calculated to be 3.70, 4.22, and

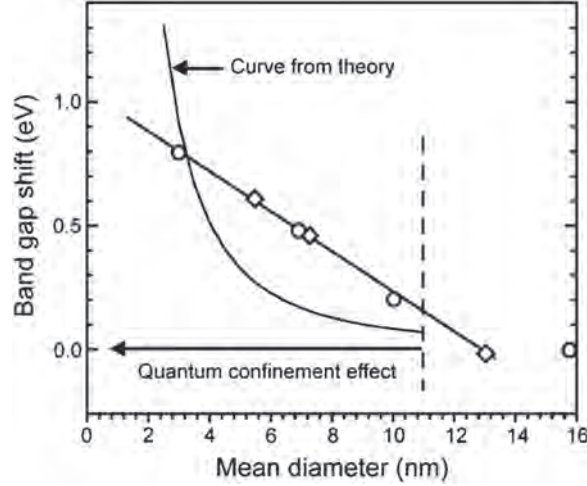


Figure 27. Relationship between the band gap shift of GaN nanoparticles and particle size. The open diamond represents the mean diameter of GaN nanoparticles obtained from HRTEM observation. The open circles are quoted from N. Preschilla et al., *Appl. Phys. Lett.* 77, 1861 (2000). Reprinted with permission from [164], N. Preschilla et al., *Appl. Phys. Lett.* 77, 1861 (2000). © 2000, Elsevier.

7.39 nm for films deposited at 1, 10, and 50 Pa, respectively. Figure 27 shows the particles size dependence of the band gap shift, together with mean diameter from high-resolution transmission electron microscopy (HRTEM) observation (marked with a white diamond). For GaN with a particle size smaller than 11 nm, QCE was shown by a band gap shift.

In contrast, a shift of band gap did not occur for GaN with a particle size of 13 nm. This result indicates that the band gap of GaN with a particle size exceeding 11 nm is the same as that of GaN bulk. Moreover, the open circles represent the size dependence of the band gap shift quoted from Ref. [161]. The measured values showed a linear increase with a decreasing of mean diameter. This study showed that the QC effect of GaN is exhibited only at particle sizes below the Bohr radius (11 nm).

5. BAND OFFSET IN QUANTUM CONFINEMENT

5.1. II–VI Materials

Until recently, it has been commonly accepted that the band gap of ZnSSe increases more or less monotonically with increasing *S* content (from 2.80 to 3.67 eV) [162]. It has been shown that the *S* incorporation generates significant effects on the band gap variation from 2.60–3.67 eV: the band gap decreases with increasing *S* content until 2.60 eV for $x = 0.2$, and then increases up to 3.67 eV for ZnS. Using the model solid theory [163], the valence and conduction band offsets (VBO and CBO) for $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_y\text{Se}_{1-y}$ heterostructures ($0 \leq x, y \leq 1$) have been calculated. The thickness and composition ($0 \leq x \leq 0.20$) are computed to get the optimum confinement of the interband electron–hole transition and the optimum output of plane oscillator strength and wave function overlap.

The ZnSe and ZnS semiconductors have the zincblende structure with a band structure that includes three valence bands at Γ . In the absence of strain and spin-orbit splitting, these bands degenerate. The spin-orbit interaction splits them into heavy and light hole bands (*hh* and *lh*) and the spin-orbit band (*so*). The hydrostatic strain shifts the average valence band energy and the conduction band while the shear component of strain splits the degeneracy of the *lh* and the *hh* valence bands when coupled to the *so* interaction. The band gap energy shifts are related to strains by making use of deformation potentials. For strain along the [001] direction of $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_y\text{Se}_{1-y}$ heterostructure, the shifts of the valence band is given by [164]

$$\delta E_{v,hh} = a_v(2\varepsilon_{||} + \varepsilon_{\perp}) - 0.5\delta E_{001} = 2a_v\varepsilon_{||}(1 - C_{12}/C_{11}) - 0.5\delta E_{001} \quad (12)$$

$$\delta E_{v, lh} = 2a_v \varepsilon_{||} (1 - C_{12}/C_{11}) - \Delta_0/2 + 0.25\delta E_{001} + 0.5[\Delta_0^2 + \Delta_0\delta E_{001} + 2.25(\delta E_{001})^2]^{1/2} \quad (13)$$

$$\delta E_{v, so} = 2a_v \varepsilon_{||} (1 - C_{12}/C_{11}) - \Delta_0/2 + 0.25\delta E_{001} - 0.5[\Delta_0^2 + \Delta_0\delta E_{001} + 2.25(\delta E_{001})^2]^{1/2} \quad (14)$$

In these equations, δE_{001} is given by $\delta E_{001} = 2b(\varepsilon_{\perp} - \varepsilon_{||})$ and $\delta E_{001} = -2b\varepsilon_{||}(2C_{12}/C_{11} + 1)$. The component of the strain tensor for $\text{ZnS}_x\text{Se}_{1-x}$, $\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon_{||}$ and $\varepsilon_{zz} = \varepsilon_{\perp}$ are given by $\varepsilon_{||} = a_{||}/a - 1$ and $\varepsilon_{\perp} = -2(C_{12}/C_{11})\varepsilon_{||}$. With $a_{||}$ and a are the substrate and the unstrained over-layer lattice constants while $a_{\perp}$ is the perpendicular over-layer lattice constant given in the $[001]$ direction, by

$$a_{\perp} = a \left\{ 1 - \frac{2C_{12}}{C_{11}} \left(\frac{a_{||}}{a} - 1 \right) \right\} \quad (15)$$

Δ_0 is the *so* splitting, a_v and b are the hydrostatic and the shear deformation potential for the valence band and C_{ij} specify the bulk elastic coefficients for the material under consideration. The first terms in the valence band shifts are the hydrostatic corrections and the second are the shear ones. Equations (12)–(14) show that the *hh* remain uncoupled to the *lh*, while the *lh* and the split-off bands are coupled by the strain and the *so*. It is noted that the hydrostatic component of the strain is the same for the *hh*, *lh* and *so*. Seeing that the ZnSSe alloy has a direct band gap along the *S* composition range, the shear strain has no effect on the energetic position of the conduction band and there is only a hydrostatic conduction band correction due to strain. This component is given by

$$\delta E_c = a_c(2\varepsilon_{||} + \varepsilon_{\perp}) \quad \text{and} \quad \delta E_c = 2a_c\varepsilon_{||}(1 - C_{12}/C_{11}) \quad (16)$$

a_c being the hydrostatic deformation potential for the conduction band. Consequently, the valence and conduction band offsets including strain effects for the $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_y\text{Se}_{1-y}$ heterojunction are given by

$$\Delta E_v = [E_v^{\text{uns}}(\text{ZnS}_x\text{Se}_{1-x}) + \delta E_v^{\text{hyd}} + \delta E_v^{\text{sh}}] - E_v^{\text{uns}}(\text{ZnS}_y\text{Se}_{1-y}) \quad (17)$$

$$\Delta E_v = \Delta E_v^{\text{uns}} + \delta E_v^{\text{hyd}} + \delta E_v^{\text{sh}} \quad (18)$$

$$\Delta E_{v, hh, ih, so} = \Delta E_v^{\text{uns}} + \delta E_{v, hh, ih, so} \quad (19)$$

$$\Delta E_c = \Delta E_v^{\text{uns}} + \Delta E_g + \delta E_c \quad (20)$$

where δE_v^{hyd} is the hydrostatic component of strain, while δE_v^{sh} is the shear component; ΔE_g^{unst} represents the band gap difference between the $\text{ZnS}_x\text{Se}_{1-x}$ and $\text{ZnS}_y\text{Se}_{1-y}$ unstrained bulk materials. Note that Eqs. (18) and (20) result from an approximation since the band offset modification due to the strain-induced charge transfers at the heterointerface is neglected. The band gap energy for the alloy $\text{ZnS}_x\text{Se}_{1-x}$ is taken [164] and given by

$$E_g^{\Gamma}(x) = 2.80 - 2.26x + 7.55x^2 - 4.43x^3 \quad (21)$$

showing an equivalent bowing parameter b i.e., $b(x) = -4.43x + 3.13$. Note that b is strongly composition dependent, leading to a nonlinear and nonmonotonic composition dependence of the band gap.

Using the above equations, band offset values for strained $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_y\text{Se}_{1-y}$ for the range $0 \leq x, y \leq 1$ have been calculated. The parameters related to the strain are listed in Table 3. The data for $\text{ZnS}_x\text{Se}_{1-x}$ alloys are evaluated from a linear interpolation of the parameters listed in Table 4. Results are depicted as a function of x and y compositions in Figures 28(a and b) for electrons. For *hh* and *lh*, the iso-energy band offsets are practically linear. The laws are as follows:

$$\Delta E_{v, hh}(x, y) = (y - x)[0.95 - 0.05x + 0.001x(x + y) - 0.01y] \quad (22)$$

$$\Delta E_{v, lh}(x, y) = (y - x)[0.788 - 0.41x + 0.01\sqrt{x}(x + y) - 0.01y] \quad (23)$$

Table 3. Spin-orbit splitting Δ_0 , the average energy of the three uppermost valence bands $E_{v,av}$, experimental values of hydrostatic deformation potential for the valence, and conduction bands a_v and a_c , experimental uniaxial deformation potential b .

	Δ_0 (eV)	$E_{v,av}$ (eV)	a_v (eV)	a_c (eV)	B (eV)	C_{11} (Nm ⁻²)	C_{12} (Nm ⁻²)
ZnSe	0.43	-8.37	1.65	4.17	-1.20	8.95	5.39
ZnS	0.07	-9.15	2.30	-4.09	-0.80	9.81	6.27

Table 4. Calculated masses for the respective x in alloy system. The data for $\text{ZnS}_x\text{Se}_{1-x}$ alloys are evaluated from a linear interpolation of the parameters listed.

	$x = 0$	$x = 0.1$	$x = 0.2$	$x = 0.8$	$x = 1$
m_e^*	0.19	0.19	0.19	0.24	0.27
m_{hh}^*	1.20	1.30	1.40	1.80	1.76
m_{lh}^*	0.15	0.15	0.15	0.19	0.23

gives good approximate analytical expressions for the hh and lh valence band discontinuities.

Following this empirical pseudopotential method for a parabolic fit of the conduction band dispersions of $\text{ZnS}_x\text{Se}_{1-x}$ alloy, the electron effective mass m^* at the minimum of the CB is given by [165]

$$m_e^*(x) = 0.193 - 0.064x + 0.160x^2 \quad (24)$$

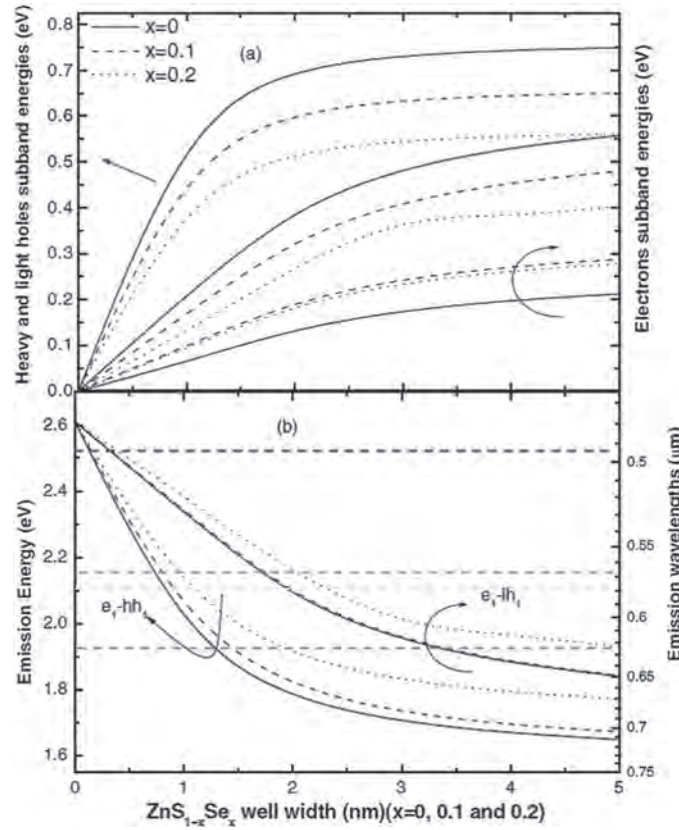


Figure 28. (a) Electron, hh and lh confinement energies for $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_{0.8}\text{Se}_{0.2}$ structure as a function of $\text{ZnS}_x\text{Se}_{1-x}$ well width with values of $x = 0$ (solid line), $x = 0.1$ (dashed line), and $x = 0.2$ (dotted line). (b) Deduced evolution of e_1-hh_1 and e_1-lh_1 transition energies versus the $\text{ZnS}_x\text{Se}_{1-x}$ layer thickness. Reprinted with permission from [165], S. A. Nasrallah et al., *J. Phys.: Condens. Matter.* 18, 3005 (2006). © 2006, Institute of Physics.

Table 5. Calculated conduction (ΔE_c) and valence ($\Delta E_{v,hh}$, $\Delta E_{v,lh}$) band offsets for strained $\text{ZnS}_x\text{Se}_{1-x}$ with $x = 0, 0.1$, and 0.2 on relaxed $\text{ZnS}_{0.8}\text{Se}_{0.2}$ substrate given in eV.

	$x = 0$	$x = 0.1$	$x = 0.2$
ΔE_c	-0.25	-0.33	-0.32
$\Delta E_{v,hh}$	0.76	0.66	0.57
$\Delta E_{v,lh}$	0.62	0.54	0.46

gives m^* for any S molar fraction. Due to the lack in the hole effective masses of ZnSSe alloys, for the hh and lh in ternary compounds we opt for linear interpolation. Values of effective masses for electrons and heavy and light holes denoted respectively m_e^* , m_{hh}^* and m_{lh}^* evaluated in units of the free electron mass m_0 , and plotted in Table 5. The results of calculation are summarized in Table 3. The conduction and valence band discontinuities for $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_{0.8}\text{Se}_{0.2}$ QW under investigation ($x = 0, 0.1, 0.2$ and $y = 0.8$) calculated are listed in Table 4.

Band gap energies for the relaxed ZnSSe and strained ZnSSe on $\text{ZnS}_{0.8}\text{Se}_{0.2}$ substrates are plotted in the whole range as a function of S molar fraction, x . The strain causes a red shift in the band gap, and furthermore the sub-band energies for $x = 0$ (solid line), 0.1 (dashed line), and 0.2 (dotted line) as a function of the well width L_w are illustrated. Figure 28(a) shows that the energy levels increase until $L_w = 2$ nm for the hh and approximately $L_w = 4$ nm for electrons and lh and then become stationary; the electron energy levels are nearly the same with $x = 0.1$ and 0.2 . As is seen, the sub-band energies increase with the L_w of $\text{ZnS}_x\text{Se}_{1-x}$ layer. Therefore, we have exhibited in (Fig. 28(b)) the emission energies for the fundamental transitions $e_1 - hh_1$ and $e_1 - lh_1$ with $x = 0$ (solid line), $x = 0.1$ (dashed line), and $x = 0.2$ (dotted line). The emission energy of the $e_1 - hh_1$ ($e_1 - lh_1$) transition undergoes a red shift with increasing L_w but increases with S composition x . The transition energy $e_1 - lh_1$ has the same value for $x = 0$ and 0.1 .

It is clear that by varying the L_w it is possible to realize $\text{ZnS}_x\text{Se}_{1-x}/\text{ZnS}_{0.8}\text{Se}_{0.2}$ -based devices that can be tuned to emit throughout the green, yellow, and orange by changing only the L_B or the S composition x . With the $e_1 - hh_1$ transition, we can achieve for S molar fraction $x = 0, 0.1, 0.2$ the green emission (2.15 eV) with $L_w = 0.7, 0.9, 1$ nm, the yellow (2.12 eV) with $L_w = 0.8, 0.9, 1.2$ nm, and the orange (1.92 eV) with $L_w = 1.3, 1.5, 2$ nm, respectively. With the $e_1 - lh_1$ transition, the same wavelengths are accomplished as follows: the green with $L_w = 1.7, 1.7, 2$ nm, the yellow with $L_w = 1.9, 1.9, 2.3$ nm, and the orange with $L_w = 3.5, 3.5 > 5$ nm, respectively. Therefore, the compositions and L_w accessible in a realistic way for epitaxial growth, $L_w = 1$ nm, $x = 0.2$; $L_w = 1.2$ nm, $x = 0.2$; and $L_w = 2$ nm, $x = 0.2$ respectively for green, yellow, and orange. To study and carry out the quantum efficiency of our designs with these sets of parameters, we have calculated

- (i) the oscillator strength of the $e - hh$ transition, which is defined by [166]

$$f_{l \rightarrow k} = \frac{2m_0}{\hbar^2} (E_l - E_k) |\langle \phi_l | z | \phi_k \rangle|^2 \quad (25)$$

where $(E_l - E_k)$ is the energy difference between the initial and final states and $|\langle \phi_l | z | \phi_k \rangle|$ is the dipole matrix element of the transition, and

- (ii) the wave function overlap, defined as $|\langle \phi_l | \phi_k \rangle|^2$. The edges of the conduction and valence bands of the simulated and optimized designs for the green, yellow, and orange wavelengths are illustrated in Figure 28.

6. BARRIERS IN QUANTUM CONFINEMENT

We pay attention to small Be content because the lattice mismatch between the well and the L_B increases with increasing Be content. The deformation potentials of BeS are unknown; therefore, conduction and valence band offsets in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs are calculated assuming a SQW condition, where a ZnS layer is sandwiched between the

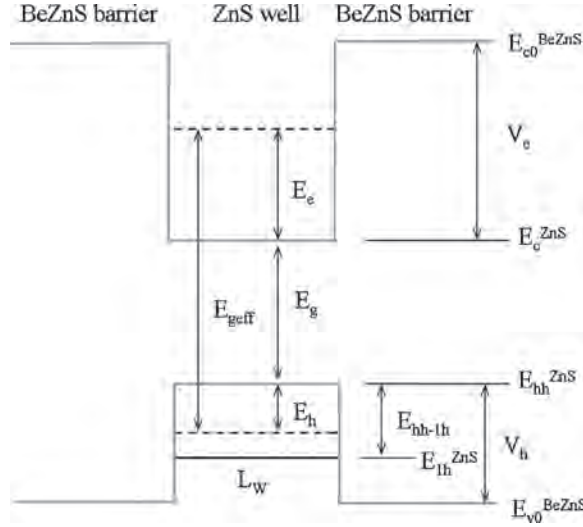


Figure 29. A schematic illustration of band lineups for ZnS/Be_xZn_{1-x}S SQWs. Conduction and valence bands are shown, together with well width.

two semi-infinite layers of Be_xZn_{1-x}S. Under this condition, induced strain applies only to the L_w . The amount of induced strain remains constant with increasing L_w . The lattice constant of ZnS is larger than that of Be_xZn_{1-x}S; the ZnS well layer is subject to compressive strain, and the top of the valence band consists of a hh band in the ZnS well layer; thus, hh band offset is calculated as a valence band offset. Figure 29 shows a schematic illustration of band lineups in ZnS/Be_xZn_{1-x}S SQW. The strain tensors in the well layer are given by

$$\varepsilon_{xx} = \varepsilon_{yy} = \varepsilon = \frac{a_w - a_b}{a_b} \quad (26)$$

$$\varepsilon_{zz} = -\frac{2C_{12}}{C_{11}}\varepsilon \quad (27)$$

$$\varepsilon_{xy} = \varepsilon_{yz} = \varepsilon_{zx} = 0 \quad (28)$$

where a_w and a_b are the lattice constants of the L_w and the L_b , respectively. The energy shifts of the hh and lh bands under the strain [167] are given by

$$dE_{hh} = E_H - E_S \quad (29)$$

$$dE_{lh} = E_H - \frac{1}{2}\Delta_{so} + \frac{1}{2}E_S + \frac{1}{2}(\Delta_{so}^2 + 2\Delta_{so}E_S + 9E_S^2)^{1/2} \quad (30)$$

with

$$E_H = -a_v(2\varepsilon + \varepsilon_{zz}) \quad (31)$$

$$E_S = -b(\varepsilon_{zz} - \varepsilon) \quad (32)$$

and Δ_{so} is the spin-orbit splitting energy and b is the shear deformation potential. The center of gravity of the valence band is shifted by the hydrostatic component and shift is given by E_H . The energy shift of the bottom of the conduction band [167] is given by

$$dE_c = a_c(2\varepsilon + \varepsilon_{zz}) \quad (33)$$

The E_{hh} , E_{lh} , and E_c band energies under a strain are defined as the summation of the energies of the top of the valence band and the bottom of the conduction band without

strain and the energy shifts of hh and lh and conduction bands, respectively. The E_{hh} , E_{lh} , and E_c band energies in ZnS well layers under a strain are given by

$$E_{hh}^{\text{ZnS}} = E_{v0}^{\text{ZnS}} + dE_{hh}^{\text{ZnS}} \quad (34)$$

$$E_{lh}^{\text{ZnS}} = E_{v0}^{\text{ZnS}} + dE_{lh}^{\text{ZnS}} \quad (35)$$

$$E_c^{\text{ZnS}} = E_{c0}^{\text{ZnS}} + dE_c^{\text{ZnS}} \quad (36)$$

where E_{v0}^{ZnS} and E_{c0}^{ZnS} are the bottom of the conduction band energy without strain and the top of the valence band energy without strain, respectively, in the ZnS well layer. The conduction and valence band offsets are calculated using

$$V_e = E_{c0}^{\text{BeZnS}} - E_c^{\text{ZnS}} \quad (37)$$

$$V_h = E_{hh}^{\text{ZnS}} - E_{v0}^{\text{BeZnS}} \quad (38)$$

We calculate the exciton binding energy using a variation method [168]. The Hamiltonian is given using the cylindrical coordinates as

$$H = \frac{-\hbar^2}{2\mu} \left(\frac{1}{r} \frac{\partial}{\partial r} r \frac{\partial}{\partial r} \right) - \frac{\hbar^2}{2m_c^*} \frac{\partial^2}{\partial z_e^2} - \frac{\hbar^2}{2m_h^*} \frac{\partial^2}{\partial z_h^2} - \frac{e^2}{\epsilon \sqrt{r^2 + (z_e - z_h)^2}} + V_e(z_e) + V_h(z_h) \quad (39)$$

To simplify the calculation, we neglect any effective-mass mismatch between the well and barrier layers, and assume a position-independent dielectric constant, thus neglecting the effects of the image charges. In addition, we assume isotropic mass to simplify the calculation. We therefore assume that the electron and hole masses in the L_B and the dielectric constant of the L_B are the same as those of the L_W . The potential wells for the electrons and for the holes are assumed to be square wells of width L_w [168],

$$V_e(z_e) = \begin{cases} 0, & |z_e| < L_w/2 \\ V_e, & |z_e| > L_w/2 \end{cases} \quad (40)$$

and

$$V_h(z_h) = \begin{cases} 0, & |z_h| < L_w/2 \\ V_h, & |z_h| > L_w/2 \end{cases} \quad (41)$$

Here, we have chosen the origin of the coordinate system to be the center of the L_W . The values of the potential well heights V_e and V_h are determined by Eqs. (37) and (38). We use a variational approach. The trial function used [169] is given by

$$\varphi_r(z_e, z_h, r) = \phi_e(z_e) \phi_h(z_h) \times \exp[-\alpha_{1s}\{r^2 + \beta_{1s}^2(z_e - z_h)^2\}^{1/2}] \quad (42)$$

with

$$\phi_e(z_e) = \begin{cases} B_e \exp[-\lambda_e |z_e|] & z_e > |L_w/2| \\ \cos(k_e z_e) & z_e \leq |L_w/2| \end{cases} \quad (43)$$

$$\phi_h(z_h) = \begin{cases} B_h \exp[-\lambda_h |z_h|] & z_h > |L_w/2| \\ \cos(k_h z_h) & z_h \leq |L_w/2| \end{cases} \quad (44)$$

where $\phi_e(z_e)$ and $\phi_h(z_h)$ are the electron and heavy-hole wave functions, respectively. k_e is determined for the energy of the first electron subband, and B_e and λ_e are obtained from k_e by requiring continuity of $\phi_e(z_e)$ and the first derivative at the interface [22]. k_h is determined for the energy of the first hole subband, and B_h and λ_h are obtained from k_h

by requiring continuity of $\hbar(z_h)$ and the first derivative at the interface [168]. The subband energy is given by

$$\frac{m_w^* \beta}{m_b^* \alpha} = \tan\left(\alpha \frac{L_w}{2}\right) \quad (45)$$

with

$$\alpha = \sqrt{\frac{2m_w^*}{\hbar^2} E} \quad (46)$$

$$\beta = \sqrt{\frac{2m_b^*}{\hbar^2} (V - E)} \quad (47)$$

where V is the potential barrier height, and E is the sub-band energy. In this system, the electron and hole masses in the $\text{Be}_x\text{Zn}_{1-x}\text{S}$ barrier layer are unknown; therefore, we assume that $m_w = m_b$. The excitonic transition energy is given by

$$E_{\text{tr}} = E_{\text{geff}} - E_x \quad (48)$$

where E_{geff} is the effective band gap energy and E_x is the exciton-binding energy. The effective band gap energy is given by $E_{\text{geff}} = E_g + E_e + E_h$. The band gap energy of the well layer is given by $E_g = E_{c,w} - E_{hh,w}$. In addition, the energy separation between hh and lh bands in the L_w is given by

$$E_{hh-lh} = |E_{hh}^{\text{ZnS}} - E_{lh}^{\text{ZnS}}| \quad (49)$$

Under the thin-well-width condition, the exciton binding energy drops rapidly with decreasing well width in SQWs [168]. The calculated result seems to be valid until the well width at which the exciton binding energy takes a maximum, with decreasing L_w . To determine the well width at which the exciton binding energy rapidly drops, we calculate the exciton binding energy from 0.5 to 20 nm. The calculated result is plotted in Figure 30. The exciton binding energy increases with x because the conduction and valence band offsets increase with increasing x . The maximum exciton binding energy is approximately 1.3 times larger than the bulk exciton binding energy of ZnS at $x = 0.1$. The exciton binding energy in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs is larger than that of $\text{Cd}_x\text{Zn}_{1-x}\text{S}/\text{ZnS}$

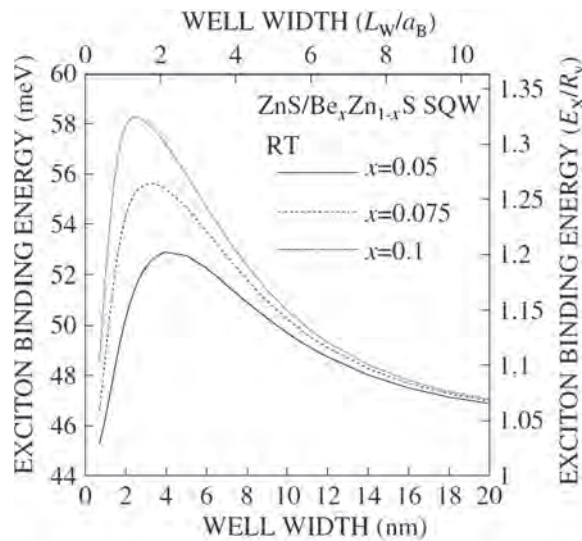


Figure 30. Exciton binding energy in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ single quantum wells (SQWs) at Be content $x = 0.05$, 0.075 , and 0.1 as function of L_w . Quantum confinement effect (QCE) is shown in terms of well width in SQW. Reprinted with permission from [171], C. Onodera et al., *Jpn. J. Appl. Phys.* 45, 5821 (2006). © 2006, The Japan Society of Applied Physics.

QWs [168]. An increase in conduction band offset effects on an increase in exciton binding energy. It seems that excitons in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs exist stably at RT. The stability of excitons in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs increases with increasing x . In addition, the exciton binding energy in the $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs at $x = 0.1$ is larger than that in $\text{ZnS}/\text{Mg}_x\text{Zn}_{1-x}\text{S}$ QWs at $x = 0.19$ [170]. This indicates that the $\text{Be}_x\text{Zn}_{1-x}\text{S}$ barrier is useful for obtaining large quasi-2D confinement using ZnS wells.

We calculate the effective band gap energy in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs at $x = 0.05, 0.075$, and 0.1 as a function of L_w at RT. The calculated results are plotted in Figure 30. The effective band gap energy clearly increases at $L_w < 8$ nm in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs. This is explained by the fact that the electron and hole sub-band energies increase with increasing x because the conduction and valence band offsets increase with increasing x . When $L_w > 8$ nm, in $\text{ZnS}/\text{Be}_x\text{Zn}_{1-x}\text{S}$ SQWs, the variation of the effective band gap energy is small with increasing x [171].

7. CORE/SHELL NANOHETEROSTRUCTURES

Small is beautiful—not only beautiful but also powerful. With the range of applications that nanoparticles find in varied fields of engineering and science, nanoparticles seem a promising option when compared to the conventional materials used. Nanoparticles are particles that have at least 1D in less than 100 nm. They have a high surface-to-volume ratio and thus their mass transfer and heat transfer properties are better compared to those of bulk materials. Core/shell NW heterostructures are of significant interest for electronics as the shell protects the active core/shell interface [173], while the band offset provides carriers and could enable modulation doping to address the difficulties in reproducible impurity doping that semiconductor nanostructures encounter. Figure 31 shows the core/shell nanostructure schematic.

Recently, core/shell CdO/ZnO and ZnO/BeO nanoparticles are finding widespread application. There is a class of core/shell nanoparticle that has its entire constituent in the nanometer range. Core/shell nanoparticles are nanostructures that have a core made of a material coated with another material. They are in the size range of 20–200 nm. The necessity to shift to core/shell nanoparticles is the improvement in the QCE properties. Taking into consideration the size of the nanoparticles, the shell material can be chosen such that the agglomeration of particle can be prevented. This implies that the monodispersity of the particles can be improved. The core/shell structure enhances the thermal and chemical stability of the nanoparticles, improves solubility, makes them less cytotoxic, and allows conjugation of other molecules to these particles. The shell can also prevent the oxidation of the core material [172, 173]. When a core nanoparticle is coated with an inorganic layer, such as silica, the inorganic layer would endow the hybrid structure with an additional function/property on top of the function/property of the core, hence synergistically emerged functions can be envisioned [174].

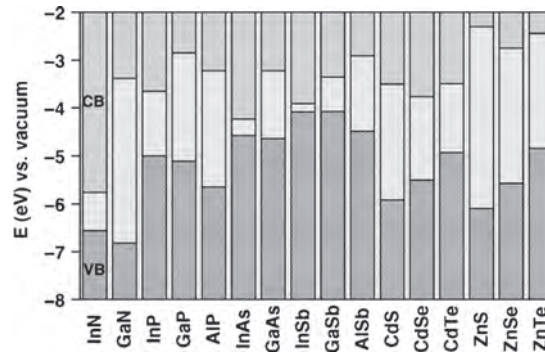


Figure 31. Schematics of core and core/shell nanoheterostructures. In this case CdO, ZnO, and BeO materials have been considered. The band gap energies of CdO, ZnO, and BeO materials are 2.24 eV, 3.37 eV, and 7.6 eV, respectively.

7.1. Core/Shell Nanocrystals

The properties of nanomaterials are diverse and cannot be generalized even though the particles under comparison might be made of similar material and composition. This rule is applicable to core/shell nanoparticles and, hence, this can be classified broadly based on the material with which the core and shell of the nanocomposite are made. Depending on the band gaps and the relative position of electronic energy levels of the involved semiconductors, the shell can have different functions in nanocrystals. Figure 32 shows a schematic of an overview of the band alignment of the bulk materials, which are mostly used in nanocrystal synthesis. Electronic energy levels of selected III–V and II–VI semiconductors are considered using the valence-band offsets from [9]. Three cases can be distinguished, denominated type-I, reverse type-I, and type-II band alignment:

- (1) In the former, the band gap of the shell material is larger than that of the core and both electrons and holes are confined in the core.
- (2) In the second, the band gap of the shell material is smaller than that of the core and, depending of the thickness of the shell, the holes and electrons are partially or completely confined in the shell.
- (3) In the latter, either the valence band edge or the conduction band edge of shell material is located in the band gap of the core. Upon excitation of nanocrystal, the resulting staggered band alignment leads to a spatial separation of hole and electron in different regions of core/shell structure.



In type-I nanocrystals, the shell is used to passivate the surface of core with the goal to improve its optical properties. The shell of nanocrystal physically separates the surface of the optically active core from its surrounding medium. Consequently, the sensitivity of optical properties to changes in the local environment of the nanocrystals surface, induced, for example, by the presence of oxygen or water molecules, is reduced. With respect to core nanocrystals, core/shell systems exhibit generally enhanced stability against photodegradation. At the same time, shell growth reduces the number of surface dangling bonds, which can act as trap states for charge carriers and thereby reduce the fluorescence. The ZnS shell significantly improves the fluorescence and stability against photobleaching. This observation is attributed to a partial leakage of the exciton into the shell material [175].

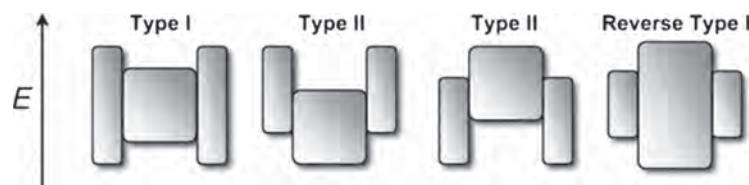


Figure 32. Schematics of energy-level alignment in different core/shell systems realized with semiconductor nanocrystals to date. The upper and lower edges of the rectangles correspond to the positions of the conduction- and valence-band edge of the core (center) and shell materials, respectively. Three cases can be distinguished, denominated type-I, reverse type-I, and type-II band alignment.

In a reverse type-I system, a material with a narrower band gap is overgrown onto the core with a wider band gap. Charge carriers are at least partially delocalized in shell and the emission wavelength can be tuned by shell's thickness. The most extensively analyzed systems of this type are CdS/HgS [176], CdS/CdSe [177], and ZnSe/CdSe [178]. The resistance against photobleaching and the fluorescence of these systems can be improved by growing a second shell of a larger band gap semiconductor on the core/shell nanocrystals [179].

In type-II systems, shell growth aims at a significant red shift of the emission wavelength of nanocrystals. The staggered-type band alignment leads to a smaller effective band gap than each one of the constituting core and shell materials. The interest of these systems is the possibility to manipulate the shell thickness and thereby tune the emission color towards spectral ranges, which are difficult to attain with other materials. Type-II nanocrystals have been developed in particular to near-infrared emission, using CdTe/CdSe or CdSe/ZnTe materials. In contrast to type-I system, the PL decay times are strongly prolonged in type-II nanocrystals due to the lower overlap of electron and hole wave functions. As one of the charge carriers (electron or hole) is located in the shell, an overgrowth of type-II core/shell nanocrystals with an outer shell of an appropriate material can be used in the same way as in a type-I system to improve the fluorescence QY and photostability.

7.1.1. GaAs/AlGaAs Core/Shell Nanostructures

GaN nanostructures have been studied more comprehensively than any other III-N materials due to their great potential for a variety of practical applications [2, 180]. Core-shell NW heterostructures are of significant interest for optoelectronics as the shell protects the active core-shell interface [181], while the band offset provides carriers [182] and could enable modulation doping to address the difficulties in reproducible impurity doping that semiconductor nanostructures encounter. The GaAs/AlGaAs system is of significant interest for high mobility electronics because it possesses an intrinsically high electron mobility [183], and the 2D conduction in GaAs/AlGaAs modulation doped field effect transistors (FETs) has demonstrated enhancement of carrier mobility over bulk systems [184]. Studies of 1D GaAs/AlGaAs conduction suggest even further improvement over 2D conduction [185], and GaAs/AlGaAs core-shell NW heterostructures would represent a unique model system to achieve this 1D carrier confinement. For high mobility electronics applications the switching speed of a high mobility transistor increases with the band offset at the GaAs/AlGaAs interface [186], therefore higher Al incorporation in the shell is needed. Moreover, defect-free and atomically sharp interfaces are essential to reduce the electron scattering, but no in-depth structural analysis of GaAs/AlGaAs core-shell NW heterostructures has been reported to date. Figure 33 shows the core/shell GaAs/AlGaAs nanostructure.

Chemical analysis confirms the core-shell morphology with a shell composed of AlGaAs and verifies the shell to be of uniform composition along the length of the NW. Structural analysis confirms the epitaxial and atomically sharp shell deposition and the entire NW heterostructure to be defect-free and single crystalline [187]. For GaAs/AlGaAs NWs with an average diameter of 80 nm [188], PL spectra measured at 10 K are displayed in Figure 34(a). The spectra of all NWs consist of a single broad peak at 1.518 eV, which corresponds closely to the free exciton emission in bulk GaAs. The large peak width and the variability in the peak position may be related to the variations in the blue shift of the exciton emission energy due to strain and compositional nonuniformities.

The emission efficiency of the core/shell NWs is over an order of magnitude higher compared to the emission efficiency of bare GaAs NWs. Representative PL spectra of the most intense core shell and of the most intense bare NW recorded at identical excitation conditions are compared in Figure 34(b). The enhancement of the low-temperature PL efficiency by passivating the GaAs surface with AlGaAs is comparable to room-temperature PL measurements on similar core/shell structures [189]. On the other hand, time-resolved (TRPL) measurements have indicated that the recombination lifetime of excitons in both

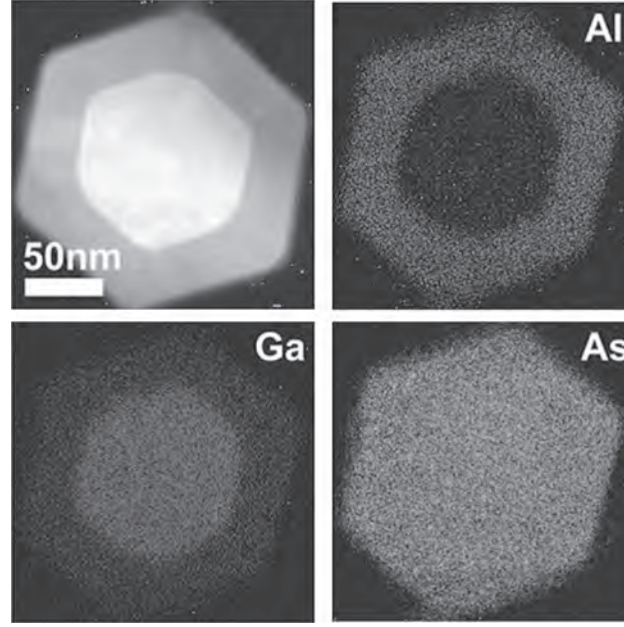


Figure 33. High-angle angular dark field (HAADF) scanning transmission electron microscopic images of a core/shell NW cross-section and corresponding transmission electron microscope equipped with energy dispersive X-ray spectroscopy (EDS) elemental mappings for Al, Ga and As. Reprinted with permission from [186], M. J. Tambe et al., *Appl. Phys. Lett.* 93, 151917 (2008). © 2008, American Institute of Physics.

the bare and core/shell GaAs NWs at 5 K is less than 80 ps, which is the system response time of our time correlated photon counting system. Even in the core/shell GaAs structures such a lifetime is more than an order of magnitude less than the lifetime observed in high-quality GaAs double heterostructures [190]. This reduced lifetime may result from defects within the GaAs NW core or at the GaAs/AlGaAs interface. Transmission electron

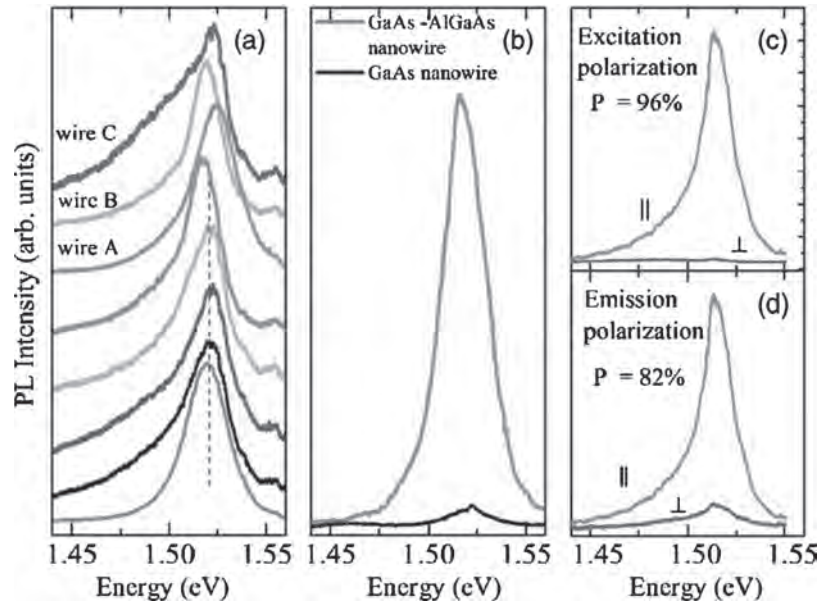


Figure 34. (a) Normalized PL spectra of GaAs/AlGaAs NWs, (b) bare GaAs and core/shell GaAs/AlGaAs NWs, (c) PL spectra of NW with excitation laser polarized parallel and normal to the NW, (d) polarized PL spectra of NW analyzed parallel and normal to the NW with excitation linearly polarized along NW direction. Reprinted with permission from [187], L. V. Titova et al., *Appl. Phys. Lett.* 89, 173126 (2006). © 2006, American Institute of Physics.

microscopy (TEM) studies of bare GaAs NW have shown the existence of twinning defects, which are known to adversely affect the optical and electronic properties of semiconductors [191]. This suggests that there is still significant room for improvement in optimizing the luminescence efficiency of these core/shell NWs.

The core/shell GaAs/AlGaAs NWs exhibit strong linear polarization anisotropy for both emission and excitation with the preferential polarization along the NW axis. In Figure 34(c) we show PL emission from NWA, excited by the laser polarized parallel and perpendicular to the NW. We find that the PL intensity is ~ 70 times stronger for laser polarized along the NW than for the laser polarized perpendicular to it. The degree of excitation polarization, defined as $P = (I_{\parallel} - I_{\perp}) / (I_{\parallel} + I_{\perp})$, is 96% for NW. A similar degrees of excitation polarization were observed in other NWs. Figure 34(d) shows the linear polarization of the PL emission from wire A. For this measurement, the 780 nm laser was linearly polarized along the NW axis, and the components of the resultant PL were analyzed parallel and perpendicular to the NW. The observed degree of wire emission polarization is 82%. This strong polarization anisotropy is similar to the one observed in InP NWs [192] and is caused primarily by the dielectric mismatch between the NW and its surroundings, which causes strong suppression of the component of the electric field inside and perpendicular to the NW [192].

QCE is unlikely to be a significant contributing factor to the polarization anisotropy since the exciton Bohr radius is smaller than the size of the GaAs core. Interestingly, Chen et al. [193] calculated that the emission from GaAs NWs should have a degree of polarization of approximately 92%, which is somewhat higher than that observed in NWs. This perhaps reflects inhomogeneities or deviations of NWs from cylindrical symmetry. This conclusion is supported by the fact that other NWs display similar behavior with the degree of polarization varying from 75% to 85%.

7.1.2. InAs/InP Core/Shell NWs

Semiconductor NWs have the double feature of providing a tool for the study of the physical properties of low-dimensional systems and, at the same time, of providing the building blocks of electronic and photonic nanoscale devices [194]. The results achieved so far are encouraging for pursuing developments in the bottom-up fabrication approach and, at the same time, for calling for investigations of the electrical and optical properties of NWs to define the conditions for future applications. The NW structure is a favorable environment to achieve 1D system, i.e., to force the motion of the electrons along only one direction, the NW axis. Indeed, if the NW diameter is small enough, the electrons will be confined by a potential well in the radial direction with a consequent energy quantization in that plane. Figure 35 shows the InAs NWs aligned along the [001] direction both schematically and experimentally.

In InAs NWs with diameter smaller than 30 nm the observed drop in the wire conductance was explained as an effect of the QC [195]. Due to the low emission efficiency of uncapped InAs NWs, study of core/shell InAs/InP structures has been approached as the shell material provides a passivation of the core surface, hindering the non-radiative electron-hole recombination through surface states, a mechanism that limits the emission efficiency of the core [196].

7.1.3. GaN/AlN Core/Shell

Recent advances in the design and control of heterostructures and superlattices in 1D nanoscale semiconductors have opened the door to new device concepts. 1D heterostructures consisting of two or more important functional materials are of prime importance for revealing unique properties and essential for developing potential nanoelectronic and optoelectronic devices. On the other hand, the controlled growth of twinned superlattices within a single nanostructure could facilitate band gap engineering and reveal novel electronic behaviors. In addition, an attractive challenge is to achieve the entire growth control within an individual nanostructure, e.g., to make highly reproducible, periodically twinned superlattices with an adjustable twin spacing. Semiconductor nanostructures offer the possibility of nearly unlimited complex bottom-up designs resulting

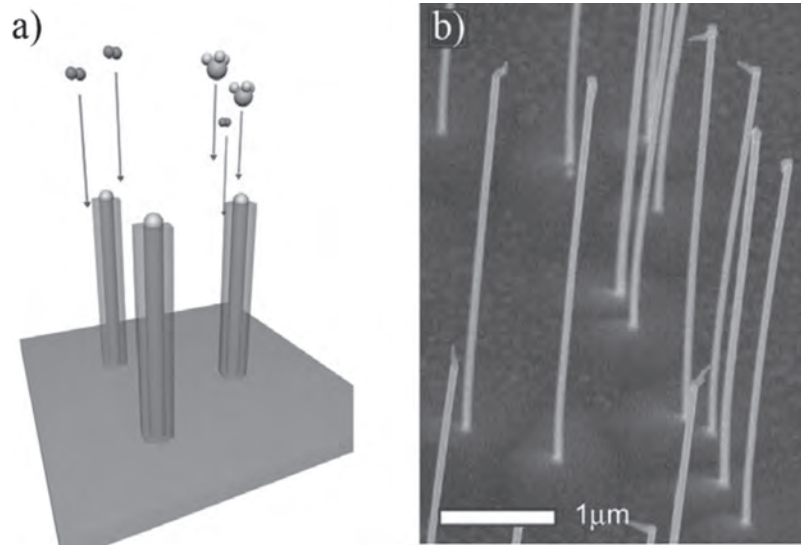


Figure 35. (a) Schematic representation of core/shell InAs/InP growth processes and (b) scanning electron microscopic image of the as-grown InAs/InP NWs (25 nm core, 20 nm shell thickness). Reprinted with permission from [196], Z. Zanolli et al., *J. Phys.: Condens. Matter* 19, 295219 (2007). © 2007, IOP Publishing Ltd.

in new device concepts [197]. Intense research on fabrication, physical properties, and applications of 1D materials has significantly been widened and deepened in recent years [198]. Such structures have been envisaged to be ideal model systems for exploring novel diverse phenomena at the nanoscale and analyzing the size and dimensionality dependence of properties toward potential applications [199]. A heterostructure is formed between two domains of dissimilar crystalline semiconductors with unequal band gaps as shown in Figure 36.

A superlattice is a structure composed of periodically alternating crystalline layers of several substances, shown in Figure 36(a). By alternating vaporization of two materials during the VLS growth, these groups have grown striped “superlattice” NWs or

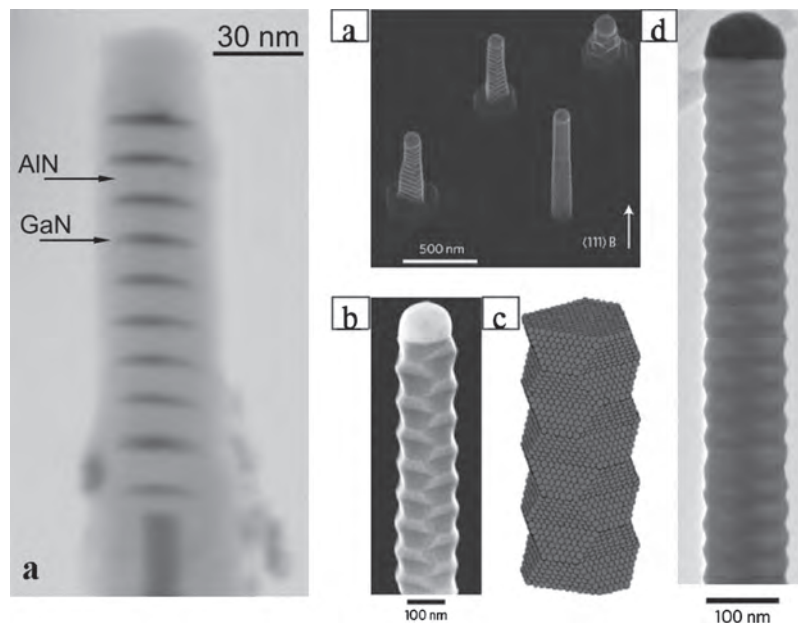


Figure 36. TEM cross-section of GaN/AlN axial NW with clear heterostructure. 3D model and their InAs nanostructures are shown with coherent twin-plane superlattices. Reprinted with permission from [197], X. Fang et al., *J. Materials Chemistry* 19, 5683 (2009). © 2009, Royal Society of Chemistry.

heterostructures consisting of up to 21 discrete bands. These pioneering works have opened an exciting new chapter in nanoscale research and have paved the way to the realization of diverse and ultimately new designs of 1D heterostructures, superlattices, and devices [200]. Figures 36(a–b) shows the core/shell nanoheterostructures with excellent heterointerfaces.

Heterostructure/superlattice formation may reveal unique properties that are essential for many potential semiconductor applications in electronics and optoelectronics [201]. One can see a straightforward analogy with the planar semiconductor industry when a complex composition and structure modulations could greatly increase the versatility and power of the building blocks in nanoscale electronic and photonic schemes [202]. In particular, 1D semiconductor heterostructures consisting of two or more components with distinct functionality often exhibit new or enhanced performances, such as emission efficiency and high electron mobility, which are the key factors for many devices [203]. In addition, recent theoretical studies indicated that superlattice NWs are promising systems for the thermoelectric applications since such structures are believed to be advantageous in reducing phonon transport and keeping high electron mobility [204].

In the case of radial nanoheterostructures, the controlled growth of one or more shells can passivate existing surface states enabling new interface properties, and introduce unique electronic and photonic functions. For example, Lieber and co-workers [205] demonstrated that the GaAs/GaP NW superlattices are suitable candidates for fabricating nanophotonic devices, whereas InAs/InP and GaN/AlN/AlGaIn radial NW heterostructures are effective in high electron mobility devices. In addition to compositional modifications along the NW length and diameter (axial and radial NW heterostructures, respectively), controlled *p*- and *n*-type doping is critical in many functional device applications. Single homogeneously-doped NWs have been utilized as basic components of FETs. Such NWs have also been proposed as important components in future complementary metal oxide semiconductor (CMOS) devices [206]. Si-based solar cells and InGaIn-based multicolor NW light-emitting diodes have been realized using radial NW heterostructures with NW core and shells being complementary doped [207, 208]. Very recently, this concept has been expanded and the first MQW core/shell NW heterostructures based on well-defined group III-N materials have been prepared. These displayed lasing over a broad range of wavelengths at room temperature. This achievement demonstrates a new level of complexity in NW heterostructures and is potentially able to yield free-standing injection nanolasers [209].

Figure 37 shows the μ -PL spectrum of a single NW containing multiple GaN segments from the sample B. As in the sample A, the peak corresponding to the emission of the GaN NW extremity is present at 3.478 eV in all NWs. The spectrum also shows additional peaks at energies that vary from wire to wire. It is difficult to give unambiguous interpretation of these peaks because of the complicated structure of the NWs. It is likely

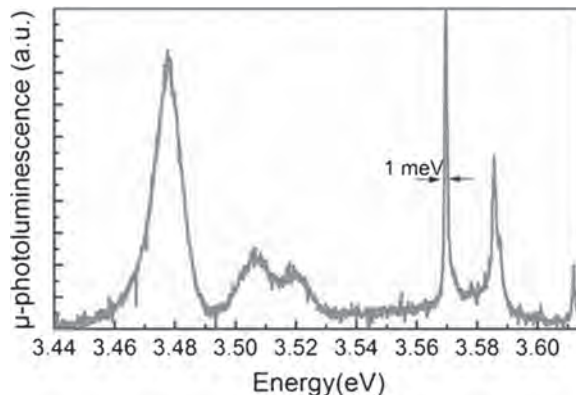


Figure 37. Normalized PL spectra of GaAs/AlGaAs NWs and PL spectra of the most intense bare GaAs and core/shell GaAs/AlGaAs NWs. Reprinted with permission from [210], M. Tcheryncheva et al., *Phys. Stat. Sol. c* 5, 1556 (2008). © 2008, Wiley-VCH Verlag GmbH & Co. KGaA.

that the broad doubled structure at 3.505 eV and 3.52 eV originates from GaN segments covered with the AlN shell, as in the case of sample A. The spectrum also shows narrow peaks with full-width at half maximum (FWHM) ~ 1 meV located at energies higher than 3.57 eV. These high-energy peaks suggest quantum confinement effects. We attribute them to the emission of the multiple thin GaN layers formed on the {1–100} planes by the lateral growth. This interpretation is supported by the fact, that these narrow peaks are not present for sample A, where there is only one thick GaN lateral layer, which is not covered with AlN [210].

7.2. II–VI Nanocrystals

The most studied core/shell structure to date is CdSe/ZnS, as evidenced by the number of publication dealing with this system. The first synthesis was described by Hines et al. [211], who overcoated 3-nm CdSe nanocrystals with 1–2 monolayers of ZnS, resulting in a QY of 50%. Recently, extremely small CdSe/ZnS nanocrystals have been synthesized by Kudera et al. [212], making the blue spectral region accessible with this system. The obtained nanocrystals emit at 470 nm with a QY of 60%. Shell growth on wurtzite CdSe core nanocrystals, generally obtained with the hot-injection method, is kinetically driven along the *c*-axis [214]. With the goal to enhance the uniformity of shell growth, zincblende CdSe nanocrystals have been used for overgrowth with ZnS by Lee et al. [216], as this structure has a more isotropic distribution of facets than the wurtzite model.

CdSe/ZnSe is a core/shell system exhibiting, in contrast to CdSe/CdS, efficient confinement of the electrons in the nanocrystal core due to the large conduction-band offset. On the other hand, only a relatively small barrier exists for holes in this system. Although the lattice mismatch is larger than in CdSe/CdS (6.3 vs. 3.9%), the common anion structure is particularly favorable for epitaxial-type shell growth. However, initially reported CdSe/ZnSe nanocrystals exhibited rather low values of the PL QY ($< 1\%$) [215]. Lee et al. [216] studied the effect of lattice distortion in the CdSe/ZnSe core/shell system on the optical spectra by varying the concentration of the ZnSe precursor solution used for the shell growth, as well as the ripening kinetics upon thermal annealing [217].

Both CdSe/CdS and CdSe/ZnSe heterostructures exhibit high fluorescence QYs and can have specific interests due to the “accessibility” of weakly confined electrons or holes, respectively. On the other hand, if the purely high stability of optical properties against photodegradation and chemical inertness of the shell material are investigated, ZnS is the shell material of choice. Although in principle it is possible to obtain green, or even blue, emission with CdSe nanocrystals of small size, their capping with ZnS has only very recently been achieved [212]. As a matter of fact, an analysis of the present state-of-the-art methods suggests that for a large variety of materials, the preparation of core/shell nanocrystals of low size dispersion and satisfying optical properties is facilitated when core nanocrystals with diameters in the range of approximately 2.5 and 5 nm are used.

One example is $\text{Cd}_{1-x}\text{Zn}_x\text{Se}$ nanocrystals: the band gap can be varied by changing *x* between the values of pure CdSe and pure ZnSe nanocrystals of the same size. This system is particularly interesting for the fabrication of efficient green emitters for use in display applications. To do so, the alloy core nanocrystals have been overgrown with a ZnS shell either using the established $\text{DEZn}/\text{bis}(\text{trimethylsilyl})\text{sulfide}$ or, more recently, by means of the air-stable monomolecular precursor zinc diethylxanthate [218]. Emission wavelengths in the blue to near-UV spectral region have been obtained by using larger band gap core materials, in particular CdS and ZnSe. It is noteworthy to mention that CdS is historically the first example of colloidal core/shell nanocrystals reported in the literature [219]. The emission wavelength range was extended to the blue range when CdS-ZnS co-colloids were activated in the same fashion.

A further shift to the near-infrared spectral region, and in particular towards 1.30–1.55 μm used in fiber-optics-based telecommunication, has been achieved by using HgTe/CdSCS nanocrystals [220]. The synthesis of this system comprised the preparation of thioglycerol-stabilized HgTe core nanocrystals, prepared in aqueous media in a similar manner as CdTe.

7.2.1. CdS/ZnS Nanoheterostructure

The PL efficiencies and electrical response of the CdS/ZnS core/shell nanorods were significantly enhanced as compared to those of the uncoated CdS nanorods, owing to the effective passivation of the surface electronic states of the CdS cores by the ZnS shell. New properties and diverse functions have been realized by assembling different types of constituents into nanocomposites with controlled structure and interface interactions. Recently, considerable research efforts have been directed on the shape and compositional control of 1D semiconductor-included nanocomposites, such as NWs with superlattice structures [220], core/shell coaxial NWs [222], biaxial or sandwich-like triaxial NWs [223], and anisotropic (e.g., dimertype and hierarchical composite materials) heterostructures [224]. In particular, core/shell nanostructures are reported most often because various mechanisms can be involved in shell growth that do not necessarily relate to epitaxy between the inorganic components, and consequently enhanced or modified properties are resulted from the particular dimensionality.

Surface passivation or functionalization has been recognized as a very advanced and convincing method to tailor the properties of nanomaterials. Surface coatings on nanostructured semiconductors can alter the charge, functionality, and reactivity of the materials and consequently enhance the functional properties due to localization of the electron-hole pairs [54]. These nanomaterials can be obtained using specialized procedures, often described as nanocrystal engineering [225]. Core/shell nanostructures constitute a very special class of nanomaterials, where a noticeable enhancement in the luminescence and conductive properties is achieved by modifying the core surfaces typically by the process of encapsulation with a shell of desired material such that the core nanocrystals withstand the process used for coating of shell material and at the same time the core and shell materials do not interdiffuse [226]. The nanostructures of luminescent materials are particularly important as the properties of these materials are greatly influenced by various surface states arising out of the higher surface-to-volume ratio.

This can be further tuned by deliberately passivating the surface of nanostructured materials using a higher band gap shell material, which effectively enhances the luminescence and electrical properties [226]. Over the years, investigations have been focused on the synthesis of novel luminescent core/shell nanomaterials such as CdSe/ZnS, CdSe/CdS, ZnO/ZnS, and TiO₂/CdS, but most of them were based on nanoparticles [227]. CdS, a mid-band gap semiconductor (2.4 eV), is an important member of the II-VI group of materials, widely used as a phosphor material in LED devices [228]. As ZnS is a phosphor material with a wide band gap energy of 3.7 eV, passivation of the CdS surface with ZnS can greatly enhance the core properties. Recently, Hsu et al. [229] reported the synthesis of CdS/ZnS core/shell coaxial NWs that showed stronger luminescence properties.

Quantum-sized semiconductor nanocrystals have raised a new increasing interest in the field of optical physics. In such structures, optical properties are emphasized by QCE. The electronic structure is modified, compared to the bulk material, when the nanocrystal radius approaches the exciton Bohr radius. Linear and nonlinear optical properties are modified as a consequence [230, 231]. Recently, attempts have been made not only to elaborate nanocrystals with well-defined structural properties, such as mean size, chemical composition, crystallography, and morphology, but also to manipulate their surfaces. The surface contains a large number of atoms so that electronic states created at the surfaces may be important and hence strongly influence the physical properties of such nanostructures. One of the methods for controlling the surface states consists of the coating of nanocrystals by a shell of a different species [3 ± 6].

Figure 39 illustrates changes in carrier localization that occur in the CdS-core heteronano-crystal with increasing shell thickness. We consider the situation for which the CdS-core size is relatively large ($R > R_c$, R_c is a critical radius for electron localization in the core) so that the lowest-energy electron level is below the core/shell energy offsets. In this case, the electron is immediately “locked” in the core for any shell thickness (an increase in H only lowers the electron 1S level further down; compare panels (a) and (b) in Figure 38. The regime of hole localization is, however, dependent upon the value of H .

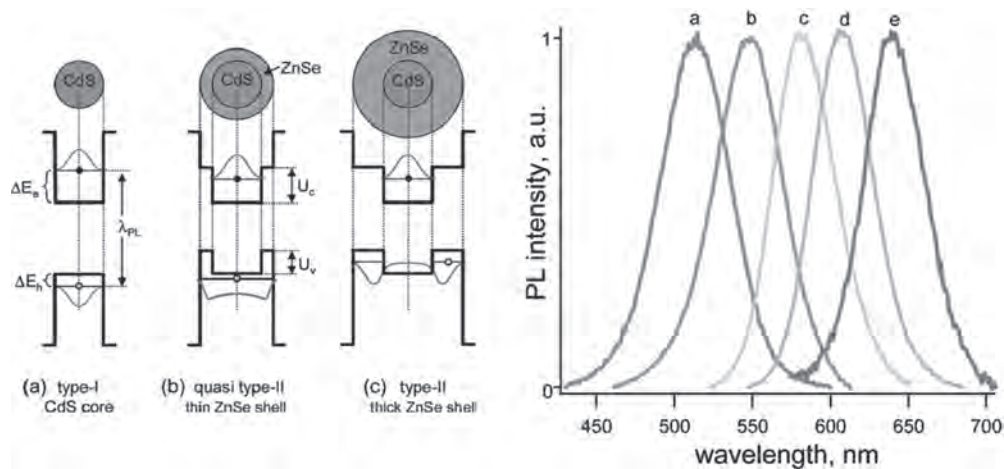


Figure 38. The different localization regimes supported by CdS/ZnSe nanocrystals as a function of shell thickness. (a) Bare CdS core, (b) core/shell nanocrystals correspond to type-II regime, and (c) heteronanocrystals with thick shells. (d) PL spectra of a series of core/shell CdS/ZnSe nanocrystals with different Cd content. Reprinted with permission from [216], S. A. Ivanov et al., *J. Am. Chem. Soc.* 129, 11708 (2007). © 2007, American Chemical Society.

For thin shells, and hence the hole the wave function is delocalized over the entire heteronanocrystal. This situation corresponds to partial charge separation (quasi type-II regime) in which only one carrier (electron in this case) is confined to a certain region of a heterostructure. The true type-II regime occurs for shell widths that are greater than a certain critical value, $H_c > H_c(R)$, which can be determined from the energy criterion: U_v , where Eh is the hole energy measured with respect to the bottom of the valence band. For $H > H_c$, the hole level shifts below the U_v offset, and hence the hole becomes shell-localized, which corresponds to the transition to a true type-II regime. In this case, the emission energy, $\hbar\omega_{LO}$, is determined by a spatially indirect transition, which couples the core-localized conduction-band state to the shell-localized valence-band state [230].

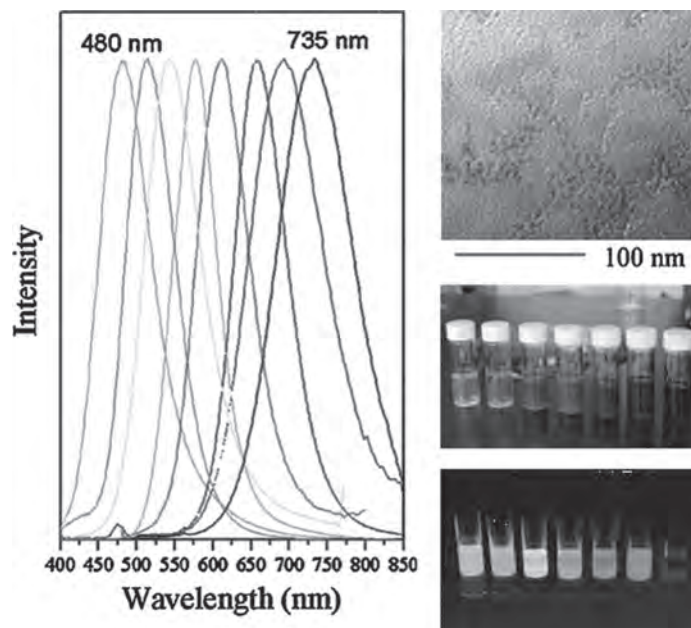


Figure 39. PL spectra of different core/shell InP/ZnS nanocrystals with different InP thickness and its transmission electron microscopy image. Different samples in tubes under room light and UV light. Reprinted with permission from [230], S. Xu et al., *J. Materials Chem.* 18, 2653 (2008). © 2008, Royal Society of Chemistry.

7.2.2. PbSe/PbS Nanocrystals

The literature on III–V semiconductor-based core/shell systems is far less extensive than in the case of II–VI compounds, as a consequence of the lack of robust synthesis methods for core nanocrystals of this family. InP is the most studied compound, as the emission of InP nanocrystals can be tuned throughout the visible and near-infrared range by changing their size [231, 232]. The PL line width is significantly broader, on the order of 50–100 nm, and band edge emission peaks related to defect-state emission occur in the spectrum. This process resulted in the removal of surface phosphorous atoms lying at the origin of trap states, which provided nonradiative recombination pathways. Concerning InP-based core/shell systems, Haubold [220] used to grow a ZnS shell on InP and observed a subsequent increase of the QY to 15% after three days and 23% after three weeks during a slow, room-temperature process. Li et al. [232] synthesized high-quality InP core nanocrystals at comparably low temperatures by adding a low-boiling-point primary alkylamine to the reaction mixture in order to activate the indium carboxylate precursor. Upon ZnS shell growth using S and Zn stearate, a fluorescence QY of up to 40% is obtained. Li et al. [232] reported a novel, low-cost method for the preparation of InP core nanocrystals based on the *in situ* generation of phosphine gas from a metal phosphide (calcium phosphide) precursor. Figure 39 shows the PbSe/PbS PL spectra and their corresponding colloidal solutions [230].

7.2.3. PbSe/PbSSe Nanocrystals

In contrast to the discussed II–VI and III–V semiconductors, exhibiting either the hexagonal wurtzite or the cubic zincblende crystal structure, the IV–VI family is characterized by the rocksalt structure. Only Pb-based nanocrystals (PbS, PbSe) have been studied in the form of core/shell systems. The bulk band gaps of these compounds are comparably small and they are, therefore, good candidates for the design of near-infrared emitters. Lifshitz et al. [234] reported the synthesis of PbSe/PbS and PbSe/PbSe_xS_{1-x} core/shell nanocrystals emitting in the range of 1–2 μm with QYs of 40–50% and 65%, respectively [182–184]. Talapin et al. [235] recently synthesized PbSe/PbS core/shell spheres and NWs, tailoring the morphology of the core nanocrystals by the addition of co-surfactants and using lead acetate in octylether and sulfur as the shell precursors. To this end, Hollingsworth et al. [236] developed PbSe/CdSe core/shell nanocrystals by the exchange of surface lead ions by Cd. A strongly improved stability of the optical properties was observed, which can further be enhanced by the subsequent overgrowth of the core/shell system with an additional ZnS shell. Figure 39 show the PbSe/PbS nanostructures colloidal solutions with the corresponding PL spectra.

7.2.4. ZnO Nanoheterostructures

Core/shell ZnO/MgZnO heterostructure nanorods with high aspect ratio were synthesized using a two-step thermal evaporation procedure, in which the core and the shell layers were formed separately at different temperatures. Microstructural characterization revealed a position dependence of the crystal structure and composition in the shell layer. The shell layer in the upper region consisted of MgO having cubic phases embedded in an amorphous oxide layer, while a Mg_{0.35}Zn_{0.65}O shell layer with a self-assembled superlattice structure of triple periodicity was formed in the middle region [237]. Figure 40 show the ZnO/MgZnO core/shell schematics.

1D ZnO nanostructures have been extensively studied because of their good physical properties, such as large excitonic binding energy, environmental stability, and low-threshold field emission. Moreover, if the ZnO is covered with wide band gap layers, such as ternary MgZnO, the excitonic binding energy can be significantly enhanced, and the electronic and optical properties in devices could be tailored through band gap engineering and the adjustment of the quantum properties. Although the realization of band gap engineering was possible for the formation of barrier layers and quantum wells in heterostructures, the difference in the crystal structures between the ZnO (wurtzite) and MgO (cubic) binaries limits the solubility of Mg atoms in ZnO.

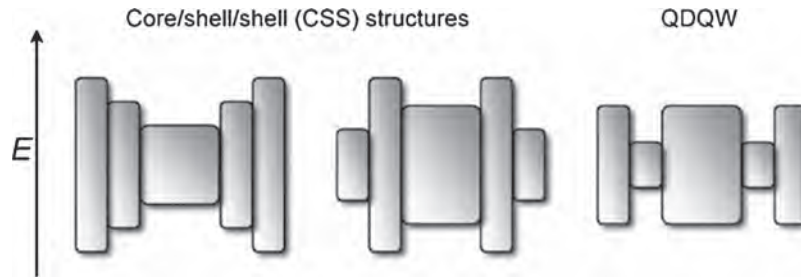


Figure 40. Schematic representation of the energy level alignment in different core/shell/shell structures and in a quantum dot quantum well (QDQW) system. The height of the rectangle represents the band gap energy and their upper and lower edges correspond to the positions of the conduction and valence band edge, respectively, of the core and shell materials. Reprinted with permission from [237], P. Reiss et al., *Small* 5, 154 (2009). © 2009, Wiley-Vch, Verlag GmbH.

In particular, composition-modulated NW/nanorod heterostructures have great potential as building blocks for the fabrication of quantum devices, such as high electron mobility transistors [238]. Composition modulation in the radial direction can efficiently confine both the carriers and emitted photons: coaxial nanorod heterostructures are expected to suppress surface-state-mediated nonradiative recombination and to decrease thermal quenching of emission intensity [239]. The shell layer thicknesses of the reported multishell nanorod heterostructures are a few tens of nm [240], which is generally too thick to confine charge carriers in the shell layer [241]. Moreover, many of the coaxial NW heterostructures show large lattice mismatches between their core and shell layer components 11% for Si/CdSe and 71% for GaP/GaN. This will result in strain-induced crystal defects at the interface, which hinders the appearance of QC [242]. Precise control of shell thickness and a negligible lattice mismatch in ZnO/Mg_{0.2}Zn_{0.8}O coaxial nanorod heterostructures enable us to successfully demonstrate the quantum confinement effect in multishell quantum structures.

The multishell nanorod heterostructures, as schematically depicted in Figure 41, were prepared by repeated alternate deposition of ZnO and Mg_{0.2}Zn_{0.8}O layers. The nanorods with various ZnO QW widths, L_W , were prepared in order to investigate the QC effect. Such distinctive coaxial nanorod structures were realized by precise control of L_W and L_B in angstrom scale. In particular, the formation of interfacial intermediate alloy layers was prevented by purging the system with pure argon whenever the reactants were delivered into the system, leading to the formation of clean and abrupt interfaces as confirmed by

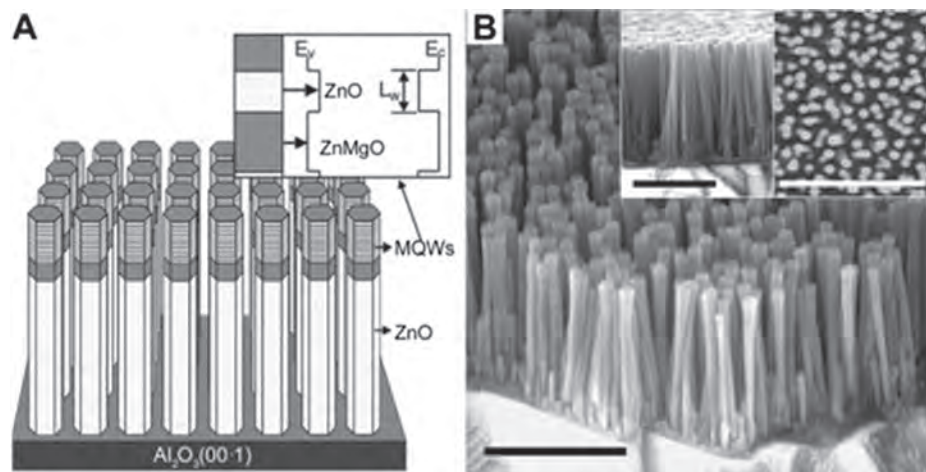


Figure 41. (a) Schematic ZnO/MgZnO multi-quantum wells and (b) the corresponding synthesized NWs. The inset shows very dense NWs with almost symmetric shapes. Reprinted with permission from [238], W. I. Park et al., *Adv. Mater.* 15, 526 (2003). © 2003, Wiley-Vch, Verlag GmbH.

high resolution transmission electron microscopy. A small lattice mismatch (1%) between ZnO and $\text{Mg}_{0.2}\text{Zn}_{0.8}\text{O}$ layers is also helpful to reduce defects in the interface [243].

7.3. Type-II Core/Shell Nanocrystals

Research on colloidal type-II systems was triggered by the seminal work of Kim et al. [244] who described the synthesis and optical properties of CdTe/CdSe and CdSe/ZnTe core/shell nanocrystals. The emission wavelength of CdTe/CdSe nanocrystals could be tuned by changing the shell thickness and the core nanocrystal size from 700 to 1000 nm. This approach is an alternative possibility to shift the emission peak to higher wavelengths, which would not be attainable by simply increasing the size of the core nanocrystal in a type-I core/shell system. On the other hand, the observed mean decay lifetime (57 ns) was significantly larger than that of the corresponding core CdTe nanocrystals (9.6 ns), and the QY was low (4%) with respect to type-I core/shell systems. This approach lead to QYs approaching 40% for small shell thicknesses below 0.5 nm [246]. Chin et al. [246] observed the formation of anisotropic (pyramids, multipods) structures during the overgrowth of spherical 2.6 nm CdTe nanocrystals with CdSe. CdSe/ZnTe nanocrystals have been prepared with CdO as the Cd precursor.

Femtosecond dynamics measurements revealed that the rate of photoinduced electron/hole spatial separation decreased with increasing core size, while remaining independent of the shell thickness [247]. ZnTe/CdTe represents the first common anion type-II system, reported by Xie et al. [248]. The same article also includes the synthesis of ZnTe nanocrystals with CdS and CdSe shells. Interestingly, the same group observed a transition from the concentric core/shell structure via pyramidal to tetrapodshaped heterostructures in the case of the ZnTe/CdSe core/shell system [249], when the shell growth was carried out at 215°C instead of 240°C [250]. The obtained core/shell particles cover an emission range of 500–650 nm and a QY of 15%, which could be improved via the incorporation of Cd in the shell.

7.4. Reverse Type-I Core/Shell Nanocrystals

Balet et al. [251] have studied the optical properties of so-called inverted core/shell nanocrystals. These nanocrystals are referred to as inverted due to the band gap of their core material (ZnSe) being larger than that of their shell material (CdSe). On the basis of the radiative recombination lifetimes recorded for nanocrystals with a fixed core size and increasing shell thickness, a continuous transition from type-I (both electron and hole wave functions are distributed over the entire nanocrystal) to type-II (electron and hole are spatially separated between the shell and the core) and back to type-I (both electron and hole primarily reside in the shell) localization regimes was observed. The samples exhibited emission in the range of 430–600 nm. By varying the CdSe shell thickness on 2.8 nm core ZnSe nanocrystals, the emission wavelength could be tuned in a broad spectral range (417–678 nm) with QYs of 40–85%. Furthermore, size focusing during the shell growth has been observed, in contrast to the usually occurring broadening of the size distribution at this stage of the reaction.

8. DOUBLE CORE/SHELL NANOSTRUCTURES

8.1. CdSe/ZnSe/ZnS Core/Shell/Shell Nanostructures

One of the motivations for the synthesis of nanocrystals containing multiple shells was the difficulty to respond simultaneously to the requirements of appropriate electronic (band gap, band alignment) and structural (lattice mismatch) parameters for most binary core/shell systems. In particular the lattice mismatch between the core and the shell material strongly limits the possibility to grow a shell with significant thickness without deteriorating the PL properties. The use of a strain-reducing intermediate shell sandwiched between the core nanocrystal and an outer shell has first been proposed in the core/shell/shell (CSS) system CdSe/ZnSe/ZnS [252]. The energy band alignment of this

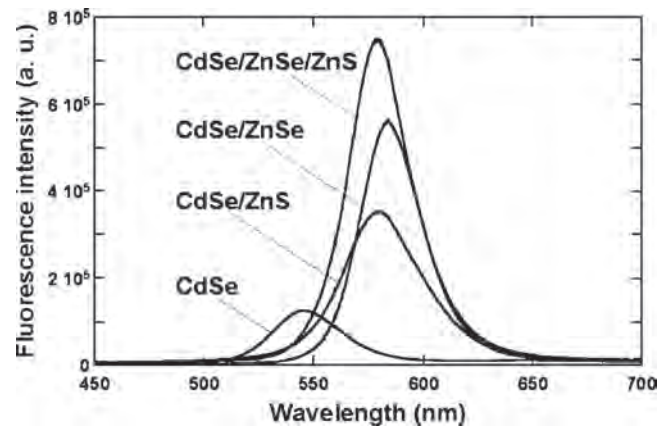


Figure 42. PL spectra of CdSe nanocrystals before and after the passivation with different types of inorganic semiconductor shells. The emission energy is increasing slowly with the increasing of shell engineering. Reprinted with permission from [252], P. Reiss et al., *Synthetic Materials* 139, 649 (2003). © 2003, Elsevier.

heterostructure corresponds to that depicted in the left panel of Figure 42. The interest of such structures lies in the combination of low strain, provided by the intermediate layer (ZnSe) serving as a “lattice adapter” and efficient passivation and charge-carrier confinement assured by the outer shell (ZnS). The CSS system offers higher stability against photo-oxidation than the core/shell system of CdSe/ZnSe, and higher QYs than in the core/shell system of CdSe/ZnS, as evidenced by Talapin and co-workers who also extended this approach to CdSe/CdS/ZnS CSS nanocrystals [214]. Uniform shell coverage was observed only for graded shells (e.g., CdS/ZnS) and was found to be critical to achieving QYs close to unity. In this context, a multishell structure consisting of CdSe core CdS/Zn_{0.5}Cd_{0.5}S/ZnS has been synthesized using the SILAR technique [253]. The monodisperse samples obtained in this study exhibit QYs of 70–85% and enhanced stability as compared to their binary core/shell counterparts. The same arguments apply for InAsP/InP/ZnSe nanocrystals, a near-infrared-emitting III–V-semiconductor-based CSS heterostructure [254]. In another approach, InAs NCs were overcoated with a CdSe/ZnSe double shell, exhibiting high QYs up to 70% in the spectral region of 800–1600 nm [255].

As mentioned before, the QYs of type-II core/shell systems are comparably low, and a limiting factor is the fact that one of the charge carriers is located in the shell material, only passivated by organic surfactant molecules. A logical step consists, therefore, in the addition of a large-band gap outer layer, resulting in a core/shell/shell (CSS) system. Such a heterostructure has been realized by the synthesis of CdSe/ZnTe/ZnS nanocrystals [256, 257]. A different type of band alignment has been achieved in CdSe/CdTe/ZnTe. Here, the intermediate CdTe shell serves as a barrier layer for electrons (CdSe) and holes (ZnTe), resulting in a strongly increased radiative lifetime, estimated as 10 ms [240].

8.2. Core/Double Shell CdSe/ZnSe/ZnS Nanomaterials

The low lattice mismatch of CdSe/ZnSe and ZnSe/ZnS motivated to prepare a three-component core/shell system in which ZnSe constitutes an inner shell between a CdSe core and a ZnS outer shell: Cd/ZnSe/ZnS nanocrystals. Thereby, the advantages of both pure CdSe/ZnSe and CdSe/ZnS systems can in principle be combined. Furthermore, the complete three-step synthesis can be accomplished without the use of any organometallic reagents. During ZnS shell growth on the surface of CdSe/ZnSe nanocrystals almost no formation of separate ZnS nanocrystals takes place, as the emission peak at 400 nm is practically non-existent. In Figure 42, the PL spectra of CdSe core nanocrystals and of the three different core/shell systems are compared.

In each case, the same concentrations and volumes for the core nanocrystal dispersions and the shell precursor solutions were used. The amount of shell precursors necessary

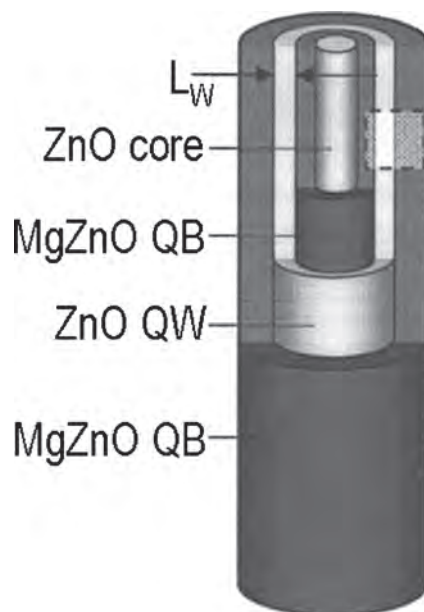


Figure 43. A schematic simple core/shell ZnO/MgZnO NWs cross-section.

for the formation of 5 ML-thick ZnSe and ZnS shells has been calculated on the basis of bulk lattice parameters. For the double shell system the ratio of ZnSe/ZnS precursors was 3/1, while keeping the same total amount of ZnS (VI) precursors as for the single shell systems. In Figure 42, only the final stages of the shell growth, i.e., after complete injection of the precursor solutions, are presented. Deposition of a ZnSe or ZnS shell results in an increase of the QY, due to a better passivation of the core nanocrystal surface; yet the fluorescence of the CdSe/ZnS system amounts to only 84% of that of CdSe/ZnSe, because of structural defects. On the contrary, deposition of a ZnS outer shell onto CdSe nanocrystals which are already covered with a relatively thin ZnSe layer (1–2 ML) significantly improves the fluorescence efficiency by a factor of 1.7 as compared to CdSe/ZnS and by 1.4 as compared to CdSe/ZnSe [257]. While shell growth is in all cases accompanied by a bathochromic shift of the emission peak originating from partial exciton leakage into the shell, the emission line width, being 33–34 nm at FWHM, is not affected.

The procedure avoids at all stages the use of 2 absolute values for the QY have been measured as $17.6 \pm 1.5\%$ (CdSe/ZnSe), $14.8 \pm 1.5\%$ (CdSe/ZnS) and $25.3 \pm 0.6\%$ (CdSe/ZnSe/ZnS). On the contrary, no efforts to optimize the QY have been undertaken in order to keep strict comparability of the different samples and to study exclusively effects arising from the shell material of organometallic reagents and allows the facile preparation of three types of highly luminescent nanocrystals of low polydispersity, namely CdSe/ZnSe, CdSe/ZnS, and CdSe/ZnSe/ZnS. The latter is a new double-shell system that combines the advantages of the two other types: low lattice mismatch (CdSe/ZnSe) and good confinement of the charge carriers in the core (CdSe/ZnS).

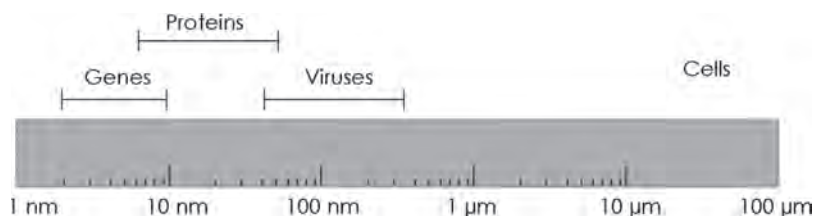


Figure 44. Nanoscale scaling in ranging from 1 nanometer to 100 micrometer. It shows that the proteins, genes, and viruses are very small, in the range of 1–300 nm, while cells are 100 μm .

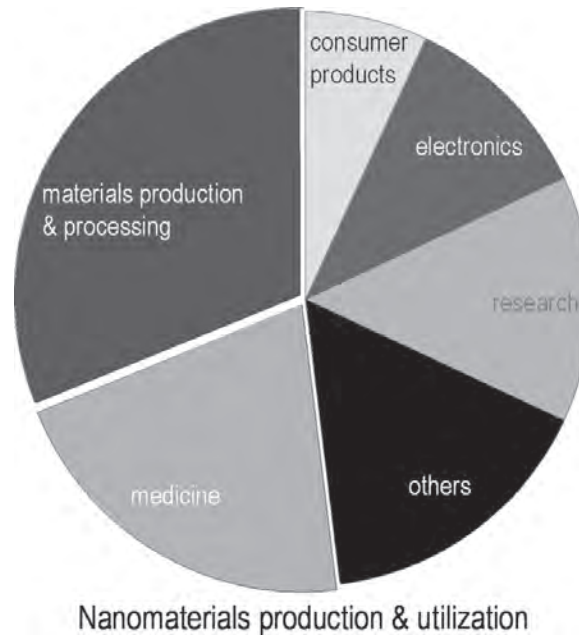


Figure 45. Materials consumption in different sectors. Materials production and processing takes a large area.

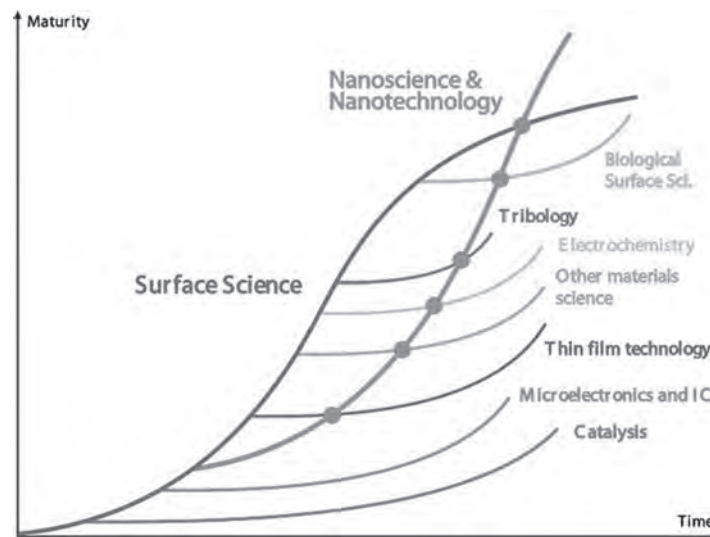


Figure 46. The historical development of surface science following an S-shaped curve, which has served as a basis for the emergence of a large variety of other fields. Today nanoscience and nanotechnology influence virtually any field of science and technology, as illustrated by the crossing points. Reprinted with permission from [256], M. Zach et al., *Current Opinion in Solid State Mater. Science* 10, 132 (2006). © 2006, Elsevier.

In consequence, this system exhibits the highest fluorescence efficiency of the three core/shell systems investigated [257].

9. APPLICATION AND OUTLOOK

There is a general feeling that the surge in interest and funding for nanoscience will bring a new understanding of physical phenomena at the nanoscale and should bring benefits in the resulting nanotechnology. Beyond the familiar Si chips that form the heart of our computers, mobile phones, and music players, there are the so-called compound semiconductors such as GaAs that will play the dominant role in nanoelectronics and

optoelectronics. The latter includes LEDs, LDs, and infrared detectors that are commonplace in supermarket checkout counters, remote controls, automobile brake lights, traffic lights, other lighting displays, and a clean environment. Within this family of materials, the wide band gap semiconductors such as II-VI materials systems have found utility in applications requiring blue/green/UV light emission, transistors capable of high temperature, high power operation, solar-blind UV detection, water sterilization, and clean environment. The large surface area of the ZnO nanostructures makes them attractive for gas and chemical sensing, and the ability to control their nucleation sites makes them candidates for microlasers or memory arrays. One important commercial application for UV photodetectors is for flame sensors that can be used to monitor the status of pilot lights and burner flames in large industrial furnaces and other applications for the control of high-temperature systems. In these applications, a solar-blind detector is of use because stray sunlight or artificial room lighting could cause a control system to falsely believe that the pilot light is on when it is actually off.

The synthesis and consequences of ZnO nanostructures present extreme challenges to the physicists and chemists since diverse nanostructures show different properties. To discuss and analyze these nanostructures from primary to the final stages, new physical techniques and even new physics may be necessary for the latest nanomaterials science. The synthesis of complex molecules requires patience, stamina, and a profound knowledge of reaction mechanisms. The drive for shorter and more efficient synthesis procedures will continue to challenge the resourcefulness of synthetic chemists.

Field emission is seen as one of the most promising applications for carbon-based films. The most attractive forms of carbon for this application are carbon nanotubes, which are capable of emitting high currents. Controlled deposition of nanotubes on a substrate has recently become possible. There is concern however for the long-term stability of the films. Investigations have shown that the film can degrade due to resistive heating, bombardment from gas molecules by the emitted electrons, or arcing. Electrostatic deflection or mechanical stress can cause a change in the local shape of the emitter and a decrease in its effectiveness.

Applications in nanotube flat-panel displays have been anticipated and a demonstration model has been produced by Samsung. Field-emitting diode (FED) displays will counter the drawbacks of liquid crystal flat screens, such as low-image quality and a restricted field of view. The viability of nanotubes for such applications are in question, problems in the correct deposition of the tubes, phosphor lifetime, and charging of the spacers need to be overcome.

The literature search showed most activity for carbon nanotubes over the last three years, while interest in nanoparticles and fullerenes decreased. The model category for application of carbon nanomaterials in the literature is energy storage. The low level of activity in fullerene research is reflected in the list of current and recent research projects, where they are not a major subject of investigation. Other important materials with respect to publications are carbon nanofilms and carbon-based nanocomposites, with diamond-like carbon layers being an important material. Nanocomposites and nanofilms are very important areas in research, and the impression is that there is currently a growing interest in developing new materials in this category.

In view of the developments in the domain of core/shell semiconductor nanocrystals in the past decade, it can be speculated that a large variety of new heterostructures with exciting and, in some cases, unprecedented features will be synthesized by chemical routes in the next few years. The ability to precisely control the shell thickness will further boost advances in the preparation of non-blinking nanocrystals, core/shell-shell, and other complex structures such as quantum dots and quantum wells. Significant progress has been achieved in the field of anisotropic shell growth on spherical CdSe nanocrystals, leading to new rod-shaped core/shell nanostructures, which combine unique optical properties (high QY, polarized emission) with appealing self-assembly properties. The developed synthesis is a method using small-core NCs as seeds in the so-called seeded-growth approach opens up the way for a generation of new heterostructures, including nanorods and branched structures such as bi-, tri-, or tetra- multipods.

Similar to the case of II–VI compounds, it can be expected that a growing number of core/shell systems based on III–V semiconductors will be developed, inspired by the huge amount of research carried out in the field of III–V nanostructures. On the one hand, the growth of such new core/shell systems with high-quality optical properties provides intriguing motivation for fundamental research. On the other hand, the use of alternative core materials is mandatory for most commercial applications, in view of the low acceptability of cadmium, lead, mercury, or arsenide-containing compounds.

Finally, the association of semiconductors with other materials such as metals or oxides in the same core/shell heterostructure allows for the design of nanocrystals combining different physical properties, for example, fluorescence, magnetism, different decay lifetimes, and so on. In such a manner, novel functional building blocks can be generated for applications in fields ranging from optoelectronics to information technology to health-care. In most of these examples, the core/shell nanocrystals are used as a platform for further surface functionalization, providing the ability to tune their assembly properties on given substrates or their solubility in various media, or to induce their binding to other molecules or macromolecules. The historical development of surface science has followed an S-shaped curve, which has served as a basis for the emergence of a large variety of other fields, each of them following its own S-curve. Today, nanoscience and nanotechnology influence virtually every field of science and technology, as illustrated by the crossing points.

There are lots of attractive nanostructures with the minimum diameter of 6~10 nm. To reduce this nanostructures dimension, interdisciplinary programs of materials science and technology are necessary, to promote the formation of fundamental associations for the improvement and development of new techniques, and of novel synthesis technique, nanomaterials and their properties. The nanomaterials research is still young and requires interdisciplinary training—a combination of studies including physics, chemistry, engineering, materials science, and biology.

The union of distinct scientific disciplines is revealing the leading edge of nanotechnology. Fifteen to 20 years ago, the interdisciplinary activity of geneticists, biologists, immunologists, and organic chemists created a more diverse toolbox now known as life science. In parallel, physicists, chemists, polymer chemists, and engineers were creating the foundation for the small world of nanomaterials science. In less than a decade, materials science and life science together are unraveling the mysteries of controlling, on a molecular level, the structure of matter.

Particles, complexes, tubes, coatings, active surfaces and devices are being explored on the nanoscale. Assembly of nature's building blocks (e.g., carbon, nucleic acids, lipids, and peptides) along with the combination of different materials (e.g., CdSe/ZnS, ZnO, Au, Ag, light-harvesting dendrimers, and thin films) are leading to insightful understanding and the creation of new scientific tools. Chemists and physicists have been manipulating matter on the molecular level for centuries. When one looks at the absolute elegance of the nanometer scale biological system, however, one is compelled to create, understand, manipulate, and control systems with equal elegance.

ABBREVIATIONS

1D	One dimensional
BGR	Band gap renormalization
CBED	Covergent electron diffraction
CBO	Conduction band offset
CNT	Carbon nanotubes
CVD	Chemical vapor deposition
DC	Dielectric continuum
DNA	Deoxyribo nucleic acid
DVD	Digital diver disk
EHP	Electron-hole plasma
FED	Field effect diode

FET	Field effect transistor
LD	Laser diode
LED	Light emitting diode
MBE	Molecular beam epitaxy
ML	Monolayers
NB	Nanobelt
NR	Nanorod
NT	Nanotubes
NW	Nanowire
PL	Photoluminescence
PVD	Physical vapor deposition
QCE	Quantum confinement effect
QCSE	Quantum confined Stark effect
QD	Quantum dot
QW	Quantum well
QWW	Quantum well wire
QY	Quantum yield
SEM	Scanning electron microscopy
SVT	Slovothermal technique
TEM	Transmission electron microscopy
TRPL	Time-resolved photoluminescence
VBO	Valence band offset
VLS	Vapor–liquid–solid
VSS	Vapor–solid–solid

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REFERENCES

1. N. Taniguchi, Japan Society of Precision Engineering, Tokyo, Japan, 1974.
2. K. E. Drexler, "Engines of Creation." p. 298, Anchor Books, New York, 1986.
3. W. D. Callister, "Fundamentals of Materials Science and Engineering." 2nd Edition, Wiley & Sons.
4. R. Kurzweil, "The Singularity is Near: When Humans Transcend Biology." Viking, New York, 2005.
5. K. Autumn, Y. A. Liang, S. T. Hsieh, W. Zesch, W. P. Chan, T. W. Kenny, R. Fearing, and R. J. Full, *Nature* 405, 681 (2000).
6. J. F. Kukowska-Latallo, K. A. Candido, Z. Cao, S. S. Nigavekar, I. J. Majoros, T. P. Thomas, L. P. Balogh, M. K. Khan, and J. R. Baker, *Cancer Res.* 65, 5317 (2005).
7. M. Kim, W.-J. Ha, J.-W. Anh, H.-S. Kim, S.-W. Park, and D. Lee, *J. Alloys. Comp.* 484, 28 (2009).
8. R. W. Meulenbergh, J. R. I. Lee, A. Wolcott, J. Z. Zhang, L. J. Terminello, and T. van Buuren, *ACS Nano* 3, 325 (2009).
9. J. Yang, G. Wang, H. Liu, J. Park, and X. Cheng, *Mater. Chem. Phys.* 115, 204 (2009).
10. S. Neeleshwar, C. L. Chen, C. B. Tsai, Y. Y. Chen, C. C. Chen, S. G. Shyu, and M. S. Seehra, *Phys. Rev. B* 71, 21307 (2005); G. N. Karanikolos, N. L. Law, R. Mallory, A. Petrou, P. Alexandridis, and T. J. Mountziaris, *Nanotechnology* 17, 3121 (2006).
11. W. Chan, D. J. Maxwell, X. Gao, R. E. Bailey, M. Han, and S. Nie, *Cur. Opinion Biotech.* 13, 40 (2002).
12. P. T. Tran, E. R. Goldman, G. P. Anderson, J. M. Mauro, and H. Mattoussi. *Phy. Status So. (B)* 229, 427 (2002).
13. X. Gao, Y. Cui, R. M. Levenson, L. W. K. Chung, and S. Nie, *Nat. Biotechnol.* 22, 969 (2004).
14. A. M. Smith, X. Gao, and S. Nie, *Photochem. Photobio.* 80, 377 (2004).
15. J. Westwater, D. P. Gosain, S. Tomiya, and S. Usui, *J. Vac. Sci. Tech. B* 15, 554 (1997).
16. Y. F. Zhang, Y. H. Tang, C. Lam, N. Wang, C. S. Lee, I. Bello, and S. T. Lee, *J. Cryst. Growth* 212, 115 (2000).
17. J. Liu, P. Fei, J. Zhou, R. Tummala, and Z. L. Wang, *Appl. Phys. Lett.* 92, 173105 (2008); R. Ghosh and D. Basak, *J. Appl. Phys.* 98, 086104 (2005).
18. X. Bao, C. Soci, D. Susac, J. Bratvold, D. P. R. Aplin, W. Wei, C.-Y. Chen, S. A. Dayeh, K. L. Kavanagh, and D. Wang, *Nano Lett.* 8, 3755 (2008); Y. Lei, L. D. Zhang, G. W. Meng, G. H. Li, X. Y. Zhang, C. H. Liang,

- W. Chen, and S. X. Wang, *Appl. Phys. Lett.* 78, 1125 (2001); C. H. Liang, G. W. Meng, Y. Lei, F. Philipp, and L. D. Zhang, *Adv. Mater.* 13, 1330 (2001).
19. P. Nguyen, H. T. Ng, J. Kong, A. M. Cassell, R. Quinn, J. Li, J. Han, M. McNeil, and M. Meyyappan, *Nano Lett.* 3, 925 (2003).
 20. Z. W. Pan, Z. Dai, and Z. L. Wang, *Science* 291, 1947 (2001).
 21. Y. Jiang, X. M. Meng, J. Liu, Z. Y. Xie, C. S. Lee, and S. T. Lee, *Adv. Mater.* 15, 323 (2003); D. Moore, C. Ronning, C. Ma, and Z. L. Wang, *Chem. Phys. Lett.* 385, 8 (2004).
 22. L. F. Dong, J. Jiao, M. Coulter, and L. Love, *Chem. Phys. Lett.* 376, 653 (2003).
 23. C. Ma, Y. Dong, D. Moore, X. Wang, and Z. L. Wang, *J. Am. Chem. Soc.* 126, 708 (2004); C. Ma and Z. L. Wang, *Adv. Mater.* 17, 1 (2005).
 24. Y. Jiang, X. M. Meng, W. C. Yiu, J. Liu, J. X. Ding, C. S. Lee, and S. T. Lee, *J. Phys. Chem. B* 108, 2784 (2004).
 25. X. D. Wang, Y. Ding, C. J. Summers, and Z. L. Wang, *J. Phys. Chem. B* 108, 8773 (2004).
 26. T. W. Odom, J.-L. Huang, P. Kim, and C. M. Lieber, *Nature* 391, 62 (1998).
 27. J. W. G. Wildoer, L. C. Venema, A. G. Rinzler, R. E. Smalley, and C. Dekker, *Nature* 391, 59 (1998).
 28. M. F. Lin and K. W. K. Shung, *Phys. Rev. B* 51, 7592 (1995).
 29. L. Chico, L. X. Benedict, S. G. Louie, and M. L. Cohen, *Phys. Rev. B* 54, 2600 (1996).
 30. B. Gil, "Group III Nitride Semiconductor Compounds." Clarendon, Oxford, 1998; S. Nakamura and S. F. Chichibu, "Introduction to Nitride Semiconductor Blue Lasers and Light Emitting Diodes." Taylor and Francis, London, 2000; A. Nakamura, *AAPPS Bull.* 10, 2 (2000).
 31. S.-K. Lee, H.-J. Choi, P. Pauzauskie, P. Yang, N.-K. Cho, H.-D. Park, E.-K. Suh, K.-Y. Lim, and H.-J. Lee, *Phys. Stat. Sol. (b)* 241, 2775 (2004).
 32. M. Mucha and P. Jungwirth, *J. Phys. Chem. B* 107, 8271 (2003); L. Zhang, *Turk. J. Phys.* 31, 85 (2007).
 33. C.-L. Hsiao, L.-W. Tu, M. Chen, T.-F. Young, C.-T. Chia, and Y.-M. Chang, *Appl. Phys. Lett.* 90, 043102 (2007); P. J. Pauzauskie, D. Talaga, K. Seo, P. Yang, and F. L. Labarthe, *J. Am. Chem. Soc.* 127, 17146 (2005).
 34. J. J. Shi, *Sol. Stat. Commun.* 127, 51 (2003).
 35. S. M. Komirenko, K. W. Kim, M. A. Strosio, and M. Dutta, *Phys. Rev. B* 59, 5013 (1999).
 36. L. Wendler, R. Haupt, and V. G. Grigoryan, *Physica B: Con. Matter* 167, 101 (1990).
 37. P. Bordone and P. Lugli, *Phys. Rev. B* 49, 8178 (1994).
 38. I. Dharssi and P. N. Butcher, *J. Phys.: Condens. Matter* 2, 119 (1990).
 39. B. Zhu, *Phys. Rev. B* 38, 7694 (1988); A. K. Sood, J. Menendez, M. Cardona, and K. Ploog, *Phys. Rev. Lett.* 54, 2115 (1985).
 40. P. Lambin, P. Senet, and A. A. Lucas, *Phys. Rev. B* 44, 6416 (1991).
 41. J. Gleize, F. Demangeot, J. Frandon, M. Kuball, B. Daudin, and N. Grandjean, *Phys. Stat. Sol. (a)* 183, 157 (2001).
 42. Z. Y. Zeng, Y. Xiang, and L. D. Zhang, *Eur. Phys. J. B* 17, 699 (2000); Z. Y. Zeng, Y. Xiang, and L. D. Zhang, *Appl. Phys. Lett.* 77, 2015 (2000).
 43. L. J. Lauhon, M. S. Gudlksen, D. Wang, and C. M. Lieber, *Nature* 420, 57 (2002); S. A. Dayeh, N. H. Mack, J. Y. Huang, and S. T. Picraux, *Appl. Phys. Lett.* 99, 023102 (2011).
 44. Z. Y. Zeng, Y. Xiang, and L. D. Zhang, *European Phys. J. B.: Condens. Mater. Com. Sys.* 17, 699 (2000); R. J. Nicholas, M. A. Brummell, and J. C. Portal, *Surf. Sci.* 113, 290 (1982).
 45. E. Tangarife and C. A. Duque, *Phys. Stat. Sol. B* 247, 1778 (2010).
 46. S. Aktas, S. E. Okan, and H. Akbas, *Superlatt. Microstruct.* 30, 129 (2001); K. Chang and J. B. Xia, *Phys. Rev. B* 57, 780 (1998).
 47. W. Huang and F. Jain, *J. Appl. Phys.* 87, 7354 (2000).
 48. Z. Y. Zeng, Y. Xiang, and L. D. Zhang, *European Phys. J. B.: Condens. Matter. Com. Sys.* 17, 699 (2000); H. W. Kroto, J. R. Heath, S. C. O'Brien, R. F. Curl, and R. E. Smalley, *Nature* 318, 162 (1985); L. Zhang and J. Shi, *Semicond. Sci. Technol.* 20, 592 (2005).
 49. S. Iijima, *Nature* 354, 56 (1991).
 50. G. Binnig and H. Rohrer, *Rev. Mod. Phys.* 59, 615 (1987).
 51. Y.-R. Jang, C. Jo, and J. I. Lee, *IEEE Trans. Magnet.* 41, 3118 (2005).
 52. H. J. Joyce, Q. Gao, H. H. Tan, C. Jagadish, Y. Kim, M. A. Fickenscher, S. Perera, T. B. Hoang, L. M. Smith, H. E. Jackson, J. M. Yarrison-Rice, X. Zhang, and J. Zou, *Adv. Func. Mater.* 18, 1 (2008).
 53. C. D. Geddes, A. Parfenov, I. Gryczynski, and J. R. Lakowicz, *Chem. Phys. Lett.* 380, 269 (2003).
 54. A. Mishra, L. V. Titova, T. B. Hoang, H. E. Jackson, L. M. Smith, J. M. Yarrison-Rice, Y. Kim, H. J. Joyce, Q. Gao, H. H. Tan, and C. Jagadish, *Appl. Phys. Lett.* 91, 263104 (2007).
 55. K. Chang, *The New York Times*, New York City, NY, (2005).
 56. Z. L. Wang and Z. C. Kang, "Functional and Smart Materials: Structural Evolution and Structure Analysis." First Edition, Plenum Press, New York City, New York, 1998.
 57. M. G. Bawendi, *J. Chem. Phys.* 96, 946 (1992).
 58. A. P. Alivisatos, A. L. Harris, N. J. Levinos, M. L. Steigerwald, and L. E. Brus, *J. Chem. Phys.* 89, 4001 (1988).
 59. L. E. Brus, *J. Chem. Phys.* 80, 4403 (1984).
 60. W. C. Chan and S. Nie, *Science* 281, 2016 (1998).
 61. K. Hiruma, M. Yazawa, K. Haraguchi, T. Katsuyama, and H. Kakibayashi, *J. Appl. Phys.* 74, 3162 (1993).
 62. K. A. Dick, K. Deppert, T. Martensson, B. Mandl, L. Samuelson, and W. Seifert, *Nano Lett.* 5, 761 (2005).
 63. X. Duan, J. Wang, and C. M. Lieber, *Appl. Phys. Lett.* 76, 1116 (2000).

64. H. T. Ng, J. Han, T. Yamada, P. Nguyen, Y. P. Chen, and M. Meyyappan, *Nano Lett.* 4, 1247 (2004).
65. J. Zou, M. Paladugu, G. J. Auchterlonie, Q. Gao, H. J. Joyce, H. H. Tan, and C. Jagadish, *Small* 3, 389 (2007).
66. H. J. Joyce, Q. Gao, H. H. Tan, C. Jagadish, Y. Kim, X. Zhang, Y. Guo, and J. Zou, *Nano Lett.* 7, 921 (2007).
67. M. Borgstrom, K. Deppert, L. Samuelson, and W. Seifert, *J. Cryst. Growth* 260, 18 (2004).
68. M. A. Verheijen, G. Immink, M. T. Borgstrom, and E. M. Bakkers, *J. Am. Chem. Soc.* 128, 1353 (2006).
69. M. Mattila, T. Hakkarainen, M. Mulot, and H. Lipsanen, *Nanotechnology* 17, 1580 (2006); J. Taagardh, A. I. Persson, J. B. Wagner, D. Hessman, and L. Samuelson, *J. Appl. Phys.* 101, 123701 (2007).
70. A. Mishra, L. V. Titova, T. B. Hoang, H. E. Jackson, L. M. Smith, J. M. Yarrison-Rice, Y. Kim, H. J. Joyce, Q. Gao, H. H. Tan, and C. Jagadish, *Appl. Phys. Lett.* 91, 263104 (2009); H. Pettersson, J. Tragardh, A. I. Persson, L. Landin, D. Hessman, and L. Samuelson, *Nano Lett.* 6, 4 (2006).
71. J. Bao, D. C. Bell, F. Capasso, J. B. Wagner, T. Mårtensson, J. Trägårdh, and L. Samuelson, *Nano Lett.* 8, 836 (2008); J. Hu, T. W. Odom, and C. M. Lieber, *Acc. Chem. Res.* 32, 435 (1999).
72. Z. Zanolli, L. E. Froberg, M. T. Bjork, M.-E. Pistol, and L. Samuelson, *Thin Solid Films* 515, 793 (2005).
73. C.-Y. Yeh, S.-H. Wei, and A. Zunger, *Phys. Rev. B* 50, 2715 (1994).
74. M. Murayama and T. Nakayama, *Phys. Rev. B* 49, 4710 (2004).
75. L. Hedin, *Phys. Rev.* 139, A796 (1965).
76. S. F. Chichibu, "Introduction to Nitride Semiconductor Blue Lasers and Light Emitting Diodes." Taylor and Francis, London, 2000; A. Nakamura, *AAPPS Bull.* 102 (2000).
77. F. F. Maia, J. A. K. Freire, V. N. Freire, G. A. Farias, and E. F. da Silva, *Appl. Surf. Sci.* 237, 261 (2004).
78. L. Hedin and S. Lundqvist, "Solid State Physics." (F. Seitz, D. Turnbull, and H. Ehrenreich, Eds.), Vol. 23, p. 1, Academic, New York, 1969.
79. A. Mishra, L. V. Titova, T. B. Hoang, H. E. Jackson, L. M. Smith, J. M. Yarrison-Rice, Y. Kim, H. J. Joyce, Q. Gao, H. H. Tan, and C. Jagadish, *Appl. Phys. Lett.* 91, 263104 (2007).
80. Z. Zanolli, M.-E. Pistol, L. E. Fröberg, and L. Samuelson, *J. Phys.: Condens. Matter* 19, 295219 (2007); M. Mattila, T. Hakkarainen, M. Mulot, and H. Lipsanen, *Nanotechnology* 17, 1580 (2006).
81. S. Iijima, *Nature* 354, 56 (1991).
82. C. R. Murray, C. R. Kagan, and M. G. Bawendi, *Science* 270, 1335 (1995).
83. W. Han, S. Fan, Q. Li, and Y. Hu, *Science* 277, 1287 (1997).
84. W. Han, P. Redlich, F. Ernst, and M. Ruhle, *Appl. Phys. Lett.* 76, 652 (2000).
85. X. Duan and C. M. Lieber, *J. Am. Chem. Soc.* 122, 188 (2000).
86. J. Y. Li, X. L. Chen, Z. Y. Quio, Y. G. Cao, and Y. C. Lan, *J. Cryst. Growth* 213, 408 (2000).
87. X. Chen, J. Li, Y. Cao, Y. Lan, H. Li, M. He, C. Wang, Z. Zhang, and Z. Qiao, *Adv. Mater.* 12, 1432 (2000).
88. S. Y. Bae, H. W. Seo, J. Park, H. Yang, and B. Kim, *Chem. Phys. Lett.* 376, 445 (2003); T. Kuykendall, S. Aloni, I. J. Plante, and T. Mokari, *Int. J. Photoenergy* 1 (2009).
89. X. Chen, J. Xu, R. M. Wang, and D. Yu, *Adv. Mater.* 15, 419 (2003).
90. H. Zhang, X. Ma, Y. Ji, J. Xu, and D. Yang, *Chem. Phys. Lett.* 377, 654 (2003).
91. H. Zhang, D. Yang, X. Ma, Y. Ji, J. Xu, and D. Que, *Nanotechnology* (in press).
92. J. Zhan, X. Yang, D. Wang, S. Li, Y. Xie, Y. Xia, and Y. Qian, *Adv. Mater.* 12, 1348 (2000).
93. Y. Xie, J. Huang, B. Li, Y. Liu, and Y. Qian, *Adv. Mater.* 12, 1523 (2000); J. M. Green, J. Lawrance, and J. Jiao, *J. Nanomaterials* (2008).
94. M. V. Artemyev, V. Sperling, and U. Woggon, *J. Appl. Phys.* 81, 6975 (1997).
95. L. E. Brus, *J. Chem. Phys.* 80, 4403 (1984); M. G. Bawendi, A. R. Kortan, M. L. Steigerwald, and L. E. Brus, *J. Chem. Phys.* 91, 7282 (1989).
96. A. P. Alivisatos, A. L. Harris, N. J. Levinos, M. L. Steigerwald, and L. E. Brus, *J. Chem. Phys.* 89, 4001 (1998).
97. M. G. Bawendi, P. J. Carroll, W. L. Wilson, and L. E. Brus, *J. Chem. Phys.* 96, 946 (1992); A. P. Alivisatos, A. L. Harris, N. J. Levinos, M. L. Steigerwald, and L. E. Brus, *J. Chem. Phys.* 89, 4001 (1998).
98. M. G. Bawendi, P. J. Carroll, W. L. Wilson, and L. E. Brus, *J. Chem. Phys.* 96, 946 (1992); D. J. Norris and M. G. Bawendi, *J. Chem. Phys.* 103, 5260 (1995).
99. V. P. Kunets, *Phys. Stat. Sol. B* 209, 179 (1998); C. Ma, D. Moore, Y. Ding, J. Li, and Z. L. Wang, *Int. J. Nanotech.* 1, 431 (2004).
100. F. Vigue, P. Venegues, S. Vezian, M. Laugt, and J. Faurie, *Appl. Phys. Lett.* 79, 194 (2001).
101. J. C. H. Spence and J. M. Zuo, "Electron Microdiffraction." Plenum, New York, NY, 1992.
102. C. Ma, D. Moore, J. Li, and Z. L. Wang, *Adv. Mater.* 15, 228 (2003); <http://esi-topics.com/fmf/2005/july05-ZhongLinWang.html>.
103. Z. L. Wang, ESI Special Topics 2005; <http://esi-topics.com/fmf/2005/july05-ZhongLinWang.html>.
104. S. K. Hong, E. Kurts, J. H. Chang, T. Hanada, M. Oku, and T. Yao, *Appl. Phys. Lett.* 78, 165 (2001).
105. M. A. Haase, J. Qiu, J. M. Depuydt, and H. Cheng, *Appl. Phys. Lett.* 59, 1272 (1991).
106. M. T. Bjork, B. J. Ohlsson, T. Sass, A. I. Persson, C. Thelander, M. H. Magnusson, K. Deppert, L. R. Wallenberg, and L. Samuelson, *Nano Lett.* 2, 87 (2002).
107. M. S. Gudiksen, L. J. Lauhon, J. F. Wang, D. C. Smith, and C. M. Lieber, *Nature* 415, 617 (2002).
108. Y. Ding, C. Ma, and Z. L. Wang, *Adv. Mater.* 16, 1740 (2004).
109. Z. L. Wang, X. Y. Kong, and J. M. Zuo, *Phys. Rev. Lett.* 91, 185502 (2003).
110. C. Ma, D. Moore, J. Li, and Z. L. Wang, *Adv. Mater.* 15, 228 (2003).
111. C. Ma, Y. Ding, D. Moore, X. D. Wang, and Z. L. Wang, *J. Am. Chem. Soc.* 126, 708 (2004).
112. X. T. Zhang, K. M. Ip, Z. Liu, Y. P. Leung, Q. Li, and S. K. Hark, *Appl. Phys. Lett.* 84, 2641 (2004).

113. J. H. Fendler, *Chem. Mater.* 8, 1616 (1996).
114. Y. J. Xiong, Y. Xie, J. Yang, R. Zhang, C. Z. Wu, and G. Du, *J. Mater. Chem.* 12, 3712 (2002); G. N. Karanikolas, N. L. Law, R. Mallory, A. Petrou, P. Alexandridis, and T. J. Mountziaris, *Nanotech.* 17, 3121 (2006).
115. M. Remskar, *Adv. Mater.* 16, 1497 (2004).
116. Y. H. Leung, W. M. Kwok, A. B. Djurisic, D. L. Phillips, and W. K. Chan, *Nanotech.* 16, 579 (2005).
117. Y. Gu, I. L. Kuskovsky, M. Yin, S. O'Brien, and G. F. Neumark, *Appl. Phys. Lett.* 85, 3833 (2004); R. D. Dupius and P. D. Dapkus, *Appl. Phys. Lett.* 32, 473 (1978).
118. Y. Gu, I. L. Kuskovsky, M. Yin, S. O'Brien, and G. F. Neumark, *Appl. Phys. Lett.* 85, 3833 (2004).
119. Y. W. Heo, M. Kaufman, K. Pruessner, K. N. Siebein, D. P. Norton, and F. Ren, *Appl. Phys. A* 80, 263 (2005).
120. E.-S. Jang, J. Y. Bae, J. Yoo, W. I. Park, D.-W. Kim, G.-C. Yi, T. Yatsui, and M. Ohtsu, *Appl. Phys. Lett.* 88, 023102 (2006).
121. B. P. Zhang, N. T. Binh, K. Wakatsuki, Y. Segawa, Y. Yamada, N. Usami, M. Kawasaki, and H. Koinuma, *Appl. Phys. Lett.* 84, 4098 (2004).
122. W. I. Park, S. J. An, J. L. Yang, G.-C. Yi, S. Hong, T. Joo, and M. Kim, *J. Phys. Chem. B* 108, 15457 (2004).
123. Y. Ding, X. Y. Kong, and Z. L. Wang, *Phys. Rev. B* 70, 235408 (2004).
124. Q. Chang, C. Chang, X. Zhang, H. Ye, G. Shi, W. Zhang, Y. Wang, X. Xin, and Y. Song, *Opt. Commun.* 274, 201 (2007).
125. H. Kohler, *Sol. Stat. Commun.* 11, 1687 (1972); J. H. Kim, Y. C. Hong, and H. S. Uhm, *Jpn. J. Appl. Phys.* 46, 4351 (2007).
126. F. A. Benko and F. P. Koffyberg, *Sol. Stat. Commun.* 57, 901 (1986).
127. D. M. Carballeda-Galicia, R. Castaneda-Perez, O. Jimenez-Sandoval, S. Jimenez-Sandoval, G. Torres-Delgado, and C. I. Zuniga-Romero, *Thin Solid Films* 371, 105 (2000).
128. B. J. Lokhande and M. D. Uplane, *Mater. Res. Bull.* 36, 439 (2001).
129. T. K. Subramanyam, S. Uthanna, and B. S. Naidu, *Mater. Lett.* 35, 214 (1998).
130. N. Matsuura, D. J. Johnson, and D. T. Amm, *Thin Solid Films* 295, 260 (1997).
131. A. Gulino, G. Tabbi, and A. A. Scalisi, *Chem. Mater.* 15, 3332 (2003).
132. S.-H. Kan, T. Mokari, E. Rothenberg, and U. Banin, *Nature Mater.* 2, 155 (2003).
133. A. D. Yoffe, *Adv. Phys.* 42, 173 (1993).
134. B. L. Evans and P. A. Young, *Proc. R. Soc. Lond. Ser. A* 298, 74 (1967).
135. W. U. Huynh, J. J. Dittmer, and A. P. Alivisatos, *Science* 295, 2425 (2002).
136. X. Duan, Y. Huang, Y. Cui, J. Wang, and C. M. Lieber, *Nature* 409, 66 (2001).
137. M. Kazes, D. Y. Lewis, Y. Ebenstein, T. Mokari, and U. Banin, *Adv. Mater.* 14, 317 (2002).
138. A. D. Yoffe, *Adv. Phys.* 42, 173 (1993); A. D. Yoffe, *Adv. Phys.* 50, 1 (2001).
139. O. L. Míćić, J. Sprague, Z. Lu, and A. J. Nozik, *Appl. Phys. Lett.* 68, 3150 (1996).
140. O. I. Míćić, S. P. Ahrenkiel, and A. J. Nozik, *Appl. Phys. Lett.* 78, 4022 (2001).
141. S.-H. Kan, T. Mokari, E. Rothenberg, and U. Banin, *Nature Mater.* 2, 155 (2003); W. E. Buhro and V. L. Colvin, *Nat. Mater.* 2, 138 (2003).
142. X. Duan, Y. Huang, Y. Cui, J. Wang, and C. M. Lieber, *Nature* 409, 66 (2001).
143. S. K. Prasad, "Advanced Technology, Encyclopedia of Nanoscience." p. 214, Publishing House Pvt. Ltd., 2008.
144. S.-H. Kan, T. Mokari, E. Rothenberg, and U. Banin, *Nature Mater.* 2, 155 (2003); W. E. Buhro and V. L. Colvin, *Nature Mater.* 2, 138 (2003).
145. L.-S. Li, J. Hu, W. Yang, and A. P. Alivisatos, *Nano Lett.* 1, 349 (2001).
146. J. Hu, L. Li, W. Yang, L. Manna, L. Wang, and A. P. Alivisatos, *Science* 292, 2060 (2001).
147. J. Wang, M. S. Gudiksen, X. Duan, Y. Cui, and C. M. Lieber, *Science* 293, 1455 (2001).
148. F. Patolsky, G. Zheng, and C. M. Lieber, *Anal. Chem.* 78, 4260 (2006).
149. C. S. Lao, P. X. Gao, R. S. Yang, Y. Zhang, Y. Dai, and Z. L. Wang, *Chem. Phys. Lett.* 417, 359 (2005).
150. P. X. Gao, C. S. Lao, W. L. Hughes, and Z. L. Wang, *Chem. Phys. Lett.* 408, 174 (2005).
151. C.-Y. Wu, H.-C. Hsu, H.-M. Cheng, S. Yang, and W.-F. Hsieh, *J. Cryst. Growth* 287, 189 (2006).
152. T. B. Hur, Y. H. Hwang, and H. K. Kim, *Appl. Phys. Lett.* 86, 193113 (2005).
153. A. I. Efros and A. L. Efros, *Sov. Phys. Semicond.* 16, 772 (1982).
154. L. Brus, *J. Phys. Chem.* 90, 2555 (1986); X. Jie, L. Wei, H. Yongjie, W. Yue, Y. Hao, and C. M. Lieber, *Nature* 441, 489 (2006).
155. Q. Fang, L. Yat, S. Gradecak, D. Wang, C. J. Barrelet, and C. M. Lieber, *Nano Lett.* 4, 1975 (2004).
156. T. B. Hoang, L. V. Titova, J. M. Yarrison-Rice, H. E. Jackson, A. O. Govorov, Y. Kim, H. J. Joyce, H. H. Tan, C. Jagadish, and L. M. Smith, *Nano Lett.* 7, 588 (2007).
157. S. C. Dahlberg and W. A. Orr, *Surf. Sci.* 67, 226 (1977).
158. B. Boudart, B. Mari, and B. Prevot, *Mater. Sci. Eng. B* 20, 109 (1993).
159. M. Passlack, M. Hong, R. L. Opila, J. P. Mannaerts, and J. R. Kwo, *Appl. Surf. Sci.* 104–105, 441 (1996).
160. D. Englund, H. Altug, and J. Vuckovic, *Nature Phys.* 2, 484 (2006); D. Englund, H. Altug, I. Fushman, and J. Vuckovic, *Appl. Phys. Lett.* 91, 071126 (2007).
161. Open source <http://www.ohio.edu/people/diao/papers/qd.pdf>.
162. D. W. Palmer, Properties of the Semiconductors, <http://www.semiconductors.co.uk> (2002).
163. C. G. Van de Walle, *Phys. Rev. B* 39, 1871 (1989).
164. N. Preschilla, S. Major, and N. Kumar, *Appl. Phys. Lett.* 77, 1861 (2000); P. Perlin, E. Litwin-Staszewska, B. Suchanek, W. Knap, J. Camassel, T. Suski, R. Piotrkowski, I. Grzegory, S. Porowski, E. Kaminska, and J. C. Chervin, *Appl. Phys. Lett.* 68, 1114 (1996).

165. S. A. BenNasrallah, S. B. Afia, H. Belmabrouk, and M. Said, *Eur. Phys. J. B* 43, 3 (2005); S. A. Nasrallah, N. Sfina, N. Bouarissa, and M. Said, *J. Phys.: Condens. Matter* 18, 3005 (2006).
166. W. Q. Chen and T. G. Andersson, *J. Appl. Phys.* 73, 4484 (1993).
167. S. Abdi-Ben Nasrallah, N. Sfina, N. Bouarissa, and M. Said, *J. Phys. D: Appl. Phys.* 18, 3005 (2006).
168. R. L. Greene, K. K. Bajaj, and D. E. Phelps, *Phys. Rev. B* 29, 1807 (1984).
169. J.-W. Wu, *Sol. Stat. Commun.* 69, 1057 (1989).
170. R. T. Senger and K. K. Bajaj, *Phys. Stat. Sol. B* 241, 1896 (2004).
171. C. Onodera, T. Shoji, Y. Hiratate, and T. Taguchi, *Jpn. J. Appl. Phys.* 45, 5821 (2006).
172. N. Sounderya and Y. Zhang, *Recent Patents on Biomedical Engineering*, Bentham Science Publishers Ltd. Edition, 2008 Vol. 1, p. 34.
173. L. M. Manocha, "Composites with Nanomaterials." (K. E. Geckeler, Ed.), American Scientific Publishers, California, 2006.
174. C. Onodera, T. Shoji, Y. Hiratate, and T. Taguchi, *Jpn. J. Appl. Phys.* 45, 5821 (2006).
175. B. S. E. Mihai, N. Chunming, and C. S. Erik, US20070122101A1 (2007); L. M. Manocha, "Composites with Nanomaterials." (K. E. Geckeler, Ed.), American Scientific Publishers, California 2006.
176. D. Dorfs and A. Eychmuller, *Z. Phys. Chem.* 220, 1539 (2006).
177. M. A. Hines and P. Guyot-Sionnest, *J. Phys. Chem.* 100, 468 (1996).
178. B. O. Dabbousi, J. RodriguezViejo, F. V. Mikulec, J. R. Heine, H. Mattoussi, R. Ober, K. F. Jensen, and M. G. Bawendi, *J. Phys. Chem. B* 101, 9463 (1997).
179. A. Mews, A. Eychmuller, M. Giersig, D. Schooss, and H. Weller, *J. Phys. Chem.* 98, 934 (1994).
180. D. Battaglia, J. J. Li, Y. J. Wang, and X. G. Peng, *Angew. Chem. Int. Ed.* 42, 5035 (2003); H. M. Kim, D. S. Kim, D. Y. Kim, T. W. Kang, Y.-H. Cho, and K. S. Chung, *Appl. Phys. Lett.* 81, 2193 (2002).
181. X. H. Zhong, R. G. Xie, Y. Zhang, T. Basche, and W. Knoll, *Chem. Mater.* 17, 4038 (2005); G. Liang, J. Xiang, N. Kharche, G. Klimeck, C. M. Lieber, and M. Lundstrom, *Nano Lett.* 7, 642 (2007).
182. Y. Li, J. Xiang, F. Qian, S. Gradeak, Y. Wu, H. Yan, D. A. Blom, and C. M. Lieber, *Nano Lett.* 6, 1468 (2006).
183. S. Adachi, *J. Appl. Phys.* 58, R1 (1985).
184. R. Dingle, H. L. Stormer, A. C. Gossard, and W. Wiegmann, *Appl. Phys. Lett.* 33, 665 (1978).
185. T. J. Thornton, M. Pepper, H. Ahmed, D. Andrews, and G. J. Davies, *Phys. Rev. Lett.* 56, 1198 (1986).
186. L. D. Nguyen, L. E. Larson, and U. K. Mishra, *Proc. IEEE* 80, 494 (1992); ray spectroscopy (EDS) elemental mappings for Al, Ga, and As, M. J. Tambe, S. K. Lim, M. J. Smith, L. F. Allard, and S. Gradeak, *Appl. Phys. Lett.* 93, 151917 (2008).
187. L. V. Titova, T. B. Hoang, H. E. Jackson, L. M. Smith, J. M. Yarrison-Rice, and Y. Kim, *Appl. Phys. Lett.* 89, 173126 (2006).
188. L. V. Titova, T. B. Hoang, H. E. Jackson, and L. M. Smith, *Appl. Phys. Lett.* 89, 173126 (2006).
189. J. Wang, M. S. Gudiksen, X. Duan, Y. Cui, and C. M. Lieber, *Science* 293, 1455 (2001).
190. J. C. Johnson, H. Q. Yan, R. D. Schaller, L. H. Haber, R. J. Saykally, and P. D. Yang, *J. Phys. Chem. B* 105, 11387 (2001).
191. A. D. Yoffe, *Adv. Phys.* 42, 173 (1993).
192. X. Duan, J. Wang, and C. M. Lieber, *Appl. Phys. Lett.* 76, 1116 (2000).
193. C. Chen, S. Shehata, C. Fradin, R. LaPierre, C. Couteau, and G. Weihs, *Nano Lett.* 7, 2584 (2007); M. S. Gudiksen, J. Wang, and C. M. Lieber, *J. Phys. Chem. B* 106, 4036 (2002).
194. J. Hu, T. W. Odom, and C. M. Lieber, *Acc. Chem. Res.* 32, 435 (1999).
195. C. Thelander, M. T. Bjork, M. W. Larsson, A. E. Hansen, L. R. Wallenberg, and L. Samuelson, *Sol. Stat. Commun.* 131, 573 (2004).
196. Z. Zanolli, L. E. Froberg, M. T. Bjork, M.-E. Pistol, and L. Samuelson, *Thin Solid Films* 515, 793 (2007); Z. Zanolli, M.-E. Pistol, L. E. Froberg, and L. Samuelson, *J. Phys.: Condens. Matter* 19, 295219 (2007).
197. C. M. Lieber and Z. L. Wang, *MRS Bull.* 32, 99 (2007); X. Fang, Y. Bando, U. K. Gautam, T. Zhai, S. Gradeak, and D. Golberg, *J. Materials Chemistry* 19, 5683 (2009).
198. X. S. Fang, Y. Bando, U. K. Gautam, C. H. Ye, and D. Golberg, *J. Mater. Chem.* 18, 509 (2008); X. S. Fang, Y. Bando, G. Z. Shen, C. H. Ye, U. K. Gautam, P. M. F. J. Costa, C. Y. Zhi, C. C. Tang, and D. Golberg, *Adv. Mater.* 19, 2593 (2007).
199. F. Qian, Y. Li, S. Gradeak, D. Wang, C. J. Barrelet, and C. M. Lieber, *Nano Lett.* 4, 1975 (2004); F. Qian, S. Gradeak, Y. Li, C. Wen, and C. M. Lieber, *Nano Lett.* 5, 2287 (2005).
200. A. J. Mieszawska, R. Jalilian, G. U. Sumanasekera, and F. P. Zamborini, *Small* 3, 722 (2007).
201. M. T. Bjork, B. J. Ohlsson, T. Sass, A. I. Persson, C. Thelander, M. H. Magnusson, K. Deppert, L. R. Wallenberg, and L. Samuelson, *Nano Lett.* 2, 87 (2002).
202. C. M. Lieber, *Nano Lett.* 2, 81 (2002).
203. U. K. Gautam, X. S. Fang, Y. Bando, J. H. Zhan, and D. Golberg, *ACS Nano* 2, 1015 (2008).
204. L. D. Hicks and M. S. Dresselhaus, *Phys. Rev. B* 47, 16631 (1993); Y. M. Lin and M. S. Dresselhaus, *Phys. Rev. B* 68, 075304 (2003).
205. C. M. Lieber, X. C. Jiang, Q. H. Xiong, S. Nam, F. Qian, and Y. Li, *Nano Lett.* 7, 3214 (2007); Y. Li, J. Xiang, F. Qian, S. Gradeak, Y. Wu, H. Yan, D. A. Blom, and C. M. Lieber, *Nano Lett.* 6, 1468 (2006).
206. International technology roadmap for semiconductors, www.itrs.net/reports.
207. X. Fang, Y. Bando, U. K. Gautam, T. Zhai, S. Gradeak, and D. Golberg, *J. Mater. Chem.* 19, 5683 (2006).
208. E. C. Garnett and P. Yang, *J. Am. Chem. Soc.* 130, 9224 (2008).

209. F. Qian, Y. Li, S. Gradecak, H. G. Park, Y. J. Dong, Y. Ding, Z. L. Wang, and C. M. Lieber, *Nature Mater.* 7, 701 (2008).
210. M. Tchernycheva, C. Sartel, G. Cirlin, L. Travers, G. Patriarche, L. Largeau, O. Mauguin, J.-C. Harmand, Le Si Dang, J. Renard, B. Gayral, L. Nevou, and F. Julien, *Phys. Stat. Sol. C* 5, 1556 (2008).
211. M. A. Hines and P. Guyot-Sionnest, *J. Phys. Chem.* 100, 468 (1996); O. Madelung, M. Schulz, and H. Weiss, "Numerical Data and Functional Relationships in Science and Technology, New Series, Group III: Crystal and Solid State Physics." Vol. III/17b, Springer, Berlin, 1982.
212. S. Kudara, M. Zanella, C. Giannini, A. Rizzo, Y. Q. Li, G. Gigli, R. Cingolani, G. Ciccarella, W. Spahl, W. J. Parak, and L. Manna, *Adv. Mater.* 19, 548 (2007).
213. S. Jun and E. Jang, *Chem. Commun.* 4616 (2005).
214. P. Reiss, M. Protiere, and L. Li, *Small* 5, 154 (2009); D. V. Talapin, I. Mekis, S. Gotzinger, A. Kornowski, O. Benson, and H. Weller, *J. Phys. Chem. B* 108, 18826 (2004).
215. Y. W. Lin, M. M. Hsieh, C. P. Liu, and H. T. Chang, *Langmuir* 21, 728 (2005).
216. Y. J. Lee, T. G. Kim, and Y. M. Sung, *Nanotechnology* 17, 3539 (2006); S. A. Ivanov, A. Piryatinski, J. Nanda, S. Tretiak, W. O. Wallace, and V. I. Klimov, *J. Am. Chem. Soc.* 129, 11708 (2007).
217. Y. M. Sung, K. S. Park, Y. J. Lee, and T. G. Kim, *J. Phys. Chem. C* 111, 1239 (2007).
218. J. S. Steckel, P. Snee, S. Coe-Sullivan, J. R. Zimmer, J. E. Halpert, P. Anikeeva, L. A. Kim, V. Bulovic, and M. G. Bawendi, *Angew. Chem. Int. Ed.* 45, 5796 (2006).
219. M. Protiere and P. Reiss, *Small* 3, 399 (2007).
220. S. Haubold, M. Haase, A. Kornowski, and H. Weller, *Chem. Phys. Chem.* 2, 331 (2001).
221. M. S. Gudiksen, L. J. Lauhon, J. Wang, D. C. Smith, and C. M. Lieber, *Nature* 415, 617 (2002).
222. Y. Zhang, K. Suenaga, C. Colliex, and S. Iijima, *Science* 281, 973 (1998).
223. Z. L. Wang, Z. R. Dai, R. P. Gao, Z. G. Bai, and J. L. Gole, *Appl. Phys. Lett.* 77, 3349 (2000).
224. Y. Zheng, L. Zheng, Y. Zhan, X. Lin, Q. Zheng, and K. Wei, *Inorg. Chem.* 46, 6980 (2007).
225. J. H. Choy, E. S. Jang, J. H. Won, J. H. Chung, D. J. Jang, and Y. W. Kim, *Adv. Mater.* 15, 1911 (2003).
226. Y. B. Li, Y. Bando, T. Sato, and K. Kurashima, *Appl. Phys. Lett.* 81, 144 (2002).
227. R. C. Wang, C. P. Liu, J. L. Huang, and S. J. Chen, *Appl. Phys. Lett.* 88, 023111 (2006).
228. E. R. Segnit and A. E. Holland, *J. Am. Ceram. Soc.* 48, 412 (1965); W. Yang, S. S. Hullavarad, B. Nagaraj, I. Takeuchi, R. P. Sharma, T. Venkatesan, R. D. Vispute, and H. Shen, *Appl. Phys. Lett.* 82, 3424 (2003).
229. Y.-J. Hsu and S.-Y. Lu, *Chem. Commun.* 15, 1350 (2005); W. I. Park, S. J. An, J. L. Yang, G. C. Yi, S. Hong, T. Joo, and M. Kim, *J. Phys. Chem. B* 108, 15457 (2004).
230. I. Levin, A. Davydov, B. Nikoobakht, N. Sanford, and P. Mogilevsky, *Appl. Phys. Lett.* 87, 103110 (2005); S. Xu, J. Ziegler, and T. Nann, *J. Materials Chem.* 18, 2653 (2008).
231. D. V. Talapin, N. Gaponik, H. Borchert, A. L. Rogach, M. Haase, and H. Weller, *J. Phys. Chem. B* 106, 12659 (2002).
232. L. Li, M. Protiere, and P. Reiss, *Chem. Mater.* 20, 2621 (2008).
233. M. Brumer, A. Kigel, L. Amirav, A. Sashchiuk, O. Solomesch, N. Tessler, and E. Lifshitz, *Adv. Funct. Mater.* 15, 1111 (2005).
234. E. Lifshitz, M. Brumer, A. Kigel, A. Sashchiuk, M. Bashouti, M. Sirota, E. Galun, Z. Burshtein, A. Q. Le Quang, I. Ledoux-Rak, and J. Zyss, *J. Phys. Chem. B* 110, 25356 (2006).
235. D. V. Talapin, H. Yu, E. V. Shevchenko, A. Lobo, and C. B. Murray, *J. Phys. Chem. C* 111, 14049 (2007).
236. J. M. Pietryga, D. J. Werder, D. J. Williams, J. L. Casson, R. D. Schaller, V. I. Klimov, and J. A. Hollingsworth, *J. Am. Chem. Soc.* 130, 4879 (2008).
237. C. H. Ahn, S. K. Mohanta, B. H. Kong, and H. K. Cho, *J. Phys. D: Appl. Phys.* 42, 115106 (2009); P. Reiss, M. Protiere, and L. Li, *Small* 5, 154 (2009).
238. W. I. Park, G.-C. Yi, M. Kim, and S. J. Pennycook, *Adv. Mater.* 15, 526 (2003).
239. S. W. Jung, W. I. Park, G.-C. Yi, and M. Kim, *Adv. Mater.* 15, 1358 (2003).
240. D. Li, Y. Wu, R. Fan, P. Yang, and A. Majumdar, *Appl. Phys. Lett.* 83, 3186 (2003).
241. X. Wang, C. J. Summers, and Z. L. Wang, *Adv. Mater.* 16, 1215 (2004).
242. C. Thelander, T. Mårtensson, M. T. Björk, B. J. Ohlsson, M. W. Larsson, L. R. Wallenberg, and L. Samuelson, *Appl. Phys. Lett.* 83, 2052 (2003).
243. Y. Wu, J. Xiang, C. Yang, W. Lu, and C. M. Lieber, *Nature* 430, 61 (2004); W. I. Park, S. J. An, G.-C. Yi, and M. Kim, *Fourth IEEE Conf. Proc. Nanotech.* p. 83, New York (2004).
244. S. Kim, B. Fisher, H. J. Eisler, and M. Bawendi, *J. Am. Chem. Soc.* 125, 11466 (2003).
245. K. Yu, B. Zaman, S. Romanova, D. S. Wang, and J. A. Ripmeester, *Small* 1, 332 (2005).
246. P. T. K. Chin, C. D. M. Donega, S. S. Bavel, S. C. J. Meskers, N. Sommerdijk, and R. A. J. Janssen, *J. Am. Chem. Soc.* 129, 14880 (2007).
247. C. Y. Chen, C. T. Cheng, C. W. Lai, Y. H. Hu, P. T. Chou, Y. H. Chou, and H. T. Chiu, *Small* 1, 1215 (2005).
248. R. G. Xie, X. H. Zhong, and T. Basche, *Adv. Mater.* 17, 2741 (2005).
249. D. J. Milliron, S. M. Hughes, Y. Cui, L. Manna, J. B. Li, L. W. Wang, and A. P. Alivisatos, *Nature* 430, 190 (2004).
250. R. G. Xie, U. Kolb, and T. Basche, *Small* 2, 1454 (2006).
251. L. P. Balet, S. A. Ivanov, A. Piryatinski, M. Achermann, and V. I. Klimov, *Nano Lett.* 4, 1485 (2004).
252. J. Bleuse, S. Carayon, and P. Reiss, *Physica E* 21, 331 (2004); P. Reiss, S. Carayon, J. Bleuse, and A. Pron, *Synthetic Materials* 139, 649 (2003).
253. R. G. Xie, U. Kolb, J. X. Li, T. Basche, and A. Mews, *J. Am. Chem. Soc.* 127, 7480 (2005).

- 254. S. W. Kim, J. P. Zimmer, S. Ohnishi, J. B. Tracy, J. V. Frangioni, and M. G. Bawendi, *J. Am. Chem. Soc.* 127, 10526 (2005).
- 255. A. Aharoni, T. Mokari, I. Popov, and U. Banin, *J. Am. Chem. Soc.* 128, 257 (2006).
- 256. C. T. Cheng, C. Y. Chen, C. W. Lai, W. H. Liu, S. C. Pu, P. T. Chou, Y. H. Chou, and H. T. Chiu, , *J. Mater. Chem.* 15, 3409 (2005); M. Zach, C. Hagglund, D. Chakarov, and B. Kasemo, *Current Opinion in Solid State Mater. Science* 10, 132 (2006).
- 257. P. Reiss, S. Carayon, J. Bleuse, and A. Pron, *Synthetic Metals* 139, 649 (2003).